

Airy asymptotics for tree-child networks: fixed combining degree, and the binary total count, deficit law and inverse

AI-assisted research report, merged from four manuscripts

Sources dated 2 October 2026

Abstract

For every fixed integer $d \geq 2$, we prove a positive limiting amplitude and expansions to every finite Airy order for maximally reticulated d -combining tree-child networks and for their word diagonal a_n ; at $d = 2$ this diagonal is OEIS A213863. A shifted partial-sum recurrence gives an exact contractive Jacobi reduction; global spectral confinement, a parity singular gap, summable adjoint drift and Catalan boundary smoothing establish the positive limit, and polynomial–Airy profiles with a zero-limit bootstrap transfer every finite formal order to the exact counts (Part I, source 16). For the binary case $d = 2$, Part II (source 69) gives a detailed second route with explicit coefficients, transfers every finite order to the total number of binary tree-child networks through the recently proved tableau-to-network identity, derives weighted all-orders expansions of the reticulation deficit around its Poisson(1/2) limit, an exact eventual total-variation identity with a sharp rate, and explicit Lambert W inversions with integer brackets. Ternary-combining coefficients are given; for total counts at $d \geq 3$ only the leading equivalent is claimed. The amplitudes have convergent exact definitions; numerical diagnostics are not certified enclosures. This is an AI-assisted research report, unrefereed and not formalized; it claims no external peer review or publication.

[5 October 2026, batch 102.] Two further manuscripts, written independently of Parts I–II, prove the main theorem again. Part III (source 104) works for every fixed d in an exact gamma-product gauge and adds closed forms of the first two coefficients for every d , the first of them negative for $d \geq 4$, and, resting on its reading of a published proof, the first total-count correction at $d = 3$ with relative remainder $O(n^{-1/2})$; it also gives $d = 4$ prefixes. Part IV (source 102) treats $d = 2$ in a factorial gauge and adds the fifth and sixth logarithmic coefficients, the successive ratio through n^{-3} , and an $O(n^{-2/3})$ ratio and total-variation rate that does not use the identity on which Part II’s transfer rests.

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1 Provenance, scope and notation

This section was written when the two manuscripts were merged into this report (batch 77, 2 October 2026). Elsewhere, text added in writing is marked “[Write phase: ...]”.

1.1 Sources

The report is built from two manuscripts of batch 77 of the repository’s incoming research reports. *[5 October 2026: and, since batch 102, from two further manuscripts, sources 104 and 102; see the last two rows of the table and the note below it.]* Both archives arrived in commit 096ee7b87 and were placed in this directory by commit 34f1acd4b (batch 77P3). Neither manuscript names a repository commit, so neither has a repository pin.

Source	Archive and main file	Shipped as	Printed as
16 (base)	d-combining-asymptotics-reproducibility.zip, d-combining-asymptotics.tex (406 lines, 10-page PDF)	article.tex (rewritten), files 16-dcomb-*	Part I (Sections 2–13)
69	tree-child-asymptotics-reproducibility.zip, article/tree-child-asymptotics.tex (1,233 lines, 23-page PDF)	files 69-binary-*; manuscript not shipped	Part II (Sections 14–20)
104	Fixed_Degree_Tree_Child_Networks_All_Orders_Source.zip, general_tree_child_amplitudes.tex (926 lines, 22-page PDF)	files 104-fixdeg-*; manuscript not shipped	Part III (Sections 21–32)
102	A213863_Tableaux_and_Tree_Child_Networks_Source.zip, tree_child_amplitude_sources/tree_child_amplitude.tex (903 lines, 21-page PDF)	files 102-table aux-*; manuscript not shipped	Part IV (Sections 33–41)

Source 16 proves the theorem for every fixed combining degree $d \geq 2$. Source 69 proves the binary case $d = 2$ independently and adds results that exist only at $d = 2$. Source 16’s integrated audit pins its manuscript at SHA-256 70a1aab9..., the delivered file. Source 16’s first audit also hashed the binary manuscript it compared against, 86217958...; that is exactly the delivered manuscript of source 69, whose own verification record pins the same hash. So source 16 was checked against the text printed here as Part II. Source 16 calls source 69 “the binary research report” and uses it “for comparison and algebraic regression” only; source 69 does not cite source 16.

[5 October 2026, batch 102.] Two bundle manuscripts were added as Parts III and IV: Reports 104 and 102 of the session bundle, numbered here as sources 104 and 102 after their file prefixes. Both arrived in commit 60f54ea06 and were placed by 6ab1f1979; neither names a repository commit, so neither has a pin. Their archives were closed on 1 October 2026 (Pacific time), before the batch-77 archives of sources 16 and 69 arrived in 096ee7b87, so both prove the main theorem independently of Parts I–II; neither cites them, and source 102 does not cite source 104. Theorem 2.1 was printed here first; sources 104 and 102 are credited as independent further routes (Sections 21.1 and 33.1). Source 104 is dated 2 October 2026 with the author line “Research prepared for Vladimir / Report 104”; source 102 is dated 2 October 2026 with the author line “Research prepared for private review” (title page and PDF metadata), disclosed here.

1.2 How the two manuscripts are merged

1. *Base.* The general- d theorem of source 16 is the more general statement, so source 16 is Part I and is printed in full. Its abstract opens Part I, and source 69’s abstract opens Part II.
2. *The binary proof as a second route.* Sections 2–8 of source 69 prove Theorem 2.1 at $d = 2$ by the same argument, section by section. They also contain binary-only explicit material: the explicit quasimode and eigenvalue through ε^5 , the scalar coefficients $s_2, \dots, s_7$, the endpoint expansion, and the fourth logarithmic coefficient. They are printed in full as Section 15, a detailed second proof at $d = 2$. Displays that are literally identical in the two sources are printed once, in Part I: the polynomial–Airy identity (17), the triangular equation (18), the convolution estimate (24) and the Catalan smoothing bound (26). Part II points to them.
3. *Binary-only results.* Source 69’s sharpened theorem with explicit coefficients (Theorem 14.1), the transfer to total counts (Section 16), the inversion (Section 17), the deficit law and total variation (Section 18), its numerical diagnostics and its literature status are printed in full.
4. *Citations.* Source 16’s bibliography entry for “the binary report” (it had no locator) is replaced by references to Part II. The two bibliographies are merged. Source 69’s `\href` citations inside the text are now bibliography citations.
5. *Environments.* Sections 9 and 11 of source 69 were delivered as converted Markdown, with theorems as bold paragraphs. They are now theorem and proof environments. Their equation tags, T1–T24 and D1–D14, are kept, and each tag now has a label (source 69 has no tag D12).
6. *Additions and corrections.* Remark 5.1 spells out source 16’s one-paragraph compactness step from the derivations shipped with both sources. Remark 11.1 relates both inversions to the repository’s inversion volume. Source 69’s sentence about an “extended-precision” run is corrected in Section 19, because that run is vacuous on platforms where NumPy’s `longdouble` is double precision. Source 16’s first open question is re-scoped in Section 12, since Part II settles it at $d = 2$. Its description of a “companion archive” is corrected in Section 13 to the files actually shipped.
7. *Notation.* Symbols that the two sources use for different notions, or that one source uses twice, are renamed as listed in Section 1.4. No formula changed otherwise.
8. *[5 October 2026, batch 102.] Parts III and IV.* Source 104 proves the theorem for every fixed d and is the more general of the two new sources, so it is Part III, printed in full; source 102 ($d = 2$, same architecture, another gauge) is Part IV, printed with its binary-specific material in full and with pointers to Part III where its displays are literally Part III’s at $d = 2$ (the formal system and algorithm, the residual estimate, the forced-product proof). Each Part opens with its source’s abstract and a section on provenance, credit, a map onto Parts I–II, and its own notation table; each closes with a “Further questions and research” section. Nothing in Parts I–II was renumbered; dated notes record where the new Parts answer, sharpen or bear on them.

1.3 What is claimed and what is not

Claimed, for each fixed $d \geq 2$: a positive amplitude γ_d , defined by convergent limits, and expansions to every finite order in $n^{-1/3}$ for the word diagonal a_n and for the maximal-reticulation counts $T_{n,n-1}^{(d)}$. For $d \geq 3$ the total count $T_n^{(d)}$ gets only its leading equivalent. The ternary coefficients are

explicit, and there is Lambert W inversion with integer brackets. At $d = 2$, Part II adds explicit coefficients, every finite order for the total number TC_n of binary tree-child networks (with no new constant), weighted all-orders expansions of the reticulation deficit, the exact eventual identity $d_{\text{TV}} = \frac{3}{2}(1/\mathcal{R}_n - e^{-1/2})$ with its expansion, moment corrections, and explicit inverses.

Not claimed, by either source: uniformity in d as $d \rightarrow \infty$; convergence of any infinite correction series; exponentially small or transseries corrections beyond all finite orders; higher-order total-count corrections for $d \geq 3$; a $d \geq 3$ OEIS accession (none was verified, assigned or submitted); a closed form, certified digits or an effective error constant for any amplitude; a certified threshold algorithm or an unconditional ceiling rule; novelty beyond a bounded literature search. Neither source imports the other's argument: source 16 calls the binary report “only a historical comparison” and does not import the binary Pons–Batle identity. Each source's audits are independent research review, not external peer review, publication, a computer-verified proof or a numerical certification, and neither source was published or submitted. In the repository the report is AI-assisted and unrefereed. No statement of it has a Lean or Rocq declaration, and its placement in the repository confers no formal status.

[5 October 2026, batch 102.] Parts III and IV add, for every fixed d , closed forms of $C_{d,1}$ and $C_{d,2}$ in $\mathbb{Q}[B]$; at $d = 3$, the first total-count correction with relative remainder $O(n^{-1/2})$, conditional on source 104's reading of a published proof (Remark 28.2); at $d = 2$, L_5 , L_6 , the successive ratio through n^{-3} and an identity-free $O(n^{-2/3})$ ratio and total-variation rate. The non-claims above are those of sources 104 and 102 as well: neither claims uniformity in d , convergence of a correction series, an exponentially small sector, certified amplitude digits, an all-order total expansion (at $d \geq 4$ only the leading total equivalent), an OEIS accession for $d \geq 3$, or exhaustive priority.

1.4 Notation across the two Parts

Part I keeps source 16's notation and Part II keeps source 69's, except for the renamings below. [5 October 2026, batch 102: Parts III and IV have their own notation tables, Sections 21.3 and 33.3; they use the letters below for the host's notions and rename the sources' clashing letters, notably their Airy zero α , which is z here.] A letter kept in both Parts means the same notion in both, with $d = 2$ in Part II. At $d = 2$: $\Lambda = 3$, $\Gamma = 12$, $c = 2/3$, $\ell = (2/3)^{2/3}z = 2^{2/3}B/3$, $B = 3^{1/3}z$, $\eta = 1/6$, $\rho = -2/3$, $\alpha = -5/3$, $\gamma = \gamma_2$, $\kappa = \kappa_2$ and $c_r = C_{2,r}$. Here $z = -2.338107410459767\dots$ is the largest zero of Ai in both sources.

Symbol here	Meaning	Source symbol; the false reading to avoid
d	combining degree ($d = 2$ in Part II)	source 69's generic-inverse d is $\mathbf{p}$, its degree bound d_R is ν_R , its $d_N(j)$ is $D_N(j)$; d_{TV} is total variation
$D_N(j)$	$A_{n,k}/(\Lambda^n(n!)^{d-1})$, $N = n + k$, $j = n - k$	source 69: $d_N(j) = A_{n,k}/(3^n n!)$, the same at $d = 2$
$p_N(j)$	downward weight (10); at $d = 2$ it is $(3N + j - 2)/(3(N + j))$	source 69: $a_{N,j}$, which is not the sequence a_n
$b_{n,m}$; $b_{N,j}$	the array (6) of Part I; the Jacobi edges $\sqrt{p_N(j)}$	$b_{n,k}$ in Remark 16.1 is the notation of the cited papers, quoted
$\beta_n(m)$	deficit ratio $A_{n,n-m}/a_n$ (Part II)	source 69: $b_n(m)$; not $b_{n,m}$
$q_{n,m}$; $\omega_{n,m}$, $\Omega_m(t)$	Part I's normalized array; Part II's deficit coefficient $(3n - m - 1)a_{n-1}/a_n$ and its formal series	source 69: $q_{n,m}$ and $Q_m(t)$; neither is Part I's $q_{n,m}$ or the gauge ratio $Q_N(j)$

Symbol here	Meaning	Source symbol; the false reading to avoid
$G_N(j); \mathcal{G}(t)$	gauge; $\exp \sum L_r t^r$ in (T8)	source 69: $G(t)$
$P_N; P_r(m); P, Q$	eigenvalue product; deficit polynomials (T11); polynomials in (17)	letters kept as in the shipped scripts
$C_{d,r}, c_r; C_r(m); C_*$	expansion coefficients ($c_r = C_{2,r}$); deficit correction polynomials (D2); the multiplier A_∞/q_0	C_* is C in both sources; C, C_d, C_R are generic constants
$L; L_r; \Delta$	the operator $\partial_x^2 - cx - \ell$; diagonal logarithmic coefficients (T4); the Duhamel window $\lfloor N^{2/3} \rfloor$	both sources write the window as L
$\mathfrak{p}, \mathfrak{r}, \mathbf{K}, \mathbf{L}_j, \mathfrak{a}, \gamma_\star$	factorial power, polynomial exponent, constant, coefficients, linear coefficient and amplitude of the smooth inversion model	source 16: p, r, K, L_j, a, A ; source 69's generic inverse in sans-serif, see Section 17
$\ell; \text{lr}_n(m)$	$c^{2/3}z$; likelihood ratio $\pi_n(m)/\pi_{1/2}(m)$	source 69: $\ell_n(m)$
$\alpha; \xi_N$	network exponent $-d(3d-1)/(2(d+1))$; ground coefficient of the error e_N	both sources: α_N
$\rho; \sigma; \text{res}_{n,m}$	exponent of n ; threshold error exponent; deficit residual (T14)	source 69: $\rho, \rho_{n,m}$
$\eta; \delta_Y$	$(d-1)^2/(2(d+1))$; threshold radius	source 69: η
$\Gamma; \Gamma(\cdot); \gamma_T$	the constant 4Λ ; Euler's Gamma function in (49) and in $\log \Gamma(t+1)$; $\sqrt{e}\gamma/12$	source 69: Γ_T
$a_d; a_2; \mathfrak{c}_d; a'$	parity-gap constant; the same at $d=2$; confinement constant; cutoff constant	source 69's gap constant is c_1 , not the coefficient $c_1 = B^2/18$; $\mathfrak{c}_d$ is c_d in source 16 and c_0 in source 69; a' is source 69's c_2
$T_{n,k}^{(d)}; T_{v,k}, T_v = \text{TC}_v$	networks with n leaves and k reticulations; the binary counts $T^{(2)}$ with v leaves	Part II writes TC_n and T_v for the same total
y	logarithm of a target value	source 69: T
τ_s	coefficients of $\mathcal{R}_n/\sqrt{e}$ (T5)	source 69: r_s ; not the residual r_N
$\mathcal{R}_n$	$T_{n+1}/((n+1)!a_n)$	source 69: R_n ; R is a truncation order
$X_n; X$	reticulation deficit; a Poisson(1/2) variable	source 69: M_n, M ; not the operator M_N
$\pi_n(m); \pi_{1/2}(m)$	law of X_n ; the Poisson(1/2) mass function	source 69: $p_n(m), \pi(m)$; π alone is the number
μ_n	$\lfloor \log n \rfloor$	source 69: h_n ; not h_r or h_N
$\mathcal{E}_n(m); \Upsilon_R, \Upsilon_J; \nu_R$	deficit error; its growth base; polynomial degree in (T14)	source 69: $e_n(m), D_R, D_J, d_R$
t	$n^{-1/3}$ in (T8); $(N/2)^{-1/3}$ in the coefficient-field arguments (equal at $N=2n$); a real index in Section 17	as in the sources

1.5 Relation to other reports of the repository

Airy amplitudes of trees and automata. The report `a082161-airy-amplitudes` in this collection (the sibling directory `../a082161-airy-amplitudes` of this report's directory, written in the same batch) proves positive amplitudes and all finite orders for relaxed and compacted trees and for minimal automata, by the same family of lemmas. Its relaxed-binary Part (source 62 of batch 77)

uses the polynomial–Airy identity $(P'' + 4(x + a)Q' + 2Q)F + (2P' + Q'')F'$. That is (17) with $c = 2$ and $\ell = 2a$, together with the same triangular inverse, normalization and zero-limit projection bootstrap. The objects, recurrences and sequences are different, the two families of manuscripts do not cite each other, and the reports are kept separate. Neither is an input to the other. [5 October 2026, batch 102: sources 104 and 102 belong to the same delivery series as bundle Reports 100, 101 and 103, which 6ab1f1979 placed as Part V of that report; they share its lemma family (positive Jacobi products, the killed half-line kernel, the forced-product bootstrap). The reports stay separate.]

Inversion. The Lambert W seeds, Newton refinements and integer brackets of Sections 11 and 17 are instances of the inversion machinery of the volume *Transseries and inversion* [7] (Remark 11.1). No novelty is claimed for them.

No other repository report treats tree-child networks, A213863 or d -combining networks.

Part I

Fixed combining degree $d \geq 2$ (source 16)

Source 16 (archive d-combining-asymptotics-reproducibility), printed in full; its title was “Airy asymptotics for fixed combining degree tree child networks. Its abstract reads: For every fixed integer $d \geq 2$, we prove a positive limiting amplitude and expansions to every finite Airy order for maximally reticulated d -combining tree-child networks. A shifted partial-sum recurrence gives an exact contractive Jacobi reduction with a finite-product edge weight. Global spectral confinement, a parity singular gap, summable adjoint drift and Catalan boundary smoothing establish the positive limit. Polynomial–Airy profiles and a zero-limit bootstrap transfer every finite formal order to the exact counts. We provide ternary-combining coefficients and Lambert W inversion. For total counts at $d \geq 3$, only the leading equivalent is claimed. The amplitude has a convergent exact definition; numerical diagnostics are not certified enclosures. This research report includes reproducible algebra and an independent mathematical audit, but does not claim external peer review or publication.

2 Results and enumeration context

Fix an integer $d \geq 2$. Every implicit constant may depend on d and the requested finite order; no uniformity for unbounded d is asserted. Set

$$\Lambda = \frac{(d+1)^{d-1}}{(d-1)!}, \quad \Gamma = 4\Lambda, \quad c = \frac{2(d-1)}{d+1}, \quad (1)$$

$$\ell = c^{2/3}z, \quad \eta = \frac{(d-1)^2}{2(d+1)}, \quad \rho = -\eta - \frac{1}{2}, \quad (2)$$

$$B = 3z \left(\frac{d-1}{d+1} \right)^{2/3}, \quad \alpha = -\frac{d(3d-1)}{2(d+1)}, \quad (3)$$

where $z = -2.338107410459767\dots$ is the largest negative zero of Ai . Let $T_{n,k}^{(d)}$ count tree-child networks with n labeled leaves and k reticulations under the root convention of Chang et al. [3]. Their exact maximal-word sequence is denoted here by a_n , with $a_0 = 1$.

Theorem 2.1. *There is $\gamma_d > 0$ and uniquely constructible $C_{d,r} \in \mathbb{Q}[B]$ such that, for each integer $M \geq 0$,*

$$a_n = \gamma_d (n!)^{d-1} \Gamma^n e^{Bn^{1/3}} n^\rho \left(1 + \sum_{r=1}^M C_{d,r} n^{-r/3} + O_{d,M}(n^{-(M+1)/3}) \right). \quad (4)$$

Moreover $T_{n,n-1}^{(d)} = n! a_{n-1}$ has an expansion to every finite order with leading term

$$T_{n,n-1}^{(d)} \sim \frac{\gamma_d}{\Gamma} (n!)^d \Gamma^n e^{Bn^{1/3}} n^\alpha. \quad (5)$$

For $d \geq 3$, the total count $T_n^{(d)}$ has the same leading equivalent as (5).

The all-orders assertion means a family of finite Poincaré expansions. Neither convergence of the infinite series nor higher-order total-count corrections is claimed. [5 October 2026, batch 102: Theorem 22.1 of Part III (source 104) proves this theorem again for the word counts by a different gauge, with the closed form (87) of $C_{d,1}$; its Corollary 22.3 gives, at $d = 3$, the first total-count correction with relative remainder $O(n^{-1/2})$, resting on source 104's reading of the proof of [3, Lemma 3.24] (Remark 28.2).]

Chang et al., Theorem 1.8, prove the corresponding two-sided Θ scale. Their Corollary 1.11 gives $T_n^{(d)} / T_{n,n-1}^{(d)} \rightarrow 1$ for $d \geq 3$; Remark 1.12 explicitly notes that their method does not give the leading equivalent. The lower half of their two-sided bound is the only external enumeration-asymptotic input to our positive-amplitude argument.

3 Exact recurrence and index conversion

Proposition 3.7 and equation (17) of [3] give

$$b_{n,m} = \binom{dn+m-2}{d-1} \sum_{j=1}^m b_{n-1,j}, \quad b_{1,1} = 1, \quad a_n = \sum_{m=1}^n b_{n,m}, \quad (6)$$

where entries outside $1 \leq m \leq n$ vanish. Their Theorem 3.4 gives $T_{n,n-1}^{(d)} = n! a_{n-1}$. Define

$$q_{n,m} = \frac{b_{n,m}}{\binom{dn+m-2}{d-1}}, \quad A_{n,k} = q_{n+1,k+1} \quad (0 \leq k \leq n).$$

Subtracting adjacent partial sums proves exactly

$$\begin{aligned} A_{n,k} &= A_{n,k-1} + \binom{dn+k-1}{d-1} A_{n-1,k}, & 1 \leq k \leq n, \\ A_{n,0} &= \prod_{i=1}^n \binom{di-1}{d-1}, & A_{0,0} = 1, \quad A_{n-1,n} = 0. \end{aligned} \quad (7)$$

The right boundary is explicitly

$$q_{n+1,n+1} = q_{n+1,n} = \sum_{j=1}^n b_{n,j} = a_n, \quad A_{n,n} = A_{n,n-1} = a_n \quad (n \geq 1).$$

At $d = 2$, this is the A213863 triangle. For $d = 3$, the diagonal starts

$$1, 1, 25, 2305, 482825, 183500625, 111174597625, 98734811198625.$$

No $d \geq 3$ OEIS accession is asserted.

The source uses the leaf variable v . Put $v = n + 1$ in its two-sided bound and divide by $(n + 1)!$. The factorial becomes $(n!)^{d-1}(n + 1)^{d-1}$, the exponential base acquires the constant Γ , and $\exp(B((n + 1)^{1/3} - n^{1/3})) \rightarrow 1$. Since

$$\alpha + d - 1 = -\frac{d^2 - d + 2}{2(d + 1)} = \rho,$$

both halves of the source bound give

$$a_n = \Theta_d \left((n!)^{d-1} \Gamma^n e^{Bn^{1/3}} n^\rho \right). \quad (8)$$

4 Exact contractive Jacobi reduction

Put $N = n + k$, $j = n - k$ and

$$D_N(j) = \frac{A_{n,k}}{\Lambda^n (n!)^{d-1}} \quad (0 \leq j \leq N, j \equiv N \pmod{2}),$$

with zero elsewhere. Equation (7) gives

$$D_N(j) = D_{N-1}(j + 1) + p_N(j) D_{N-1}(j - 1), \quad (9)$$

$$p_N(j) = \prod_{h=1}^{d-1} \left(1 - \frac{2(j+h)}{(d+1)(N+j)} \right), \quad D_0 = e_0. \quad (10)$$

The downward term is absent at $j = 0$. The upper physical edge $j = N$ reproduces the initial column, so there is no omitted boundary source. Also

$$a_n = \Lambda^n (n!)^{d-1} D_{2n}(0). \quad (11)$$

For $N, j \geq 1$, every factor of (10) lies in $[1/(d+1), 1]$. The lower bound is equivalent to $dN + (d-2)j \geq 2h$, which holds for $h \leq d-1$. Each factor has positive N derivative

$$\frac{2(j+h)}{(d+1)(N+j)^2}.$$

Define $G_N(0) = 1$ and $G_N(j)^2 = \prod_{k=1}^j p_N(k)$. On $0, \dots, N+1$, let S_N have edges $b_{N,j} = \sqrt{p_N(j)}$, and zero-pad outside. For $N \geq 2$ put $Q_N(j) = G_{N-1}(j)/G_N(j)$ there and zero outside; let Q_1 project onto e_0 . Then

$$v_N = G_N^{-1} D_N = S_N Q_N v_{N-1}. \quad (12)$$

The product formula defines G at the artificial endpoints as well. Monotonicity makes Q_N a positive contraction. If J is half-line adjacency and λ_N the Perron eigenvalue of S_N , then

$$0 \leq S_N Q_N \leq J \text{ entrywise}, \quad \|S_N Q_N\| \leq \lambda_N. \quad (13)$$

For $N = 1$, $p_1(1) = 1/\Lambda$, and (12) gives $v_1 = e_1/\sqrt{\Lambda}$, as required.

5 Global Airy spectrum and parity gap

Let $\varepsilon = N^{-1/3}$, $x_j = (j+1)\varepsilon$, and $F(x) = \text{Ai}(z + c^{1/3}x)$. Then $F(0) = 0$ and $F'' = (cx + \ell)F$. With $b_j = b_{N,j}$, the exact form is

$$\begin{aligned} \langle v, (2I - S_N)v \rangle &= \sum_{j=0}^N b_{j+1} |v_{j+1} - v_j|^2 + (2 - b_1) |v_0|^2 \\ &\quad + \sum_{j=1}^N (2 - b_j - b_{j+1}) |v_j|^2 + (2 - b_{N+1}) |v_{N+1}|^2. \end{aligned} \quad (14)$$

Split $2 - b_1 = 1 + (1 - b_1)$ to retain the fictitious Dirichlet interval. The fixed- d edge lower bound and

$$1 - b_j \geq \frac{1 - p_N(j)}{2} \geq \frac{j+1}{(d+1)(N+j)}$$

give scaled confinement

$$\varepsilon^{-2} \langle v, (2I - S_N)v \rangle \geq \mathfrak{c}_d \sum_j x_j |v_j|^2$$

for a positive constant $\mathfrak{c}_d$ and all sufficiently large N . Interpolate with nodal values $\varepsilon^{-1/2}v_j$ and zero at $x = 0$. Bounded scaled energy gives local H^1 bounds, and confinement makes the L^2 tails uniformly small. Thus local compactness gives global strong L^2 compactness. The first interval forces zero trace. On compact scaled windows the edges tend to one and $\varepsilon^{-2}(2 - b_j - b_{j+1}) \rightarrow cx_j$ uniformly. The limiting lower form is

$$\int_0^\infty (|f'|^2 + cx|f|^2) dx.$$

Sampling compactly supported smooth Dirichlet functions supplies the reverse upper bound. The min-max principle therefore gives convergence of each fixed low eigenvalue to that of $-\partial_x^2 + cx$ (Remark 5.1 gives the details). If $z = z_1 > z_2 > \dots$ are the Airy zeros,

$$\lambda_N = 2 + \ell\varepsilon^2 + o(\varepsilon^2), \quad \lambda_N - \lambda_{2,N} \sim c^{2/3}(z - z_2)\varepsilon^2. \quad (15)$$

The spectrum is symmetric under parity. Hence a full spectral-radius gap would be false. Between opposite parity subspaces, singular values are the positive eigenvalues; removal of the Perron singular direction leaves norm at most $\lambda_N(1 - a_d N^{-2/3})$.

Remark 5.1 (The compactness step in detail; write phase). The paragraph above compresses an argument that source 16's shipped derivation notes (`16-dcomb-expanded-proof.md`, Section 3) and source 69 (Section 15 below) state at greater length. The steps are collected here; nothing in them is new.

1. *Interpolation.* For v in the domain of S_N let f be piecewise linear with $f(0) = 0$ and $f(x_j) = \varepsilon^{-1/2}v_j$. Each of the $d-1$ factors of (10) is at least $1/(d+1)$, so the edges satisfy $b_j \geq (d+1)^{-(d-1)/2}$. So the first sum in (14), multiplied by ε^{-2} , bounds $\int_\varepsilon^\infty |f'|^2$. The retained unit term of $2 - b_1 = 1 + (1 - b_1)$ gives $\varepsilon^{-2}|v_0|^2$, which is $\int_0^\varepsilon |f'|^2$ on the fictitious first interval. Bounded scaled energy therefore bounds f in $H^1(0, \infty)$, with zero trace at 0.
2. *Norms.* On each lattice interval $\int f^2$ differs from $\varepsilon f(x_j)^2$ by at most a constant times $\varepsilon |f(x_{j+1}) - f(x_j)| (|f(x_j)| + |f(x_{j+1})|)$. By Cauchy-Schwarz and the gradient bound, the sum of these differences is $O(\varepsilon^{1/2})$ when $\|v\|$ and the scaled energy are bounded. Hence $\|f\|_{L^2}^2 = \|v\|^2 + o(1)$: these are the Riemann sums of source 69.

3. *Tails.* The confinement bound gives $\sum_j x_j |v_j|^2 \leq \mathfrak{c}_d^{-1} \times (\text{scaled energy})$, so $\int_X^\infty f^2 = O(1/X)$ uniformly in N .
4. *Compactness.* Rellich's theorem on each $[0, X]$, a diagonal argument and the uniform tails give a subsequence that converges strongly in $L^2(0, \infty)$, and weakly in H^1 , to some f with $f(0) = 0$. This is “local compactness gives global strong L^2 compactness”.
5. *Lower bound.* On compact windows the edges tend to one and $\varepsilon^{-2}(2 - b_j - b_{j+1}) \rightarrow cx_j$ uniformly. Weak lower semicontinuity of the gradient term and Fatou's lemma for the nonnegative potential term then give $\liminf \varepsilon^{-2} \langle v, (2I - S_N)v \rangle \geq \int_0^\infty (|f'|^2 + cx|f|^2)$.
6. *Upper bound and min-max.* The samples $v_j = \varepsilon^{1/2} \phi(x_j)$ of a fixed $\phi \in C_c^\infty(0, \infty)$ have scaled form and norm converging to those of ϕ . Applied to k -dimensional spaces of such ϕ , this bounds the k -th scaled eigenvalue from above in the limit. Conversely, normalized eigenvectors of the k lowest scaled eigenvalues have bounded energy. By the steps above they converge strongly along subsequences, and orthonormality survives in the limit, so the lower bound applies. Each fixed low eigenvalue of $\varepsilon^{-2}(2I - S_N)$ therefore converges to the corresponding eigenvalue $-c^{2/3}z_k$ of $-\partial_x^2 + cx$ on the Dirichlet half-line.

6 Finite quasimodes and quantitative drift

The exact scaled left edge is

$$p_-(\varepsilon, x) = \prod_{h=1}^{d-1} \left(1 - \frac{2(x\varepsilon^2 + (h-1)\varepsilon^3)}{(d+1)(1 + x\varepsilon^2 - \varepsilon^3)} \right), \quad p_+(\varepsilon, x) = p_-(\varepsilon, x + \varepsilon). \quad (16)$$

Writing $k_0 = (d-1)(d-2)/(d+1)$ gives $p_- = 1 - cx\varepsilon^2 - k_0\varepsilon^3 + O(\varepsilon^4)$. Thus

$$\sqrt{p_-}F(x - \varepsilon) + \sqrt{p_+}F(x + \varepsilon) = 2F + \varepsilon^2(F'' - cxF) - (c/2 + k_0)\varepsilon^3F + O(\varepsilon^4).$$

Since $c/2 + k_0 = 2\eta$, the next scalar coefficient is -2η .

All finite frozen corrections exist by a polynomial inverse. For $L = \partial_x^2 - cx - \ell$,

$$L(PF + QF') = (P'' + 2(cx + \ell)Q' + cQ)F + (2P' + Q'')F'. \quad (17)$$

To solve $L(PF + QF') = RF + TF'$, eliminate $P' = (T - Q'')/2$:

$$-Q'''/2 + 2(cx + \ell)Q' + cQ = R - T'/2. \quad (18)$$

On polynomials, this is triangular with nonzero diagonal $c(2j+1)$ on x^j . A new scalar eigenvalue coefficient changes Q by a nonzero constant; choose it to force $Q(0) = 0$. The integration constant of P then forces $P(0) + Q'(0) = 0$. These are the boundary and derivative-normalization conditions.

Construct the frozen profiles through degree five and cut off smoothly at $j \leq 2\sqrt{N}$, identically one up to $\sqrt{N}$. Taylor remainders have a fixed polynomial–Airy envelope times ε^6 ; cutoff errors are exponentially small in $N^{1/4}$. After sample normalization, the residual is $O_d(\varepsilon^6)$. The global gap identifies the eigenvalue and bounds the eigenvector error by $O_d(\varepsilon^4)$. In particular,

$$\lambda_N = 2 + \ell N^{-2/3} - 2\eta N^{-1} + O_d(N^{-4/3}), \quad (19)$$

$$\|\psi_N - \psi_{N-1}\| = O_d(N^{-1}), \quad \psi_N(0) = \sqrt{c} N^{-1/2} (1 + O_d(N^{-1/3})). \quad (20)$$

For the first drift bound, normalized explicit samples have continuous- N derivative $O(N^{-1})$, and the quasimode errors are smaller. The endpoint follows from

$$I := \int_0^\infty F^2 = c^{-1/3} \text{Ai}'(z)^2, \quad F'(0)/\sqrt{I} = \sqrt{c}, \quad (21)$$

choosing the positive sign. The pointwise error is bounded by the smaller global quasimode error. Logarithmic differentiation of (10) and summation over the gauge product give, on the cutoff support,

$$0 \leq 1 - Q_N(j) \leq C_d \frac{j(j+d)}{N^2}, \quad -\log G_N(j) \leq C_d \frac{j(j+d)}{N}.$$

Consequently, by polynomial–Airy moments and contraction on the approximation error,

$$\|(I - Q_N)\psi_N\| + \|(I - Q_N)\psi_{N-1}\| = O_d(N^{-4/3}). \quad (22)$$

The inverse gauge is uniformly bounded on the cutoff support.

7 A convergent positive scalar amplitude

Set

$$P_N = \prod_{i=1}^N \lambda_i, \quad u_N = v_N/P_N, \quad M_N = S_N Q_N/\lambda_N.$$

Then $u_N = M_N u_{N-1}$ and $\|u_N\| \leq 1$. The Perron vector has equal mass on both parities, by taking inner products in each part of $S_N \psi_N = \lambda_N \psi_N$. Let g_N, h_N be $\sqrt{2}$ times its restrictions to current and preceding parity respectively. They have unit norm and $S_N h_N = \lambda_N g_N$. From (20)–(22),

$$\begin{aligned} \|h_N - g_{N-1}\| &= O(N^{-1}), \\ \|(I - Q_N)h_N\| + \|(I - Q_N)g_{N-1}\| &= O(N^{-4/3}). \end{aligned} \quad (23)$$

Write $u_N = A_N g_N + w_N$, $w_N \perp g_N$. The parity singular gap gives

$$\|w_N\| \leq (1 - a_d N^{-2/3}) \|w_{N-1}\| + C_d N^{-1}, \quad \|w_N\| = O_d(N^{-1/3}).$$

We use throughout the elementary convolution bound

$$\sum_{k=N_0}^N k^{-r} \prod_{i=k+1}^N (1 - a_d i^{-2/3}) = O_{d,r}(N^{2/3-r}). \quad (24)$$

Split at $N/2$: the first portion is exponentially small in $N^{1/3}$ times a polynomial, and the later portion is a geometric sum of scale $N^{2/3}$.

The exact adjoint identity $M_N^* g_N = Q_N h_N$ gives

$$A_N = t_N A_{N-1} + \langle Q_N h_N - g_{N-1}, w_{N-1} \rangle, \quad t_N = \langle Q_N h_N, g_{N-1} \rangle.$$

The unit-vector identity $\langle h, g \rangle = 1 - \|h - g\|^2/2$ and (23) give $t_N = 1 + O(N^{-4/3})$, while the other inner-product factor has norm $O(N^{-1})$. Hence

$$A_N = A_\infty + O_d(N^{-1/3}), \quad A_\infty \geq 0. \quad (25)$$

Nonnegativity follows from positivity of u_N and g_N . Strict positivity requires boundary recovery. For $U(N, k) = M_N \cdots M_{k+1}$, entrywise domination and the Catalan identity give, for $0 \leq \Delta \leq N^{2/3}$,

$$\|e_0^T U(N, N - \Delta)\|_2 \leq \frac{\|e_0^T J^\Delta\|_2}{\lambda_{N-\Delta+1} \cdots \lambda_N} \leq C_d(\Delta + 1)^{-3/4}. \quad (26)$$

Indeed $\|e_0^T J^\Delta\|_2^2 = (J^{2\Delta})_{0,0} = \binom{2\Delta}{\Delta}/(\Delta + 1)$, and (19) bounds the denominator correction by $\exp(C_d \Delta N^{-2/3})$. The exact w_N recurrence has forcing $A_{N-1} M_N g_{N-1} - A_N g_N$ of norm $O(N^{-1})$. Duhamel over $\Delta = \lfloor N^{2/3} \rfloor$ yields

$$|w_N(0)| \leq C L^{-3/4} N^{-1/3} + C N^{-1} \sum_{l < \Delta} (l + 1)^{-3/4} = O_d(N^{-5/6}). \quad (27)$$

Summing (19) logarithmically gives

$$P_N = \kappa_d 2^N \exp((3\ell/2)N^{1/3}) N^{-\eta} (1 + O_d(N^{-1/3})), \quad \kappa_d > 0, \quad (28)$$

with convergent definition

$$\log \kappa_d = \frac{\ell}{2} \zeta(2/3) - \eta \gamma_E + \sum_{N=1}^{\infty} \left[\log(\lambda_N/2) - \frac{\ell}{2} N^{-2/3} + \frac{\eta}{N} \right]. \quad (29)$$

For even N , combining (20), (25), (27) and (28) gives

$$\begin{aligned} D_N(0) &= \kappa_d \sqrt{2c} A_\infty 2^N e^{(3\ell/2)N^{1/3}} N^\rho \\ &\quad + O_d\left(2^N e^{(3\ell/2)N^{1/3}} N^{\rho-1/3}\right). \end{aligned} \quad (30)$$

If $A_\infty = 0$, this contradicts the lower half of (8) through (11). Therefore

$$\gamma_d = \kappa_d A_\infty \sqrt{c} 2^{-\eta} > 0. \quad (31)$$

This proves a convergent amplitude with a relative $O_d(n^{-1/3})$ error; no numerical fit is used.

8 Every finite formal order

Return to (9) and seek

$$Z_N(j) = H_N \sum_{r \geq 0} \varepsilon^r F_r(x_j), \quad \frac{H_N}{H_{N-1}} \sim 2 \left(1 + \sum_{m \geq 2} s_m \varepsilon^m \right), \quad (32)$$

where $F_0 = F$ and $F_r(0) = F'_r(0) = 0$ for $r \geq 1$. At the preceding time the arguments are $(x \pm \varepsilon)(1 - \varepsilon^3)^{-1/3}$, and profile powers acquire $(1 - \varepsilon^3)^{-r/3}$. At degree m the new equation has the form

$$L F_{m-2} - 2s_m F + R_m F + T_m F' = 0,$$

where R_m, T_m are known polynomials. The same inverse (18) determines uniquely the new scalar and normalized profile. In particular

$$s_2 = \ell/2, \quad s_3 = -\eta - 1/6.$$

For an independent check, the order-three forcing is $-k_0 F + (c + 2/3)x F'$. Since $\int x F F' = -\frac{1}{2} \int F^2$, solvability gives $2s_3 = -k_0 - c/2 - 1/3$.

Choose a genuine finite scalar expression

$$\log H_N = N \log 2 + \frac{3\ell}{2} N^{1/3} + (-\eta - 1/6) \log N + \sum_{r=1}^J h_r N^{-r/3}. \quad (33)$$

The coefficient h_r enters logarithmic one-step matching at degree ε^{r+3} with nonzero coefficient $-r/3$. Its multiplicative leading constant is fixed to one at every truncation. Thus all scalar orders exist without logarithmic resonances.

For any prescribed $p > 1$, retain enough terms, cut off at $j \leq 2\sqrt{N}$, and *restrict* Z_N to $j \equiv N \pmod{2}$, setting the other coordinates to zero. Put $z_N = G_N^{-1} Z_N / P_N$. Then

$$\|z_N\| = O_d(1), \quad r_N := z_N - M_N z_{N-1} = O_{d,p}(N^{-p}) \quad \text{in } \ell^2. \quad (34)$$

Here the inverse gauge is bounded on the cutoff support and, on fixed scaled windows, is $1 + O_d(N^{-1/3}(1+x)^2)$. Finite Taylor remainders for the product coefficient, dilation and profiles have fixed polynomial–Airy envelopes times the required inverse power. The cutoff error is exponentially small in $N^{1/4}$. Finally

$$H_N / P_N \sim \kappa_d^{-1} N^{-1/6},$$

which cancels the parity sample norm $N^{1/6} \sqrt{I/2}$. The same estimates give the common positive limit

$$\langle g_N, z_N \rangle \longrightarrow q_0 := \kappa_d^{-1} \sqrt{I/2}. \quad (35)$$

9 Transfer to the exact sequence

Put $C_* = A_\infty / q_0$ and $e_N = u_N - C_* z_N$. Then $e_N = M_N e_{N-1} - C_* r_N$, and its ground projection tends to zero. Write $e_N = \xi_N g_N + W_N$, $W_N \perp g_N$. The already proved gap and adjoint estimates give

$$\begin{aligned} \|W_N\| &\leq (1 - a_d N^{-2/3}) \|W_{N-1}\| + C N^{-1} |\xi_{N-1}| + C N^{-p}, \\ |\xi_N - \xi_{N-1}| &\leq C N^{-4/3} |\xi_{N-1}| + C N^{-1} \|W_{N-1}\| + C N^{-p}. \end{aligned} \quad (36)$$

If $\xi_N = O(N^{-q})$ for $q \geq 0$, convolution gives $\|W_N\| = O(N^{-q-1/3} + N^{2/3-p})$. Summing the second inequality backwards from its zero limit improves this to

$$\xi_N = O(N^{-q-1/3} + N^{1-p}).$$

Starting at $q = 0$, finitely many repetitions imply $\|e_N\| = O(N^{1-p})$ and $\|W_N\| = O(N^{2/3-p})$. Apply Duhamel and (26) over $\Delta = \lfloor N^{2/3} \rfloor$:

$$|e_N(0)| \leq C L^{-3/4} N^{1-p} + C N^{-p} \sum_{l < \Delta} (l+1)^{-3/4} = O(N^{1/2-p}).$$

The leading normalized endpoint is of order $N^{-1/2}$, so the relative error is $O(N^{1-p})$. Since p is arbitrary, every finite formal order transfers to the exact sequence with the same positive multiplier. This proves (4).

For the coefficient field, use $t = (N/2)^{-1/3}$ and $y = (j+1)t$. Then

$$\tilde{F}'' = \left(\frac{d-1}{d+1}y + \frac{B}{3} \right) \tilde{F}.$$

The transformed recurrence, polynomial inverses, time dilation and scalar matching have rational coefficients for fixed integer d . The endpoint derivative cancels after normalization. Hence every $C_{d,r}$ lies in $\mathbb{Q}[B]$. [5 October 2026, batch 102: the first two are now explicit for every d (Part III, (87) and (147)): $C_{d,1} = B^2(8(d+1)^2 - 27(d-1)^2)/(810(d-1)^2)$, negative for $d \geq 4$, and $C_{d,2} = C_{d,1}^2/2 + \frac{B}{3}(\frac{1}{d+1} - \frac{(d+1)^2}{27(d-1)^2})$.]

10 Network counts and ternary coefficients

The exact leaf shift $T_{n,n-1}^{(d)} = n!a_{n-1}$ transfers every finite order in (4). The factorial shift contributes $n^{-(d-1)}$, the base contributes Γ^{-1} , and the remaining factors have finite Taylor expansions. Therefore

$$T_{n,n-1}^{(d)} = \frac{\gamma_d}{\Gamma} (n!)^d \Gamma^n e^{Bn^{1/3}} n^\alpha \left(1 + \sum_{r=1}^M E_{d,r} n^{-r/3} + O_{d,M}(n^{-(M+1)/3}) \right). \quad (37)$$

For $d \geq 3$, the published ratio $T_n^{(d)}/T_{n,n-1}^{(d)} \rightarrow 1$ gives only the leading total-count equivalent. A separate uniform deficit argument is needed for higher total-count corrections; the binary Pons–Batle identity is not imported here. [Write phase: at $d = 2$ that identity, now proved, gives every finite order of the total count; this is Part II, Theorem 16.2.] [5 October 2026, batch 102: at $d = 3$ Part III gives the first correction, $T_n^{(3)} = (\gamma_3/32)(n!)^3 32^n e^{Bn^{1/3}} n^{-3}(1 + \frac{B^2}{162}n^{-1/3} + O(n^{-1/2}))$, conditional on source 104's reading of a published proof (Proposition 28.1, Remark 28.2); for $d \geq 4$ only the leading equivalent is known here.]

For $d = 3$, $\Gamma = 32$, $\rho = -1$ and $B = 3z/2^{2/3}$. The exact polynomial recursion through degree six yields

$$\begin{aligned} \log \frac{a_n}{\gamma_3(n!)^2 32^n e^{Bn^{1/3}} n^{-1}} &= \frac{B^2}{162} n^{-1/3} + \frac{11B}{324} n^{-2/3} \\ &+ \left(\frac{4B^3}{6561} - \frac{3}{16} \right) n^{-1} + O(n^{-4/3}). \end{aligned} \quad (38)$$

In time N , these logarithmic coefficients are $\ell^2/36$, $11\ell/108$ and $(16\ell^3 - 729)/1944$. At $d = 2$, replay instead gives $\ell^2/4$, 0 and $-2/9$, agreeing with the binary report (Part II: with $N = 2n$ and $\ell = 2^{2/3}B/3$ they are the coefficients $B^2/18$, 0 , $-1/9$ of (45) in Theorem 14.1). These are symbolic identities, not fitted coefficients. [5 October 2026, batch 102: source 104 derives (38) independently, in another gauge, with identical values (Part III, (150)), and adds the multiplicative third coefficient $C_{3,3} = B^6/25509168 + 43B^3/52488 - 3/16$.]

11 Asymptotic inversion

For either a_n or $T_{n,n-1}^{(d)}$, let the factorial power be $\mathfrak{p} = d-1$ or d , the polynomial exponent be $\mathfrak{r} = \rho$ or α , and the amplitude be $\gamma_\star = \gamma_d$ or γ_d/Γ . Stirling's expansion gives the logarithmic smooth model

$$f(x) = \mathfrak{p}x \log x + (\log \Gamma - \mathfrak{p})x + Bx^{1/3} + (\mathfrak{r} + \mathfrak{p}/2) \log x + \mathsf{K} + \sum_{j \geq 1} \mathsf{L}_j x^{-j/3}, \quad (39)$$

where $K = \log \gamma_\star + (\mathfrak{p}/2) \log(2\pi)$, and Stirling terms are included in L_j . This denotes any required finite smooth truncation, not a differentiated unknown remainder of the discrete sequence.

[Write phase: the sans-serif letters $\mathfrak{p}, \mathfrak{r}, K, L_j, \mathfrak{a}$ and the amplitude $\gamma_\star$ are source 16's p, r, K, L_j, a and A , renamed because $p_N(j)$, the decay exponent p , the residual r_N , $A_{n,k}$ and A_N carry those letters in this Part; see Section 1.4.]

For $y = \log(\text{target})$, put $\mathfrak{a} = \log \Gamma - \mathfrak{p}$. The seed

$$x_0 = \frac{y/\mathfrak{p}}{W_0(e^{\mathfrak{a}/\mathfrak{p}} y/\mathfrak{p})} \quad (40)$$

solves $\mathfrak{p}x_0 \log x_0 + \mathfrak{a}x_0 = y$ exactly. Since $f'(x) = \mathfrak{p} \log x + \log \Gamma + o(1) > 0$, Newton or Taylor reversion gives every finite inverse order. The first correction is

$$x = x_0 - \frac{Bx_0^{1/3} + (\mathfrak{r} + \mathfrak{p}/2) \log x_0 + K}{\mathfrak{p} \log x_0 + \log \Gamma} + O_d \left(\frac{x_0^{-1/3}}{\log x_0} \right). \quad (41)$$

The error includes the first omitted Airy term; quadratic reversion is smaller by logarithmic factors. More explicitly, increasing the smooth-model truncation and iterating Newton's map makes its root error smaller than any prescribed finite algebraic order. A logarithmic truncation error $O(x^{-R/3})$ produces a root uncertainty $O(x^{-R/3}/\log x)$ by the mean-value theorem applied to that explicit smooth model.

The sequence is strictly increasing eventually: for the diagonal, (7) gives $a_n \geq \binom{(d+1)n-2}{d-1} a_{n-1}$, and the network leaf factor preserves this property. Integer thresholds must be bracketed using neighboring integer counts or rigorous upper and lower approximations. An unconditional ceiling rule is invalid when an approximate root lies within its uncertainty of an integer. The present amplitude existence proof does not supply certified amplitude digits or remainder constants, so (41) alone is not a certified numerical threshold algorithm.

Remark 11.1 (Repository machinery; write phase). The inversion of this section, and that of Section 17 at $d = 2$, are instances of the inversion machinery of the repository volume *Transseries and inversion* [7]; no novelty is claimed for them. With $u = \log x_0 + \mathfrak{a}/\mathfrak{p}$ the seed equation $\mathfrak{p}x_0 \log x_0 + \mathfrak{a}x_0 = y$ becomes $u + \log u = \log(y/\mathfrak{p}) + \mathfrak{a}/\mathfrak{p}$, which is the dominant-block equation of its Theorem `p0:thm:lambert-core` with both coefficients equal to one, solved on the principal branch W_0 . The finite-order Newton or Taylor reversion around the seed is the reversion scheme of its `p0:thm:perturbed-inversion`, and the neighbouring-integer brackets here and in (83) are its separation condition `p0:thm:staircase(2)`, whose real-number core is formalized in Lean as `staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`. That Lean statement concerns the bracket for a given real error bound; nothing about the counts of this report is formalized.

12 Source status and remaining questions

The source scope was checked against the 2024 primary paper, the 2026-listed Chang dissertation [5], and recent binary enumeration discussions already screened with the binary report (Part II, Section 20). The dissertation repeats the missing-equivalent issue (Remark 35, printed page 137). A bounded search found no later fixed- d amplitude proof. This is a bounded negative finding, not a certification of universal novelty.

A213863 is the $d = 2$ specialization. Phrase and initial-term searches did not verify a $d \geq 3$ OEIS accession; none is assigned or submitted here. [5 October 2026, batch 102: Part III, Section 30, adds the $d = 4$ diagonal $1, 1, 91, 51821, \dots$, the $d = 3$ and $d = 4$ maximal-network prefixes, and a trap: A213864 $(1, 1, 19, 1075, \dots)$ is not the $d = 3$ sequence.] Natural remaining questions are:

1. Establishing uniform deficit expansions that transfer every finite order to total counts for $d \geq 3$. [Write phase: for $d = 2$ this is done in Part II (Theorem 16.2), through the binary identity (T1); the question stays open for every $d \geq 3$, where no such identity is available in the sources.] [5 October 2026, batch 102: partly advanced at $d = 3$, first order only, conditionally on source 104's reading of a published proof (Proposition 28.1); re-scoped in Part III, Section 32, items 1–2.]
2. Obtaining certified numerical enclosures for γ_d and effective remainder constants. [Write phase: still open, also at $d = 2$; Part II gives only the uncertified diagnostic $\gamma_2 \approx 2.02264201$ of Section 19.] [5 October 2026, batch 102: still open. Parts III and IV give two further convergent representations, (124) and (205), and uncertified diagnostics: $\gamma_2 \approx 2.0226420146$ (Part IV) and the intake's fits $\gamma_3 \approx 3.784368$, $\gamma_4 \approx 3.478351$, $\gamma_5 \approx 2.384755$ (Part III, Section 31).]
3. Quantifying a joint regime in which d grows with n ; the present proof fixes d . [5 October 2026, batch 102: still open; Part III's closed forms are for fixed d .]
4. Studying exponentially small corrections beyond finite Poincaré orders. [5 October 2026, batch 102: still open; neither new source treats them.]

The exact shifted recurrence and the factorwise contractive gauge are essential. The proof has not presumed that the original two-weight recurrence automatically inherits the binary argument.

13 Reproducibility and review record

The companion archive contains the editable TeX, PDF, the expanded proof notes, exact recurrence and gauge checks, symbolic coefficient scripts and outputs, ordinary floating diagnostics, an independent audit, and a file manifest with SHA-256 hashes. No frozen binary artifact is changed. [Write phase: in the repository the expanded notes, audits, scripts and outputs are shipped beside this report with the prefix `16-dcomb-`; the PDF, the HTML rendering of the audits and the SHA-256 manifest (verified, 29 of 29 files) are not shipped. They survive in the arrival commit, and the README of this report lists every file.]

Exact finite checks compare the source b array with the shifted A array for $d = 2, \dots, 8$; an independent source-array calculation checks the normalized recurrence through $N = 20$ for $d = 2, \dots, 12$. Symbolic replay at $d = 2, 3$ cancels the frozen and nonautonomous residuals through degree six. These finite checks verify the identities and implementation, while the analytic argument establishes the asymptotic conclusions.

The independent mathematical audit found no substantive analytic obstruction and requested two precision repairs: the boundary identity in Section 3 and explicit occupied-parity restriction in Section 8. Both are incorporated here, together with the interpolation normalization. The audit is included in readable form [Write phase: as Markdown, `16-dcomb-mathematical-audit.md` and `16-dcomb-integrated-report-audit.md`]. Independent review here does not mean external peer review, publication, formal proof-assistant verification, or numerical certification.

Part II

The binary case $d = 2$: total count, deficit law and inverse (source 69)

Source 69 (archive tree-child-asymptotics-reproducibility), printed in full; its title was “Asymptotic amplitude and all orders for tree child networks”. Its abstract reads: The tableau sequence A213863, which also counts the words associated with maximally reticulated binary tree-child networks, is known to have order $n!12^n \exp(z(3n)^{1/3})n^{-2/3}$, where z is the largest negative Airy zero. We prove convergence to a positive limiting amplitude and an expansion to every finite order in $n^{-1/3}$. An exact Jacobi gauge, a global Airy spectral gap, summable adjoint drift, and a Catalan boundary estimate establish the limit. Polynomial–Airy approximate solutions and a zero-limit projection bootstrap transfer all formal orders to the exact sequence. The recent proof of the tableau-to-network identity then transfers every finite order to the total number of binary tree-child networks, by a polynomial deficit recurrence with factorially summable errors. We derive weighted all-orders expansions for the reticulation deficit and a sharp total-variation convergence rate to its Poisson limit. We give explicit coefficients and Lambert W inversions, including their precise discrete-threshold interpretation. The amplitude has an exact convergent definition; numerical diagnostics are not certified enclosures. The accompanying archive contains the complete algebraic replay and proof-audit record. This is a research report, not a claim of external peer review or publication.

14 Results at $d = 2$ and enumeration context

[Write phase: source 69 is printed from here to the end of Part II, in its own order and notation, apart from the renamings of Section 1.4. The scope statement of its delivery README, which is not shipped, is folded into Section 1.3. Its Section 1 opens as follows.]

Let $A_{n,k}$ be defined by

$$A_{n,k} = A_{n,k-1} + (2n + k - 1)A_{n-1,k}, \quad A_{n,0} = (2n - 1)!!, \quad A_{n,k} = 0 \quad (k > n), \quad (42)$$

with $(-1)!! = 1$, and put $a_n = A_{n,n}$. *[Write phase: this is (7) at $d = 2$, where $\binom{2n+k-1}{1} = 2n + k - 1$ and $\prod_{i \leq n} \binom{2i-1}{1} = (2n - 1)!!$; the diagonal a_n is Part I’s at $d = 2$.]* The diagonal begins

$$1, 1, 7, 106, 2575, 87595, 3864040, 210455470, 13681123135, \dots$$

and is OEIS A213863 [6]. The recurrence, its boundary conditions, and its tableau interpretation are given in Lin et al. [4], equations (1)–(2). Put

$$z = -2.338107410459767\dots, \quad B = 3^{1/3}z,$$

where z is the largest real zero of Ai. Fuchs–Yu–Zhang [1] established the stretched-exponential order; for the tableau sequence it is also explicitly stated in Banderier–Wallner [2], Theorem 4.1:

$$a_n = \Theta\left(n!12^n e^{Bn^{1/3}} n^{-2/3}\right). \quad (43)$$

The lower half of this existing theorem is the only external enumeration-asymptotic input to the new convergence argument.

Theorem 14.1 (Positive amplitude and all finite orders). *There is a constant $\gamma > 0$ and uniquely constructible coefficients $c_r \in \mathbb{Q}[B]$ such that, for every integer $M \geq 0$,*

$$a_n = \gamma n! 12^n e^{Bn^{1/3}} n^{-2/3} \left(1 + \sum_{r=1}^M c_r n^{-r/3} + O_M(n^{-(M+1)/3}) \right). \quad (44)$$

Equivalently there is a logarithmic expansion to every finite order. Its first four coefficients are

$$\log \frac{a_n}{\gamma n! 12^n e^{Bn^{1/3}} n^{-2/3}} = \frac{B^2}{18} n^{-1/3} - \frac{1}{9} n^{-1} - \frac{B^2}{162} n^{-4/3} + O(n^{-5/3}). \quad (45)$$

In particular the $n^{-2/3}$ logarithmic coefficient is zero, and

$$c_1 = \frac{B^2}{18}, \quad c_2 = \frac{B^4}{648}, \quad c_3 = \frac{B^6}{34992} - \frac{1}{9}.$$

Remark 14.2 (Relation to Part I; write phase). Theorem 14.1 is Theorem 2.1 at $d = 2$, with $\gamma = \gamma_2$ and $c_r = C_{2,r}$; at $d = 2$, $\Gamma = 12$, $\rho = -2/3$ and $B = 3^{1/3}z$. Its existence part is proved twice: for every d in Part I (source 16), and at $d = 2$ in Section 15 (source 69). The two amplitude formulas agree: (31) at $d = 2$ is $\kappa_2 A_\infty \sqrt{2/3} 2^{-1/6} = (2^{1/3}/\sqrt{3}) \kappa_2 A_\infty$, which is (67). The explicit coefficients c_1, c_2, c_3 and the four logarithmic coefficients of (45) are source 69's. Source 16's symbolic replay reproduces the first three (Section 10), and every entry of source 16's shipped $d = 2$ coefficient file agrees with source 69's order-seven file, which runs one order further (checked in writing). [5 October 2026, batch 102: source 102 (Part IV) derives the four coefficients of (45) and c_1, c_2, c_3 again, independently and in another gauge, and adds two: $L_5 = B(108 - B^3)/8748$ and $L_6 = 2B^3/2187 - 1/162$ (Part IV, (217)). Source 69's own generator, rerun at order nine on a copy, gives exactly these two values (Remark 39.1). Source 104 (Part III) proves Theorem 2.1 again for every d .]

Let TC_n denote the total number of binary tree-child networks with n labelled leaves, with the root convention of [1, 3]. We will prove

$$\text{TC}_n = \frac{\sqrt{e} \gamma}{12} (n!)^2 12^n e^{Bn^{1/3}} n^{-5/3} \exp \left\{ \frac{B^2}{18} n^{-1/3} - \frac{23B}{72} n^{-2/3} + \frac{161}{288} n^{-1} + O(n^{-4/3}) \right\}, \quad (46)$$

again with every finite order available and with no additional unknown amplitude. The exact identity needed here was conjectural in the older literature; its proof in [4], together with [3], is important to this transfer.

Throughout, an all-orders expansion means a family of finite asymptotic statements. Convergence of the infinite formal series is neither needed nor asserted. The exact constant γ is defined below by convergent spectral and projection limits. This does not give a closed form or certified numerical digits.

15 Second route at $d = 2$: the binary proof in detail

[Write phase: this section is Sections 2–8 of source 69, its own proof of Theorem 14.1, printed in full as a second proof of Theorem 2.1 at $d = 2$. It follows Part I step by step: Subsections 15.1–15.7 correspond to Sections 4–9, and D_N , $p_N(j)$, $b_{N,j}$, G_N , Q_N , S_N are Part I's at $d = 2$. It is kept because it is more detailed than Part I at several steps and because it contains explicit binary

material that Part I lacks: the quasimode ϕ_2, ϕ_3 with the eigenvalue through ε^5 , the scalar coefficients $s_2, \dots, s_7$, h_1, h_2, h_3 , the endpoint expansion and the logarithmic coefficient at $N^{-4/3}$. Four displays are literally identical in the two sources and are printed only in Part I: (17), (18), (24) (with $a_d = a_2$) and (26). The gap constant written c_1 in source 69 is a_2 here, since c_1 also denotes the coefficient $B^2/18$.]

15.1 The exact contractive Jacobi reduction

For $N = n + k$ and $j = n - k$ set

$$D_N(j) = \frac{A_{n,k}}{3^n n!} \quad (0 \leq j \leq N, \ j \equiv N \pmod{2}),$$

and set it equal to zero elsewhere. Direct substitution in (42) gives

$$D_N(j) = D_{N-1}(j+1) + p_N(j)D_{N-1}(j-1), \quad p_N(j) = \frac{3N+j-2}{3(N+j)}, \quad D_0 = e_0. \quad (47)$$

The fictitious value at $j = -1$ is zero. The upper physical edge $j = N$ gives the double-factorial initial column, so this transformation includes both boundaries. In particular,

$$a_n = 3^n n! D_{2n}(0). \quad (48)$$

Define $G_N(0) = 1$ and, for $N \geq 1$,

$$G_N(j)^2 = \prod_{k=1}^j p_N(k) = 3^{-j} \frac{\Gamma(3N+j-1)\Gamma(N+1)}{\Gamma(3N-1)\Gamma(N+j+1)}. \quad (49)$$

Let S_N be the symmetric Jacobi matrix on $0, \dots, N+1$ with edges $b_{N,j} = \sqrt{p_N(j)}$ for $1 \leq j \leq N+1$, extended by zero on the rest of $\ell^2(\mathbb{N}_0)$. For $N \geq 2$ define the diagonal operator

$$Q_N(j) = \frac{G_{N-1}(j)}{G_N(j)}, \quad Q_N(j)^2 = \frac{N+j}{N} \prod_{r=2}^4 \frac{3N-r}{3N+j-r} \quad (0 \leq j \leq N+1), \quad (50)$$

and set it to zero beyond this interval. The product definition extends to the artificial endpoints; the preceding vector occupies only $j \leq N-1$. For the first step let Q_1 be projection onto e_0 . Then $v_0 = e_0$ and $v_N(j) = D_N(j)/G_N(j)$ satisfy exactly

$$v_N = S_N Q_N v_{N-1}. \quad (51)$$

This follows from $G_N(j)/G_N(j-1) = b_{N,j}$ on each edge. The first step gives $v_1 = e_1/\sqrt{3}$ as required. Since

$$\partial_N p_N(j) = \frac{2(j+1)}{3(N+j)^2} > 0,$$

Q_N is a positive diagonal contraction. Also $0 < b_{N,j} \leq 1$, so

$$0 \leq S_N Q_N \leq J \quad \text{entrywise}, \quad \|S_N Q_N\| \leq \lambda_N, \quad (52)$$

where J is the half-line adjacency matrix and λ_N is the largest eigenvalue of S_N . The finite irreducible matrix has a positive unit eigenvector ψ_N . Its spectrum is symmetric by parity, so its norm is λ_N . In particular a gap in the full spectral radius would be false; below we use singular values between opposite parities.

15.2 A global Airy limit and the parity gap

Set

$$\varepsilon = N^{-1/3}, \quad x_j = (j+1)\varepsilon, \quad c = \frac{2}{3}, \quad \ell = c^{2/3}z, \quad F(x) = \text{Ai}(z + c^{1/3}x).$$

Thus $F(0) = 0$ and $F'' = (cx + \ell)F$. The exact quadratic form is

$$\begin{aligned} \langle v, (2I - S_N)v \rangle &= \sum_{j=0}^N b_{N,j+1} |v_{j+1} - v_j|^2 + (2 - b_{N,1}) |v_0|^2 \\ &\quad + \sum_{j=1}^N (2 - b_{N,j} - b_{N,j+1}) |v_j|^2 + (2 - b_{N,N+1}) |v_{N+1}|^2. \end{aligned} \quad (53)$$

The first coefficient is $2 - b_{N,1}$, not 1. Split it into $1 + (1 - b_{N,1})$ to identify the first fictitious Dirichlet interval. Moreover,

$$1 - b_{N,j} = \frac{2(j+1)}{3(N+j)(1+b_{N,j})} \geq \frac{j+1}{3(N+j)}.$$

The edge weights are uniformly bounded below. The potential and terminal terms therefore imply, for some fixed $c_2 > 0$ and all sufficiently large N ,

$$\varepsilon^{-2} \langle v, (2I - S_N)v \rangle \geq c_2 \sum_j x_j |v_j|^2. \quad (54)$$

On every fixed x window the edges tend to 1 and

$$\varepsilon^{-2} (2 - b_{N,j} - b_{N,j+1}) \longrightarrow cx_j$$

uniformly.

Interpolate linearly with value $\varepsilon^{-1/2}v_j$ at x_j and zero at zero. A bounded scaled quadratic form gives local H^1 bounds from (53), while (54) makes the L^2 tails uniformly small. Local compactness followed by the tail bound gives strong L^2 compactness. The first interval enforces zero trace at zero. Norm convergence follows first on a compact window from the gradient bound and the Riemann sums, and then globally from the tail estimate. The limiting lower form is

$$\int_0^\infty (|f'(x)|^2 + cx|f(x)|^2) dx.$$

Sampling smooth compactly supported functions with zero trace gives the reverse upper bound. The min-max principle in both directions now gives convergence of each fixed low eigenvalue of $\varepsilon^{-2}(2I - S_N)$ to that of $-\partial_x^2 + cx$ on the Dirichlet half-line. Those eigenvalues are $-c^{2/3}z_k$, with Airy zeros $z = z_1 > z_2 > \dots$. Hence

$$\lambda_N = 2 + \ell\varepsilon^2 + o(\varepsilon^2), \quad \lambda_N - \lambda_{2,N} \sim c^{2/3}(z - z_2)\varepsilon^2. \quad (55)$$

Between opposite parity subspaces, the nonzero singular values of S_N are its positive eigenvalues. Removing the largest singular direction therefore leaves norm at most $\lambda_N(1 - a_2N^{-2/3})$. This is a global gap statement, not a consequence presumed from a local quasimode.

15.3 Finite quasimodes and quantitative drift

At $x = (j+1)\varepsilon$ the left and right edges are exactly

$$b_-^2 = 1 - \frac{2x\varepsilon^2}{3(1+x\varepsilon^2-\varepsilon^3)}, \quad b_+^2 = 1 - \frac{2(x+\varepsilon)\varepsilon^2}{3(1+x\varepsilon^2)}. \quad (56)$$

The expansion of $b_-F(x-\varepsilon) + b_+F(x+\varepsilon)$ through degree three is $2F + \varepsilon^2(F'' - cxF) - \varepsilon^3F/3$. Thus the first eigenvalue corrections are ℓ and $-1/3$. Define

$$\begin{aligned} \phi_2 &= \frac{(-3\ell + 2x)F - x(-3\ell + x)F'}{9}, & \phi_3 &= \frac{-F + xF'}{9}, \\ f_N &= F + \varepsilon^2\phi_2 + \varepsilon^3\phi_3, & \hat{\lambda}_N &= 2 + \ell\varepsilon^2 - \frac{1}{3}\varepsilon^3 + \frac{3\ell^2}{4}\varepsilon^4 + \frac{\ell}{18}\varepsilon^5. \end{aligned}$$

Both correcting profiles and their first derivatives vanish at zero. Direct substitution in (56), with $F'' = (cx + \ell)F$, cancels the residual through ε^5 .

For completeness, this finite calculation extends by an exact triangular algebra. Write $L = \partial_x^2 - cx - \ell$. Then the polynomial–Airy identity (17) holds, and solving $L(PF + QF') = RF + TF'$ by eliminating $P' = (T - Q'')/2$ gives (18), here with $c = 2/3$. *[Write phase: source 69 prints both displays; they are identical to Part I's.]* On polynomials of each bounded degree, this map is triangular with nonzero diagonal $c(2j+1)$ on x^j . *[Source 69 names the degree bound d , which here is the combining degree.]* It is invertible. The new scalar eigenvalue coefficient adds a nonzero multiple to $Q(0)$; choose it to force $Q(0) = 0$. The integration constant of P then enforces $P(0) + Q'(0) = 0$. These are precisely the boundary and derivative normalization conditions.

Take a smooth cutoff $\chi(j/N^{1/2})$, equal to one up to 1 and zero from 2 onward, and normalize the samples of f_N in ℓ^2 to obtain $\hat{f}_N$. On this support $x\varepsilon^2 = O(N^{-1/2})$. Taylor remainders are bounded by $C\varepsilon^6$ times a fixed polynomial–Airy envelope. Cutoff derivatives contribute $\exp(-a'N^{1/4})$ times powers of N . Polynomially weighted Airy functions are square-integrable, and the unnormalized sampled norm is asymptotic to $\varepsilon^{-1/2} \|F\|_{L^2}$. Thus

$$\|(S_N - \hat{\lambda}_N)\hat{f}_N\| = O(\varepsilon^6).$$

The gap (55) identifies the nearby eigenvalue as the top one, and projection on its complement gives

$$\lambda_N = \hat{\lambda}_N + O(\varepsilon^6), \quad \|\psi_N - \hat{f}_N\| = O(\varepsilon^4). \quad (57)$$

The positive sign is fixed by the leading Airy profile.

These estimates imply

$$\begin{aligned} \|\psi_N - \psi_{N-1}\| &= O(N^{-1}), \\ \|(I - Q_N)\psi_N\| + \|(I - Q_N)\psi_{N-1}\| &= O(N^{-4/3}), \\ \psi_N(0) &= \sqrt{c}N^{-1/2}(1 + O(N^{-1/3})). \end{aligned} \quad (58)$$

For the first estimate, the normalized explicit samples have continuous- N derivative $O(N^{-1})$; the errors in (57) are smaller. For the second, integration of

$$\partial_N \log p_N(j) = \frac{2(j+1)}{(N+j)(3N+j-2)}$$

and summation over the gauge product give $1 - Q_N(j) \leq Cj(j+1)/N^2$ on the cutoff support. The polynomial moments of the Airy profile and (57) prove the claimed bound, also with samples at $N-1$. Finally,

$$I := \int_0^\infty F(x)^2 dx = c^{-1/3} \text{Ai}'(z)^2, \quad F'(0)/\sqrt{I} = \sqrt{c}. \quad (59)$$

A Riemann-sum normalization error $O(\varepsilon)$ and the pointwise bound by the ℓ^2 error establish the endpoint estimate.

15.4 Convergence of a scalar amplitude

Define

$$P_N = \prod_{i=1}^N \lambda_i, \quad u_N = v_N/P_N, \quad M_N = S_N Q_N / \lambda_N.$$

Then $u_N = M_N u_{N-1}$, $\|M_N\| \leq 1$, and $\|u_N\| \leq 1$. The positive eigenvector has equal mass on the two parities: take inner products in each half of $S_N \psi_N = \lambda_N \psi_N$. If E_p denotes parity projection, the vectors

$$g_N = \sqrt{2} E_{N \bmod 2} \psi_N, \quad h_N = \sqrt{2} E_{(N-1) \bmod 2} \psi_N$$

are unit and satisfy $S_N h_N = \lambda_N g_N$. By (58),

$$\|h_N - g_{N-1}\| = O(N^{-1}), \quad \|(I - Q_N)h_N\| + \|(I - Q_N)g_{N-1}\| = O(N^{-4/3}). \quad (60)$$

Write $u_N = A_N g_N + w_N$, with $w_N \perp g_N$, and let Π_N project onto the complement of g_N on its parity. The parity singular gap gives, for every preceding-parity vector y ,

$$\|\Pi_N M_N y\| \leq (1 - a_2 N^{-2/3}) \|Q_N y\| \leq (1 - a_2 N^{-2/3}) \|y\|. \quad (61)$$

Also $\|M_N g_{N-1} - g_N\| = O(N^{-1})$, so

$$\|w_N\| \leq (1 - a_2 N^{-2/3}) \|w_{N-1}\| + C N^{-1}, \quad \|w_N\| = O(N^{-1/3}).$$

Here and below we use the elementary convolution estimate (24), with $a_d = a_2$. *[Write phase: source 69 prints this display, identical to Part I's, together with the same proof.]* Split at $N/2$: the first part is exponentially small in $N^{1/3}$ times a polynomial, and the later part is a geometric sum of scale $N^{2/3}$.

The exact adjoint identity $M_N^* g_N = Q_N h_N$ gives

$$\begin{aligned} A_N &= t_N A_{N-1} + \langle k_N, w_{N-1} \rangle, \\ t_N &= \langle Q_N h_N, g_{N-1} \rangle, \quad k_N = Q_N h_N - g_{N-1}. \end{aligned}$$

By the unit-vector identity $\langle h, g \rangle = 1 - \|h - g\|^2/2$ and (60),

$$t_N = 1 + O(N^{-4/3}), \quad \|k_N\| = O(N^{-1}).$$

Consequently $A_N - A_{N-1} = O(N^{-4/3})$, and

$$A_N = A_\infty + O(N^{-1/3}), \quad A_\infty \geq 0. \quad (62)$$

The nonnegativity follows from positivity of u_N and g_N . Strict positivity will follow after recovering the boundary coordinate.

15.5 Boundary smoothing and strict positivity

Let $U(N, k) = M_N \cdots M_{k+1}$. From (52), for $0 \leq \Delta \leq N^{2/3}$,

$$\left\| e_0^T U(N, N - \Delta) \right\|_2 \leq \frac{\left\| e_0^T J^\Delta \right\|_2}{\lambda_{N-\Delta+1} \cdots \lambda_N}.$$

The squared numerator counts nonnegative walks returning to zero:

$$\left\| e_0^T J^\Delta \right\|_2^2 = (J^{2\Delta})_{0,0} = \frac{1}{\Delta+1} \binom{2\Delta}{\Delta} \leq C 4^\Delta (\Delta+1)^{-3/2}.$$

By (57), $2^\Delta / (\lambda_{N-\Delta+1} \cdots \lambda_N) \leq \exp(C\Delta N^{-2/3}) \leq C'$. Therefore the smoothing bound (26) holds, $\left\| e_0^T U(N, N - \Delta) \right\|_2 \leq C(\Delta+1)^{-3/4}$. *[Write phase: source 69 prints this display, identical to Part I's; the walk-counting bound above is its more explicit derivation.]* The exact recurrence for w_N has forcing $A_{N-1}M_N g_{N-1} - A_N g_N$ of norm $O(N^{-1})$. Duhamel over $\Delta = \lfloor N^{2/3} \rfloor$ yields

$$|w_N(0)| \leq CL^{-3/4}N^{-1/3} + CN^{-1} \sum_{l < \Delta} (l+1)^{-3/4} = O(N^{-5/6}). \quad (63)$$

The finite eigenvalue expansion gives

$$\log \lambda_N = \log 2 + \frac{\ell}{2} N^{-2/3} - \frac{1}{6N} + O(N^{-4/3}),$$

so summation yields

$$P_N = \kappa 2^N \exp\left(\frac{3\ell}{2} N^{1/3}\right) N^{-1/6} (1 + O(N^{-1/3})), \quad \kappa > 0. \quad (64)$$

A convergent exact definition is

$$\log \kappa = \frac{\ell}{2} \zeta(2/3) - \frac{\gamma_E}{6} + \sum_{N=1}^{\infty} \left[\log(\lambda_N/2) - \frac{\ell}{2} N^{-2/3} + \frac{1}{6N} \right]. \quad (65)$$

Here γ_E is Euler's constant. For even N , $g_N(0) = \sqrt{2c} N^{-1/2} (1 + O(N^{-1/3}))$. Combining (62), (63), and (64),

$$\begin{aligned} D_N(0) &= \kappa \sqrt{2c} A_\infty 2^N e^{(3\ell/2)N^{1/3}} N^{-2/3} \\ &\quad + O\left(2^N e^{(3\ell/2)N^{1/3}} N^{-1}\right). \end{aligned} \quad (66)$$

If $A_\infty = 0$, setting $N = 2n$ and using (48) contradicts the lower half of (43). Hence $A_\infty > 0$, and

$$\boxed{\gamma = \frac{2^{1/3}}{\sqrt{3}} \kappa A_\infty > 0.} \quad (67)$$

This proves the leading equivalent, with an error $O(n^{-1/3})$ relative to it. The argument first proves a convergent scalar projection and a quantitative endpoint error; it does not replace the desired limit by compactness or by a formal Airy ansatz.

15.6 All finite formal profiles

Return to (47), before symmetrization, and seek

$$Z_{N,j} = H_N \sum_{r \geq 0} \varepsilon^r F_r(x_j), \quad F_0 = F, \quad F_r(0) = F'_r(0) = 0 \quad (r \geq 1), \quad \frac{H_N}{H_{N-1}} \sim 2 \left(1 + \sum_{m \geq 2} s_m \varepsilon^m \right). \quad (68)$$

At time $N - 1$ the arguments and powers become

$$y_{\pm} = (x \pm \varepsilon)(1 - \varepsilon^3)^{-1/3}, \quad \varepsilon_{N-1}^r = \varepsilon^r (1 - \varepsilon^3)^{-r/3},$$

and the downward coefficient is $1 - 2x\varepsilon^2/[3(1 + x\varepsilon^2 - \varepsilon^3)]$. At degree m the new unknowns occur in the form

$$LF_{m-2} - 2s_m F + R_m F + T_m F',$$

where the polynomials R_m, T_m depend only on earlier orders. Equation (18) solves this uniquely: changing s_m changes the constant term of Q by a nonzero multiple $2/c = 3$, so one can impose $Q(0) = 0$, then impose $P(0) + Q'(0) = 0$ through the additive constant of P . This constructs every finite formal order, with no unresolved solvability condition.

The first scalar coefficients and profile are

$$\begin{aligned} s_2 &= \ell/2, & s_3 &= -1/3, & s_4 &= \ell^2/24, \\ s_5 &= \ell/9, & s_6 &= -(3\ell^3 + 32)/144, & s_7 &= 49\ell^2/360, & F_1 &= -x^2 F/3. \end{aligned} \quad (69)$$

For every finite matching order choose the genuine smooth expression

$$\log H_N = N \log 2 + \frac{3\ell}{2} N^{1/3} - \frac{1}{3} \log N + \sum_{r=1}^J h_r N^{-r/3}. \quad (70)$$

Its one-step ratio matches (68) triangularly: the new coefficient h_r enters at degree ε^{r+3} with nonzero coefficient $-r/3$. The leading multiplicative constant in (70) is fixed to one for every truncation, which is essential to retain a common amplitude.

The first values are $h_1 = \ell^2/4$, $h_2 = -\ell/6$, $h_3 = 1/9$. The derivative-normalized endpoint gives

$$\frac{\sum_r \varepsilon^r F_r(\varepsilon)}{F'(0)\varepsilon} = 1 + \frac{\ell}{6} \varepsilon^2 - \frac{1}{3} \varepsilon^3 + O(\varepsilon^4).$$

Combining these yields the first three logarithmic corrections in time N :

$$\frac{\ell^2}{4} N^{-1/3} + 0 \cdot N^{-2/3} - \frac{2}{9} N^{-1}.$$

The next coefficient, obtained by the same exact recursion through degree seven, is $-\ell^2/18$ at $N^{-4/3}$. Substituting $N = 2n$ gives (45) once the formal construction is transferred to the exact sequence.

All coefficients in the n normalization lie in $\mathbb{Q}[B]$. To see this without an unproved grading assertion, run the same recursion in

$$t = (N/2)^{-1/3}, \quad y = (j+1)t.$$

The downward coefficient becomes $1 - yt^2/[3(1 + yt^2/2 - t^3/2)]$, the previous-time dilation is $(1 - t^3/2)^{-1/3}$, and the rescaled Airy equation is

$$\tilde{F}'' = (y + B)\tilde{F}/3.$$

All triangular inversions and scalar matching then use polynomials in B and divisions only by nonzero rational numbers. Derivative normalization cancels $\tilde{F}'(0)$ from every normalized endpoint coefficient.

15.7 From formal profiles to the exact sequence

Fix $p > 1$. Retain enough profile and scalar orders from the preceding section, cut off at $j \leq 2\sqrt{N}$, and restrict to the occupied parity. Put

$$z_N = G_N^{-1} Z_N / P_N.$$

Then

$$\|z_N\| = O(1), \quad r_N := z_N - M_N z_{N-1} = O(N^{-p}) \quad \text{in } \ell^2. \quad (71)$$

Here are the global controls behind this assertion. On the cutoff support,

$$-\log G_N(j) \leq Cj(j+1)/N \leq C',$$

so its reciprocal is uniformly bounded. On a fixed scaled window, $G_N^{-1} = 1 + O(N^{-1/3}(1+x)^2)$. All finite Taylor remainders from the rational coefficient, the time dilation, and the polynomial profiles have a fixed polynomial–Airy envelope times the required inverse power. Cutoff errors are exponentially small in $N^{1/4}$. Finally

$$H_N / P_N \sim \kappa^{-1} N^{-1/6},$$

which cancels the parity-restricted sampled norm $N^{1/6} \sqrt{I/2}$. This proves (71) by direct summation. The same estimates show

$$\langle g_N, z_N \rangle \longrightarrow q_0 := \kappa^{-1} \sqrt{I/2} > 0. \quad (72)$$

Every finite truncation has this same limit.

Set $C_* = A_\infty / q_0$ and $e_N = u_N - C_* z_N$. Then $e_N = M_N e_{N-1} - C_* r_N$, and its ground projection tends to zero. Decompose $e_N = \xi_N g_N + W_N$, $W_N \perp g_N$. The same parity contraction and adjoint estimates used above give

$$\begin{aligned} \|W_N\| &\leq (1 - a_2 N^{-2/3}) \|W_{N-1}\| + C N^{-1} |\xi_{N-1}| + C N^{-p}, \\ |\xi_N - \xi_{N-1}| &\leq C N^{-4/3} |\xi_{N-1}| + C N^{-1} \|W_{N-1}\| + C N^{-p}. \end{aligned} \quad (73)$$

If $\xi_N = O(N^{-q})$ for $q \geq 0$, the convolution bound gives $\|W_N\| = O(N^{-q-1/3} + N^{2/3-p})$. Substituting this into the second inequality and summing from $N+1$ to infinity, using $\xi_\infty = 0$, improves it to

$$\xi_N = O(N^{-q-1/3} + N^{1-p}).$$

Starting at $q = 0$, finitely many repetitions imply

$$\|e_N\| = O(N^{1-p}), \quad \|W_N\| = O(N^{2/3-p}). \quad (74)$$

Apply Duhamel to e_N on the last $\Delta = \lfloor N^{2/3} \rfloor$ steps. By (26) and (71),

$$\begin{aligned} |e_N(0)| &\leq C L^{-3/4} N^{1-p} + C N^{-p} \sum_{l < \Delta} (l+1)^{-3/4} \\ &= O(N^{1/2-p}) + O(N^{1/6-p}) = O(N^{1/2-p}). \end{aligned}$$

The leading endpoint is of order $N^{-1/2}$; thus the relative error is $O(N^{1-p})$. Since p is arbitrary, every prescribed finite order of the explicit profile transfers to the exact sequence. The common q_0 in (72) gives the common multiplier C_* and the same positive γ at every order. This completes the proof of Theorem 14.1.

No inverse of a shrinking spectral gap has been assumed uniformly bounded. The only condition at infinity is the ground-projection limit already proved in (62) and (72). These facts distinguish the argument from a merely formal WKB expansion.

16 Transfer to total network counts

Theorem 14.1 (equivalently Theorem 2.1 at $d = 2$) supplies the diagonal hypothesis below. We give the transfer in general form to isolate its exact combinatorial input and its uniform error argument. No separate uniform Airy-profile theorem or additional amplitude is needed.

16.1 Exact identities and source provenance

Let

$$A_{n,k} = A_{n,k-1} + (2n + k - 1)A_{n-1,k}, \quad A_{n,0} = (2n - 1)!!, \quad A_{n,k} = 0 \ (k > n), \quad a_n = A_{n,n}.$$

Write $T_{v,k}$ for binary tree-child networks with v labeled leaves and k reticulations, and $T_v = \sum_{k=0}^{v-1} T_{v,k}$.

The exact identity is

$$T_{v,k} = \frac{v!}{(v-k)!} A_{v-1,k}. \quad (\text{T1})$$

Its provenance matters. It was Pons–Batle’s conjecture and does **not** follow from the 2024 paper alone. It is now proved by Lin, Liu, Liu, Liu, and Xin [4, Theorem 1.1 and the displayed proof following Conjecture 1.3]. Their notation satisfies $2^{n-k}a_{n,k} = (n-k+1)!b_{n,k}$; combine this with Chang et al. [3, Theorem 3.4], $T_{v,k} = v!2^{-(v-k-1)}b_{v-1,k}$.

Remark 16.1 (Notation of the cited papers; write phase). In the preceding sentence $a_{n,k}$ and $b_{n,k}$ are the arrays of the cited papers, quoted as there: $a_{n,k}$ is not the sequence a_n , and $b_{n,k}$ is neither Part I’s array (6) nor the deficit ratio $\beta_n(m)$ below.

Putting $v = n + 1$ and $k = n - m$ gives

$$T_{n+1,n} = (n+1)!a_n, \quad T_{n+1} = (n+1)! \sum_{m=0}^n \frac{A_{n,n-m}}{(m+1)!}. \quad (\text{T2})$$

There is also the exact domination

$$0 \leq \frac{T_{n+1,n-m}}{T_{n+1,n}} = \frac{A_{n,n-m}}{(m+1)!a_n} \leq \frac{1}{2^m m!}, \quad 0 \leq m \leq n. \quad (\text{T3})$$

This is Fuchs–Yu–Zhang [1, equation (4), p. 5 of the PDF], or Chang et al. [3, Lemma 3.8, equation (16)]. It follows by iterating the injective free-edge insertion inequality $2(v-k-1)T_{v,k} \leq T_{v,k+1}$.

Chang et al., Proposition 3.21 and equation (26), previously proved the leading relation $T_v \sim \sqrt{e}T_{v,v-1}$. We upgrade this to all finite orders, conditional only on the diagonal expansion.

16.2 Diagonal input and the transfer theorem

Assume that there are $\gamma > 0$, $B \in \mathbb{R}$, and real constants $L_1, L_2, \dots$ such that, for every fixed $R \geq 0$,

$$a_n = \gamma 12^n n! e^{Bn^{1/3}} n^{-2/3} \exp \left(\sum_{r=1}^R L_r n^{-r/3} + O_R(n^{-(R+1)/3}) \right). \quad (\text{T4})$$

The relevant value is $B = 3^{1/3}z$, where z is the largest negative zero of Ai. The exponential version of the hypothesis is equivalent to the usual Poincare expansion with leading coefficient 1.

Theorem 16.2 (Transfer theorem). *Under (T4), there are explicitly computable constants τ_s with $\tau_0 = 1$ such that for every fixed $R \geq 0$,*

$$\frac{T_{n+1}}{(n+1)!a_n} = \sqrt{e} \left(\sum_{s=0}^R \tau_s n^{-s/3} + O_R(n^{-(R+1)/3}) \right). \quad (\text{T5})$$

They are obtained by the polynomial recursion in the polynomial construction below. In particular,

$$\tau_1 = 0, \quad \tau_2 = \frac{B}{72}, \quad \tau_3 = \frac{1}{288}, \quad \tau_4 = -\frac{13B^2 + 80L_1}{5760}. \quad (\text{T6})$$

Thus the total and maximal-reticulation sequences have precisely the same amplitude up to the factor $\sqrt{e}$, with no new unknown connection constant.

16.3 The exact deficit recurrence

Define

$$\beta_n(m) = \frac{A_{n,n-m}}{a_n} \quad (0 \leq m \leq n).$$

The boundary equality $A_{n,n} = A_{n,n-1}$ gives $\beta_n(0) = \beta_n(1) = 1$. Rearranging the triangular recurrence gives, for $1 \leq m \leq n-1$,

$$\beta_n(m+1) = \beta_n(m) - \omega_{n,m} \beta_{n-1}(m-1), \quad \omega_{n,m} = (3n-m-1) \frac{a_{n-1}}{a_n}. \quad (\text{T7})$$

Set $t = n^{-1/3}$,

$$\phi(t) = t(1-t^3)^{-1/3}, \quad \mathcal{G}(t) = \exp \left(\sum_{r \geq 1} L_r t^r \right).$$

The formal series of $4\omega_{n,m}$ is

$$\Omega_m(t) = \left(1 - \frac{m+1}{3} t^3 \right) (1-t^3)^{-2/3} \exp \left\{ \frac{B}{t} ((1-t^3)^{1/3} - 1) \right\} \frac{\mathcal{G}(\phi(t))}{\mathcal{G}(t)}. \quad (\text{T8})$$

The expression in braces is analytic at zero. Every coefficient of Ω_m is a polynomial in m (in fact affine), with coefficients in the rational algebra generated by $B, L_1, L_2, \dots$. In particular,

$$\Omega_m(t) = 1 - \frac{B}{3} t^2 + \frac{1-m}{3} t^3 + \left(\frac{B^2}{18} + \frac{L_1}{3} \right) t^4 + O(t^5). \quad (\text{T9})$$

Seek the formal solution

$$\beta_n(m) \sim 2^{-m} \sum_{r \geq 0} P_r(m) t^r. \quad (\text{T10})$$

Write $P_0(m) = m + 1$, and, for $r \geq 1$, define P_r as the unique polynomial satisfying

$$\begin{aligned} P_r(m+1) - 2P_r(m) + P_r(m-1) &= -[t^r] \left\{ \Omega_m(t) \sum_{s=0}^{r-1} P_s(m-1) \phi(t)^s \right\}, \\ P_r(0) &= P_r(1) = 0. \end{aligned} \quad (\text{T11})$$

Existence and uniqueness follow because the second-difference operator maps polynomials modulo the two-dimensional space of affine polynomials bijectively onto all polynomials. The right side involves only lower indices: $\Omega_m(0) = 1$ and $\phi(t) = t + O(t^4)$. The two initial values impose the affine part.

Every P_r is divisible by $m + 1$. Indeed $P_0(-1) = 0$; if all lower-index polynomials vanish at -1 , evaluation of (T11) at $m = 0$, using $P_r(0) = P_r(1) = 0$, gives $P_r(-1) = 0$. Consequently

$$\tau_s = \mathbb{E} \left[\frac{P_s(X)}{X+1} \right], \quad X \sim \text{Poisson}(1/2), \quad (\text{T12})$$

is a finite polynomial calculation using Poisson moments. In particular all τ_s belong to the rational algebra generated by $B, L_1, L_2, \dots$

16.4 Uniform remainder bounds

This is the step that justifies all finite orders; a mere fixed- m asymptotic and dominated convergence would only justify the leading term.

Fix R and put $p = (R + 1)/3$. Define

$$\widehat{\beta}_n(m) = 2^{-m} \sum_{r=0}^R P_r(m) n^{-r/3}.$$

For $0 \leq m \leq \lfloor \log n \rfloor$, hypothesis (T4) and ordinary finite Taylor expansion of (T8) give

$$\omega_{n,m} = \frac{1}{4} \sum_{r=0}^R [t^r] \Omega_m(t) n^{-r/3} + O_R((1+m)n^{-p}). \quad (\text{T13})$$

To see rigorously that the diagonal error does not need to be differentiable, apply (T4) separately at n and $n - 1$, expand their quotient, and bound both remainders by $O_R(n^{-p})$. One may use a still higher finite truncation of (T4) if desired; no cancellation of unknown remainders is being assumed. Because the finite P_r 's are polynomials, substituting $\widehat{\beta}$ into (T7) gives a residual

$$\text{res}_{n,m} := \widehat{\beta}_n(m+1) - \widehat{\beta}_n(m) + \omega_{n,m} \widehat{\beta}_{n-1}(m-1), \quad |\text{res}_{n,m}| \leq C_R 2^{-m} (1+m)^{\nu_R} n^{-p}. \quad (\text{T14})$$

All powers below p vanish by (T11). The estimate is uniform in the logarithmic window: Taylor remainders have polynomial dependence on m , and $n - 1 \asymp n$. Also $\omega_{n,m} \leq 1/2$ for all sufficiently large n , uniformly in this window, by (T4).

Let $\mathcal{E}_n(m) = \beta_n(m) - \hat{\beta}_n(m)$. Its two initial values vanish and

$$|\mathcal{E}_n(m+1)| \leq |\mathcal{E}_n(m)| + \frac{1}{2}|\mathcal{E}_{n-1}(m-1)| + |\text{res}_{n,m}|. \quad (\text{T15})$$

Choose a sufficiently large constant Υ_R . Induction on m gives

$$|\mathcal{E}_n(m)| \leq C'_R \Upsilon_R^m n^{-p} \quad (0 \leq m \leq \lfloor \log n \rfloor). \quad (\text{T16})$$

For completeness, perform the induction on the domain $n \geq n_0 + m$ together with the logarithmic-window condition. Both predecessor points remain in that domain. The logarithmic-window condition is also closed: $m - 1 \leq \log(n - 1)$ whenever $m \leq \log n$ and $n \geq 2$. Enlarge the lower threshold of n so $(n/(n-1))^p \leq 2$. The two propagated terms in (T15) are then bounded by $C'_R n^{-p} (\Upsilon_R^m + \Upsilon_R^{m-1})$. Choosing $\Upsilon_R \geq 3$ leaves a fixed fraction of $C'_R \Upsilon_R^{m+1} n^{-p}$ to absorb the residual, since $2^{-m}(1+m)^{\nu_R}$ is bounded by a constant times Υ_R^m . For all sufficiently large n , $n - \lfloor \log n \rfloor \geq n_0$, so the induction domain contains the whole required window. Thus no invocation of a large- n estimate below its threshold is needed.

Multiplying (T16) by $1/(m+1)!$ and summing gives exactly the desired order, with no logarithmic loss:

$$\sum_{m \leq \log n} \frac{|\mathcal{E}_n(m)|}{(m+1)!} \leq C'_R n^{-p} \sum_{m \geq 0} \frac{\Upsilon_R^m}{(m+1)!} = O_R(n^{-p}). \quad (\text{T17})$$

This observation is why exponential-in- m stability suffices; one need not prove a sharp uniform Airy estimate at every deficit.

16.5 Superalgebraically small tails

Put $\mu_n = \lfloor \log n \rfloor$. From (T3),

$$\sum_{m > \mu_n}^n \frac{\beta_n(m)}{(m+1)!} \leq \sum_{m > \mu_n} \frac{1}{2^m m!} = \exp\{-\log n \log \log n + O(\log n)\}. \quad (\text{T18})$$

The last quantity is $O_A(n^{-A})$ for every fixed $A > 0$. The same statement holds for each formal tail $\sum_{m > \mu_n} 2^{-m} |P_r(m)|/(m+1)!$, since P_r is polynomial. Thus (T17), (T18), and (T2) yield

$$\frac{T_{n+1}}{(n+1)! a_n} = \sum_{r=0}^R n^{-r/3} \sum_{m \geq 0} \frac{2^{-m} P_r(m)}{(m+1)!} + O_R(n^{-(R+1)/3}). \quad (\text{T19})$$

Since $P_r(m)/(m+1)$ is polynomial and $\sum_{m \geq 0} 2^{-m}/m! = e^{1/2}$, the inner sum is $\sqrt{e} \tau_r$ from (T12). This proves Theorem 16.2.

16.6 Explicit correction coefficients

Solving (T11) gives

$$\begin{aligned} P_1(m) &= 0, \\ P_2(m) &= \frac{B}{18}m(m-1)(m+1), \\ P_3(m) &= \frac{1}{36}m(m-1)(m-2)(m+1), \\ P_4(m) &= \frac{m(m-1)(m+1)}{1080}\{B^2m^2 - 5B^2m - 4B^2 - 60L_1\}. \end{aligned} \quad (\text{T20})$$

Using the factorial moments of Poisson(1/2) produces (T6). In logarithmic form,

$$\log \frac{T_{n+1}}{\sqrt{e}(n+1)!a_n} = \frac{B}{72}n^{-2/3} + \frac{1}{288}n^{-1} - \left(\frac{61B^2}{25920} + \frac{L_1}{72} \right) n^{-4/3} + O(n^{-5/3}). \quad (\text{T21})$$

Therefore, if $\tilde{L}_r$ denote the logarithmic corrections in

$$T_{n+1} = \sqrt{e} \gamma (n+1)! n! 12^n e^{Bn^{1/3}} n^{-2/3} \exp \left(\sum_{r=1}^R \tilde{L}_r n^{-r/3} + O_R(n^{-(R+1)/3}) \right), \quad (\text{T22})$$

then

$$\tilde{L}_1 = L_1, \quad \tilde{L}_2 = L_2 + B/72, \quad \tilde{L}_3 = L_3 + 1/288, \quad \tilde{L}_4 = L_4 - 61B^2/25920 - L_1/72. \quad (\text{T23})$$

In terms of the number of leaves $v = n + 1$, the leading forms are

$$\begin{aligned} T_v &\sim \frac{\sqrt{e} \gamma}{12} 12^v (v!)^2 e^{Bv^{1/3}} v^{-5/3}, \\ T_v &\sim \frac{\pi \sqrt{e} \gamma}{6} v^{-2/3} e^{Bv^{1/3}} \left(\frac{12}{e^2} \right)^v v^{2v}. \end{aligned} \quad (\text{T24})$$

With the factorial normalization in the first line, the first three logarithmic corrections are L_1 , $L_2 - 23B/72$, and $L_3 + 193/288$. With the second (Stirling) normalization they are L_1 , $L_2 - 23B/72$, and $L_3 + 241/288$.

17 Asymptotic inversion and integer thresholds

The following inversion statements use the positive constant γ defined in (67). It is exact, but currently lacks a certified numerical enclosure. Each smooth model is explicit once that constant is specified. No differentiation of an unknown remainder of the integer sequence is involved.

[Write phase: in this section the logarithm of the target is y (source 69: T , which is reserved here for network counts), and every letter of source 69's generic inverse is printed in sans-serif and renamed to agree with Section 11: source 69's (d, k, A, p, C, L, D, E) are $(\mathfrak{p}, \mathfrak{k}, \mathfrak{A}, \mathfrak{g}, \mathfrak{K}, \mathfrak{L}_1, \mathfrak{L}_2, \mathfrak{L}_3)$, its s, q are $\mathfrak{s}, \mathfrak{q}$, its u, v, w, r, f are $\mathfrak{u}_1, \dots, \mathfrak{u}_5$, and the variable of its verification is ϑ (source: t). Its constant Γ_T is γ_T , and in the threshold statement its $\rho, \eta, r, N(Y)$ are $\sigma, \delta_Y, x_Y, \mathcal{N}(Y)$. No formula

changed otherwise. The model h_a is the case $d = 2$ of (39) for a_n , and the inverse (76) refines (41) there; Remark 11.1 applies.]

17.1 The word count model

By (45) and Stirling's formula,

$$\begin{aligned} h_a(t) &= t \log(12t/e) + Bt^{1/3} - \frac{1}{6} \log t + C_a + \frac{B^2}{18} t^{-1/3} - \frac{1}{36} t^{-1}, \\ C_a &= \log(\gamma\sqrt{2\pi}), \quad \log a_n = h_a(n) + O(n^{-4/3}). \end{aligned} \quad (75)$$

Both the $\frac{1}{2} \log n$ and $1/(12n)$ terms from the factorial matter: the resulting logarithmic power is $-1/6$, not $-2/3$.

For $y = \log Y$, put

$$x = \frac{y}{W_0(12y/e)}, \quad s = \log(12x).$$

Then $x \log(12x/e) = y$, and the first inverse terms are

$$h_a^{-1}(y) = x - \frac{B}{s} x^{1/3} + \frac{1}{6} - \frac{(\log 12)/6 + C_a}{s} + O(x^{-1/3}/s). \quad (76)$$

The Airy displacement is positive because $B < 0$. The amplitude first appears at order $1/\log x$.

17.2 The total count model and its index shift

Let $\gamma_T = \sqrt{e} \gamma/12$. The leaf-index expansion (46), with Stirling applied to both factorials, gives

$$\begin{aligned} h_T(t) &= 2t \log(2\sqrt{3}t/e) + Bt^{1/3} - \frac{2}{3} \log t + C_T + \frac{B^2}{18} t^{-1/3} - \frac{23B}{72} t^{-2/3} + \frac{209}{288} t^{-1}, \\ C_T &= \log(2\pi\gamma_T), \quad \log TC_n = h_T(n) + O(n^{-4/3}). \end{aligned} \quad (77)$$

The logarithmic n^{-1} coefficient before Stirling is $161/288$; twice the Stirling contribution adds $1/6$, giving $209/288$. The use of TC_n , rather than TC_{n+1} , is essential. The latter index would give logarithmic power $4/3$ instead of $-2/3$.

For $y = \log Y$ define

$$x = \frac{y}{2W_0(\sqrt{3}y/e)}, \quad s = 2 \log(2\sqrt{3}x).$$

Then

$$h_T^{-1}(y) = x - \frac{B}{s} x^{1/3} - \frac{-(2/3) \log x + C_T}{s} + O(x^{-1/3}/s). \quad (78)$$

The constant-size shift from the logarithmic prefactor is $+1/3$.

17.3 A common explicit inverse through order one over the index

For parameters $p, k > 0$, consider

$$h(t) = pt \log(kt/e) + At^{1/3} + g \log t + K + L_1 t^{-1/3} + L_2 t^{-2/3} + L_3 t^{-1}. \quad (79)$$

The two specializations are

$$\begin{aligned} (\mathbf{p}, \mathbf{k}, \mathbf{A}, \mathbf{g}, \mathbf{K}, \mathbf{L}_1, \mathbf{L}_2, \mathbf{L}_3)_a &= \left(1, 12, B, -\frac{1}{6}, C_a, \frac{B^2}{18}, 0, -\frac{1}{36}\right), \\ (\mathbf{p}, \mathbf{k}, \mathbf{A}, \mathbf{g}, \mathbf{K}, \mathbf{L}_1, \mathbf{L}_2, \mathbf{L}_3)_T &= \left(2, 2\sqrt{3}, B, -\frac{2}{3}, C_T, \frac{B^2}{18}, -\frac{23B}{72}, \frac{209}{288}\right). \end{aligned}$$

Put

$$x = \frac{y}{\mathbf{p}W_0(\mathbf{k}y/(\mathbf{p}e))}, \quad \mathbf{s} = \mathbf{p} \log(\mathbf{k}x), \quad \mathbf{q} = \mathbf{g} \log x + \mathbf{K}.$$

Define

$$\begin{aligned} \mathbf{u}_1 &= -\mathbf{A}/\mathbf{s}, & \mathbf{u}_2 &= -\mathbf{q}/\mathbf{s}, \\ \mathbf{u}_3 &= -\mathbf{L}_1/\mathbf{s} + \frac{\mathbf{A}^2}{3\mathbf{s}^2} - \frac{\mathbf{p}\mathbf{A}^2}{2\mathbf{s}^3}, \\ \mathbf{u}_4 &= -\mathbf{L}_2/\mathbf{s} + \frac{\mathbf{A}(\mathbf{g} + \mathbf{q}/3)}{\mathbf{s}^2} - \frac{\mathbf{p}\mathbf{A}\mathbf{q}}{\mathbf{s}^3}, \\ \mathbf{u}_5 &= -\mathbf{L}_3/\mathbf{s} + \frac{\mathbf{g}\mathbf{q}}{\mathbf{s}^2} - \frac{\mathbf{p}(\mathbf{A}\mathbf{L}_1 + \mathbf{q}^2/2)}{\mathbf{s}^3} + \frac{\mathbf{p}\mathbf{A}^3}{3\mathbf{s}^4} - \frac{\mathbf{p}^2\mathbf{A}^3}{2\mathbf{s}^5}. \end{aligned} \tag{80}$$

Then

$$h^{-1}(y) = x + \mathbf{u}_1 x^{1/3} + \mathbf{u}_2 + \mathbf{u}_3 x^{-1/3} + \mathbf{u}_4 x^{-2/3} + \mathbf{u}_5 x^{-1} + O(x^{-4/3}/\mathbf{s}). \tag{81}$$

For verification, set $\vartheta = x^{-1/3}$ and $\delta = \mathbf{u}_1/\vartheta + \mathbf{u}_2 + \mathbf{u}_3\vartheta + \mathbf{u}_4\vartheta^2 + \mathbf{u}_5\vartheta^3$. The successive coefficients of $h(x + \delta) - y$ through degree three in ϑ are

$$\begin{aligned} &\mathbf{s}\mathbf{u}_1 + \mathbf{A}, & \mathbf{s}\mathbf{u}_2 + \mathbf{q}, \\ &\mathbf{s}\mathbf{u}_3 + \mathbf{p}\mathbf{u}_1^2/2 + \mathbf{A}\mathbf{u}_1/3 + \mathbf{L}_1, \\ &\mathbf{s}\mathbf{u}_4 + \mathbf{p}\mathbf{u}_1\mathbf{u}_2 + \mathbf{A}\mathbf{u}_2/3 + \mathbf{g}\mathbf{u}_1 + \mathbf{L}_2, \\ &\mathbf{s}\mathbf{u}_5 + \mathbf{p}\mathbf{u}_2^2/2 + (\mathbf{p}\mathbf{u}_1 + \mathbf{A}/3)\mathbf{u}_3 - \mathbf{p}\mathbf{u}_1^3/6 - \mathbf{A}\mathbf{u}_1^2/9 + \mathbf{g}\mathbf{u}_2 - \mathbf{L}_1\mathbf{u}_1/3 + \mathbf{L}_3. \end{aligned}$$

All vanish identically with (80). Since $\mathbf{q} = O(\mathbf{s})$, $\mathbf{u}_1 = O(\mathbf{s}^{-1})$, $\mathbf{u}_2 = O(1)$ and $\mathbf{u}_3, \mathbf{u}_4, \mathbf{u}_5 = O(\mathbf{s}^{-1})$, the remaining residual is $O(x^{-4/3})$. The derivative of the explicit model is asymptotic to $\mathbf{s}$, so the mean-value theorem proves (81). The symbolic replay keeps all parameters free, and the independent audit separately expands the actual smooth model.

17.4 All finite inverse orders and Newton computation

For every fixed $\sigma > 0$ the forward theorem and the transfer provide a smooth model h_σ with logarithmic error $O(n^{-\sigma})$ on the integers. One may retain $\log \Gamma(t+1)$ exactly rather than truncate Stirling. Every such model has

$$h'_\sigma(t) \sim \mathbf{p} \log(\mathbf{k}t) > 0, \quad h''_\sigma(t) \sim \mathbf{p}/t.$$

Starting at the Lambert anchor $x_0 = x$, perform

$$x_{j+1} = x_j - \frac{h_\sigma(x_j) - y}{h'_\sigma(x_j)}.$$

The initial error to the model inverse is $O(x^{1/3}/\log x)$, and each Newton step multiplies its square by $O(1/(x \log x))$. The error after j steps is therefore

$$O\left(x^{1-2^{j+1}/3}(\log x)^{1-2^{j+1}}\right). \tag{82}$$

Finitely many steps put numerical model inversion below the unavoidable forward uncertainty $O(x^{-\sigma}/\log x)$. Equivalently, the coefficient construction leading to (81) is triangular and extends to every finite order.

17.5 The exact discrete interpretation

Both counting sequences are eventually strictly increasing. Their consecutive logarithmic differences are respectively $\log(12n) + o(1)$ and $2\log(2\sqrt{3}n) + o(1)$. Define $\mathcal{N}(Y)$ as the least index with count at least Y . Suppose a model satisfies

$$|\log a_n - h(n)| \leq Kn^{-\sigma}$$

for all sufficiently large integers, with the analogous statement for total counts. Let $x_Y = h^{-1}(\log Y)$. For all sufficiently large Y , a conservative radius is

$$\delta_Y = \frac{8 \max(K, 1) x_Y^{-\sigma}}{\mathfrak{p} \log(\mathfrak{k} x_Y)}.$$

Monotonicity and the mean-value theorem give the valid bracket

$$\lceil x_Y - \delta_Y \rceil \leq \mathcal{N}(Y) \leq \lceil x_Y + \delta_Y \rceil. \quad (83)$$

An explicit asymptotic inverse can replace x_Y if its proved absolute error is added to δ_Y . When the bracket crosses no integer, it identifies the threshold. Arbitrarily close to a sequence value it can span two indices, as it must. An unqualified ceiling of a truncated real expansion is not justified. Effective numerical certification additionally requires rigorous bounds for γ and the forward remainder; the present existence proof supplies neither numeric enclosure automatically.

18 Deficit distribution and sharp total variation

The all-orders transfer above yields a quantitative refinement of the published Poisson limit. All statements in this section concern a uniformly chosen binary tree-child network with $n+1$ labelled leaves.

18.1 Exact law and notation

Let X_n be the number of missing reticulations relative to the maximum n . Set

$$\beta_n(m) = A_{n,n-m}/a_n, \quad \mathcal{R}_n = T_{n+1}/((n+1)!a_n), \quad \pi_{1/2}(m) = e^{-1/2} \frac{2^{-m}}{m!}.$$

By the exact Lin–Chang counting identity,

$$\pi_n(m) := \mathbb{P}(X_n = m) = \frac{\beta_n(m)}{(m+1)!\mathcal{R}_n}, \quad 0 \leq m \leq n, \quad (D1)$$

and $\pi_n(m) = 0$ for $m > n$. The probability mass function $\pi_{1/2}$ is Poisson with mean $1/2$.

The transfer theorem supplies polynomials P_r , with $P_0(m) = m+1$, and constants τ_r , with

$$\mathcal{R}_n \sim \sqrt{e} \sum_{r \geq 0} \tau_r n^{-r/3}, \quad \tau_r = \sum_{m \geq 0} \pi_{1/2}(m) \frac{P_r(m)}{m+1}.$$

Every P_r is divisible by $m+1$. Define the polynomials $C_r(m)$ by formal division:

$$\sum_{r \geq 0} C_r(m) t^r = \frac{\sum_{r \geq 0} P_r(m) t^r / (m+1)}{\sum_{r \geq 0} \tau_r t^r}. \quad (\text{D2})$$

Then $C_0 = 1$, and $\mathbb{E}_{\pi_{1/2}} C_r(X) = 0$ for all $r \geq 1$.

18.2 Weighted expansion to every finite order

Theorem 18.1 (Weighted expansion). *For every fixed integer $J \geq 0$ and every fixed $\theta \geq 0$,*

$$\sum_{m \geq 0} e^{\theta m} \left| \pi_n(m) - \pi_{1/2}(m) \sum_{r=0}^J C_r(m) n^{-r/3} \right| = O_{J,\theta}(n^{-(J+1)/3}). \quad (\text{D3})$$

Consequently the expansion can be integrated against any fixed function with at most exponential growth, including every polynomial. It also gives a uniform expansion of the probability generating function on each fixed complex disk.

Proof of Theorem 18.1. In the logarithmic window $0 \leq m \leq \lfloor \log n \rfloor$, the transfer proof gives

$$\beta_n(m) = 2^{-m} \sum_{r=0}^J P_r(m) n^{-r/3} + O_J(\Upsilon_J^m n^{-(J+1)/3}). \quad (\text{D4})$$

Divide by $(m+1)!\mathcal{R}_n$, use the finite expansion of $1/\mathcal{R}_n$, and sum with weight $e^{\theta m}$. The error series converges because

$$\sum_{m \geq 0} \frac{(e^\theta \Upsilon_J)^m}{(m+1)!} < \infty.$$

The other errors are polynomial times $\pi_{1/2}(m)$ and satisfy the same estimate. Outside the window, the exact domination from FYZ2021 gives

$$\pi_n(m) \leq \frac{1}{\mathcal{R}_n 2^m m!} \leq \frac{1}{2^m m!},$$

since $\mathcal{R}_n \geq 1$. After multiplication by any fixed exponential weight, this tail and each polynomial-Poisson formal tail remain $\exp\{-\log n \log \log n + O_{J,\theta}(\log n)\}$, hence smaller than every fixed algebraic order. This proves (D3). No pointwise relative-error assertion for the far tail is used. $\square$

The first two nonzero correction polynomials are

$$\begin{aligned}
C_1(m) &= 0, \\
C_2(m) &= B \left(\frac{m(m-1)}{18} - \frac{1}{72} \right), \\
C_3(m) &= \frac{m(m-1)(m-2)}{36} - \frac{1}{288}.
\end{aligned} \tag{D5}$$

Here $B = 3^{1/3}z < 0$, with z the largest negative Airy zero. The Poisson limit is Chang, Fuchs, Liu, Wallner, and Yu [3, Theorem 1.9 and equation (26)]; moment convergence is discussed in their Remark 3.23. Formula (D5) supplies higher-order corrections to those published results.

18.3 Total variation through all algebraic orders

Use the convention

$$d_{\text{TV}}(\pi_n, \pi_{1/2}) = \frac{1}{2} \sum_{m \geq 0} |\pi_n(m) - \pi_{1/2}(m)| = \sum_{m \geq 0} (\pi_n(m) - \pi_{1/2}(m))_+. \tag{D6}$$

Theorem 18.2 (Exact eventual total variation). *For every sufficiently large n , there is the exact identity*

$$d_{\text{TV}}(\pi_n, \pi_{1/2}) = \frac{3}{2} \left(\frac{1}{\mathcal{R}_n} - e^{-1/2} \right). \tag{D7}$$

Consequently its full algebraic expansion is obtained by formal inversion of the total/maximal ratio. In particular,

$$d_{\text{TV}}(\pi_n, \pi_{1/2}) = \frac{|B|}{48\sqrt{e}} n^{-2/3} - \frac{1}{192\sqrt{e}} n^{-1} + O(n^{-4/3}). \tag{D8}$$

Using $L_1 = B^2/18$, one further term is

$$d_{\text{TV}}(\pi_n, \pi_{1/2}) = \frac{|B|}{48\sqrt{e}} n^{-2/3} - \frac{1}{192\sqrt{e}} n^{-1} + \frac{167B^2}{34560\sqrt{e}} n^{-4/3} + O(n^{-5/3}). \tag{D9}$$

Proof of Theorem 18.2. The one-step free-edge insertion inequality of Fuchs–Yu–Zhang [1, Lemma 2] gives

$$\pi_n(m) \leq \frac{\pi_n(m-1)}{2m} \quad (1 \leq m \leq n).$$

Since $\pi_{1/2}(m) = \pi_{1/2}(m-1)/(2m)$, it follows that the likelihood ratio

$$\text{lr}_n(m) = \frac{\pi_n(m)}{\pi_{1/2}(m)}$$

is nonincreasing in m , including the zero values for $m > n$. This is an exact inequality, valid at every n .

The exact domination and finite support imply

$$\mathcal{R}_n \leq \sum_{m=0}^n \frac{1}{2^m m!} < \sqrt{e}.$$

Also $\beta_n(0) = \beta_n(1) = 1$, so $\text{lr}_n(0) = \text{lr}_n(1) = \sqrt{e}/\mathcal{R}_n > 1$. The deviations at $m = 0, 1$ are therefore strictly positive and sum to

$$\left(\frac{1}{\mathcal{R}_n} - e^{-1/2} \right) + \frac{1}{2} \left(\frac{1}{\mathcal{R}_n} - e^{-1/2} \right). \quad (\text{D10})$$

At $m = 2$, the fixed-deficit expansion (D5) gives

$$\text{lr}_n(2) = 1 + \frac{7B}{72} n^{-2/3} + O(n^{-1}) < 1 \quad (\text{D11})$$

for all sufficiently large n , because $B < 0$. Monotonicity then implies $\text{lr}_n(m) < 1$ for every $m \geq 2$. Thus the positive-deviation set in (D6) is exactly $\{0, 1\}$ for all sufficiently large n , which proves (D7). This proof requires no uniform pointwise sign assertion deduced from the growing-in- m error in (D4).

Finally,

$$\mathcal{R}_n = \sqrt{e} \left(1 + \frac{B}{72} n^{-2/3} + \frac{1}{288} n^{-1} + O(n^{-4/3}) \right),$$

so expanding its reciprocal in (D7) gives (D8). At order $n^{-4/3}$, the coefficient is $\frac{3}{2\sqrt{e}}(\tau_2^2 - \tau_4)$; using $\tau_2 = B/72$ and $\tau_4 = -157B^2/51840$ gives (D9). $\square$

For every n , the same likelihood-ratio monotonicity also implies the exact stochastic domination $X_n \leq_{\text{st}} \text{Poisson}(1/2)$. Indeed the decreasing likelihood ratio crosses 1 at most once, so each upper-tail sum for π_n is at most the corresponding Poisson upper-tail sum. This observation is a direct consequence of the published insertion inequality; no claim of priority is made for it.

[5 October 2026, batch 102: source 102 (Part IV, Proposition 34.3) proves the weaker bounds $\mathcal{R}_n = \sqrt{e}(1 + O(n^{-2/3}))$ and $d_{\text{TV}} = O(n^{-2/3})$ from Lemmas 3.17–3.20 of [3] alone, without the identity (T1); (D7) implies its bound for all large n (Remark 34.4).]

The use of $n + 1$ leaves is explicit throughout. Replacing n by the leaf number minus one does not change the coefficients displayed in (D8) or (D9); index-shift effects first enter at order $n^{-5/3}$.

18.4 Probability generating function and moment refinements

For every fixed $Z < \infty$, uniformly for complex $|u| \leq Z$, (D3) and the Poisson factorial-moment identities give

$$\mathbb{E}[u^{X_n}] = e^{(u-1)/2} \left\{ 1 + \frac{B}{72} (u^2 - 1) n^{-2/3} + \frac{1}{288} (u^3 - 1) n^{-1} + O_Z(n^{-4/3}) \right\}. \quad (\text{D13})$$

The remainder is locally uniform and analytic; alternatively, apply (D3) directly with polynomial weights to obtain

$$\begin{aligned}\mathbb{E}[X_n] &= \frac{1}{2} + \frac{B}{36}n^{-2/3} + \frac{1}{96}n^{-1} + O(n^{-4/3}), \\ \text{Var}(X_n) &= \frac{1}{2} + \frac{B}{18}n^{-2/3} + \frac{1}{32}n^{-1} + O(n^{-4/3}).\end{aligned}\tag{D14}$$

The mean and variance both approach $1/2$ from below at leading order. Every higher moment, exponential moment, and finite-order correction is computable from (D2).

19 Reproducibility and numerical diagnostics

The accompanying archive separates exact algebraic assertions from numerical evidence. Its main replay script checks:

1. the transformed recurrence and gauge contraction by exact rational arithmetic, including artificial endpoints;
2. the frozen and nonautonomous polynomial–Airy identities through degree seven, with zero symbolic residuals;
3. the coefficient-field argument in the independent natural coordinate system;
4. the deficit-polynomial transfer through degree six and finite exact counting identities;
5. the generic inverse through the displayed order, with both factorial powers retained symbolically;
6. independently derived coefficients, index shifts, and inverse substitutions.

These computations check the algebra, not the compactness, gap, stability or positivity arguments. The latter are proved in the preceding sections and examined in the accompanying mathematical verification documents. *[Write phase: in the repository the main replay script is `code/69-binary-reproduce.py`, and the verification documents are `69-binary-mathematical-verification.md`, `69-binary-coef-certificate.md` and `69-binary-coef-numerical-and-field-check.md`; the *README* of this report maps every delivered name to its shipped name.]* The finite source checks do not replace the all-index combinatorial identities from the cited literature.

A direct floating-point evaluation of the defining recurrence, normalized by $12^n n!$, gives useful diagnostics. Correcting by the four logarithmic terms displayed in (45) produces the following finite-index values:

n	Four-correction amplitude diagnostic
10000	2.0226419900894603
20000	2.0226420075718210
30000	2.0226420114939580
50000	2.0226420138123630

These are uncertified numerical diagnostics. They are not rigorous enclosures for γ , and their printed digits are not claimed correct digits of that limit. Exact integer checks through $n = 80$ and a separate run in the platform’s extended precision through $n = 10000$ support the implementation,

[write phase: the script uses NumPy’s *longdouble*, which is the 80-bit extended format on x86-64 Linux. The recorded run (`data/69-binary-coef-forward-numerical.json`) reports nonzero relative differences from double precision, up to 2.0×10^{-14} , so it used a genuinely extended format. On platforms where *longdouble* is double precision, as on Windows, a rerun of this comparison is vacuous and reports differences of exactly zero] but do not bound the finite- n asymptotic remainder. Likewise, direct tridiagonal eigenvalue calculations support the frozen coefficients without replacing the quasimode proof.

The exact mathematical specification of the amplitude remains (65), (62), and (67). An effective numerical enclosure would require quantitative constants for spectral remainders and projection tails, or a different certified method. [5 October 2026, batch 102: source 102’s independent diagnostic, with six logarithmic corrections, gives 2.0226420146 at $n = 2000$ (Part IV, Section 39.3), consistent with this table; it is equally uncertified.]

20 Literature status and scope of the conclusion

The source check distinguishes the following milestones.

- The order estimate (43) is already known from the 2021 literature. It is an input here, not a new result.
- Chang et al. [3] proved the leading total-to-maximal ratio $\sqrt{e}$ and the Poisson limit for the reticulation deficit. The all-finite-orders transfer above is stronger than using that leading relation alone.
- Chang’s dissertation [5] is catalogued in 2026, while the inspected PDF is dated 31 December 2025. It describes the missing multiplicative constant in the tree-child asymptotics. The distinction between the cataloguing year and the PDF date is retained.
- Lin et al. [4], version 3 dated 6 September 2026, proved the exact tableau relation used in the total-count transfer. The identity must not be retrospectively attributed as already proved by the 2024 paper alone.

The source receipts in the archive record the inspected versions and relevant locations [write phase: shipped as `69-binary-sources-source-status.md` and `data/69-binary-sources-source-status.json`]. They support the stated relationship to these papers. They are not an exhaustive guarantee that no other author has obtained a similar result in unpublished work. [5 October 2026, batch 102: sources 104 and 102 screened further 2026 works [8, 9, 10, 11, 12, 13], which concern exact identities, bounded, sparse or one-reticulation regimes, and likewise found no positive amplitude for the maximal diagonal in the literature (Sections 30 and 34.2).]

The report establishes a positive limiting amplitude, every finite algebraic Airy correction, a transfer to total binary tree-child networks, sharp distributional corrections and total-variation rates, and all-finite-order asymptotic inversion. It does not determine a closed form for γ , prove convergence of the infinite correction series, provide a beyond-all-orders transseries, or certify an unconditional integer rounding rule. No external publication or submission has been made.

Bibliography note. [Write phase: the bibliography merges those of both sources. Source 16’s entries *Chang* and *Thesis* are [3] and [5]; its entry *Binary*, the then-unpublished binary report

without a locator, is source 69 and is cited here as Part II. Source 69’s entry *OEIS*, uncited in its text, is now cited in Section 14.]

Part III

Fixed degree in the gamma gauge: explicit universal coefficients (source 104)

Source 104 (bundle Report 104, archive `Fixed_Degree_Tree_Child_Networks_All_Orders_Source`), printed in full; its title was “Positive amplitudes and all order expansions for fixed degree tree child networks: a self contained proof for every fixed combining degree”. Its abstract reads: For every fixed integer $d \geq 2$, we prove that the constrained-word counts underlying maximally reticulated d -combining tree-child networks have a strictly positive finite leading amplitude and a complete Poincaré expansion in powers of $n^{-1/3}$. An exact gamma-product normalization yields contractive positive Jacobi products for every such d , including the degrees where a simpler factorial normalization fails to be contractive. We prove the Airy spectral limit, localization, endpoint smoothing, positivity of the product amplitude, and arbitrary-depth transfer of polynomial–Airy profiles. We give a canonical convergent amplitude representation, an exact coefficient algorithm, explicit first corrections, and all-order Lambert–Newton inverses. Published total-to-maximal ratio results then give leading equivalents for total networks at every fixed degree. For $d = 3$, a quantitative consequence of the published proof transfers the first correction with relative remainder $O(n^{-1/2})$. The all-order theorem concerns word and maximal counts; an all-order expansion for total networks is not asserted. Finite exact checks accompany the proof. No certified amplitude digits, convergence of the infinite correction series, exponentially small sector, or exhaustive priority claim is made.

21 Reading Part III: provenance, credit and notation

This section was written when source 104 was added to the report (batch 102, 5 October 2026). In Part III, text added in writing is marked “[Write phase: ...]”.

21.1 Provenance and credit

Source 104 is bundle Report 104 of the repository’s incoming research reports, archive `Fixed_Degree_Tree_Child_Networks_All_Orders_Source.zip` (489,776 bytes), main file `general_tree_child_amplitudes.tex` (926 lines, 22-page PDF), dated 2 October 2026, with the author line “Research prepared for Vladimir / Report 104”. It arrived in commit `60f54ea06` and was placed in this directory by `6ab1f1979` (batch 102), with the file prefix `104-fixdeg-`. It names no repository commit, so it has no pin.

Credit order. The archive of source 104 was closed on 1 October 2026 at 23:18 Pacific time (06:18 UTC on 2 October), before the batch-77 archives of sources 16 and 69 reached the repository (`096ee7b87`, 2 October, 11:05 Pacific) and before they were placed and written here (`34f1acd4b`, `4dde85e7f`). Source 104 therefore proves Theorem 2.1 independently of Parts I–II; it neither cites nor uses them. In the repository Theorem 2.1 was printed first, from sources 16 and 69; source 104

is a third, independent route to it for every fixed d , by a different gauge, and is credited as such, not as a derivative of Parts I–II. Its literature-priority sentences (Sections 22 and 30) are true of the literature it inspected and are qualified below only where the repository already held the same theorem.

21.2 What Part III proves again and what it adds

Source 104	Relation to Parts I–II
Theorem 22.1 (positive amplitude, all orders)	second proof of Theorem 2.1 (third route counting Part II at $d = 2$): two-step elimination in an exact gamma-product gauge with a Gram factor, instead of Part I's one-step partial-sum gauge with a parity gap
Closed forms (87), (146), (147), (148) for every d ; $C_{d,1} < 0$ for $d \geq 4$	new: Part I proves $C_{d,r} \in \mathbb{Q}[B]$ and makes them explicit only at $d = 2, 3$
Amplitude formula (124)	a new convergent representation of the same γ_d as (31)
Proposition 28.1 and (90) ($d = 3$ total ratio and first total correction)	new; Part I has only the leading total equivalent for $d \geq 3$; rests on source 104's reading of a published proof (Section 32)
Ternary logarithmic coefficients (150)	second derivation of (38), identical values; the multiplicative third coefficient is new
Corollary 22.3	second proof of (5) and of the leading total equivalents
Section 29 (inverses, thresholds)	second route of Section 11; the amplitude-free rounding inverse, the explicit Newton count and the $d = 3$ total inverse are new details; no novelty claimed (Remark 11.1)
Section 30 ($d = 4$ prefixes, the A213864 trap)	new information; Part I says only that no $d \geq 3$ accession was verified
Remark on the naive factorial gauge (Section 23.3)	concerns source 104's two-step frame only; Part I is unaffected (Remark 23.1)

21.3 Notation of Part III

Part III uses the host's letters for every notion the host already names, and source 104's letters, renamed where they collide, for its own proof apparatus. The renamings are listed below; no formula changed otherwise. Source 104's proof runs in the two-step time n (the host's $N = 2n$) and in its own gauge, so its eigenvalues, vectors and carriers are not those of Parts I–II even where the role is the same. A letter not listed has the host's meaning. Here $d_- = d - 1$ and $d_+ = d + 1$.

Symbol here	Meaning	Source 104; the false reading to avoid
z, z_j d_-, d_+	Airy zeros, $z = z_1$ $d - 1, d + 1$	α, α_j ; not the network exponent α r, a ; not the index of $C_{d,r}$, the residual r_N , the sequence a_n or the gap constant a_d
Λ, Γ	$d_+^{d_-}/d_-!$, 4Λ (host's)	ℓ, g ; source 104's ℓ is not the host's $\ell = c^{2/3}z$
$B/3$ ρ, α c, ℓ	$z(d_-/d_+)^{2/3}$, a third of the host's B word and maximal exponents (host's) $2d_-/d_+, c^{2/3}z$ (host's)	ρ ; not the host's exponent ρ θ, μ χ, κ (identical meaning)

Symbol here	Meaning	Source 104; the false reading to avoid
$c_A; f_A; \mathcal{H}_A$	$(4c)^{1/3} = (8d_-/d_+)^{1/3}$; unit Dirichlet ground state (103); its operator	$b_A; f; \mathcal{H}$
F	$\text{Ai}(z + c^{1/3}x)$, the host's F	$A(x)$
$\delta_f, \zeta_f, \eta_f$	$d_-d/(2d_+)$, $3d_-(2d_+)$, $\zeta_f - 1/6$	δ, ζ, η ; not the host's $\eta = (d-1)^2/(2(d+1))$, not Riemann's $\zeta(2/3)$
$Q_\Gamma; G_\Gamma(p)$	$\prod_{i \leq d_-} \Gamma(1 + i/d_+)$; the gamma-product gauge (93)	$Q; G(p)$; not $Q_N(j)$, $G_N(j)$
$\tilde{b}(p, q); \mathcal{W}, \mathcal{V}$	$b_{p+1, q+1}$; the rotated weights	$B(p, q); W, V$; not the Airy constant B , nor W_0, W_N
$\mathcal{Y}(N, h); \omega_N^\pm(h); e_n(k)$	one-step gauged array; its weights; $\mathcal{Y}(2n, 2k)$	$y(N, h); u_N(h), v_N(h); e_n$; not the target logarithm y , Part I's u_N, v_N , Part II's $\omega_{n,m}$, or the error vector e_N
$\mathfrak{z}, \mathfrak{P}$	$d_+n + d_-k, d_+(n+k)$	z, P (local)
$A_n, B_n, C_n, F_n; L_n, U_n, D_n; M_n$	factors (95); entries of the two-step matrix M_n	$A_n, B_n, C_n, F_n; L_n, U_n, D_n; M_n$; not $A_{n,k}, D_N(j), M_N$, nor the sans-serif inversion letters of Sections 11 and 17
$\Sigma_n, \varsigma_n(j); R_n$	symmetrizer, its edge ratio; inter-time ratio (100)	$S_n, s_n(j); R_n$; not the Jacobi matrix S_N or $\mathcal{R}_n$
$J_n; J_\infty; \mathbf{b}_n(k); K_n$	symmetric two-step matrix; $J + 2I - e_0 e_0^\top$ (diagonal $1, 2, 2, \dots$); its off-diagonal; Gram factor	same; same; $\beta_n(k); K_n$; the host's J (no index) is half-line adjacency; not Part II's $\beta_n(m)$
$\lambda_n, \lambda_n^{(j)}, \psi_n, \psi_n^A$	top and j -th eigenvalue of J_n (near 4); Perron vector; normalized Airy sample	λ_n, ϕ_n, q_n ; the host's λ_N is near 2
P_n, u_n, T_n	$\prod_{j \leq n} \lambda_j$; normalized vector; one normalized step $(J_n/\lambda_n)R_n$	$\Lambda_n, w_n, T_n = H_n R_n$; the roles of the host's P_N, u_N, M_N
$p_n, p_n^\perp, p_\infty; C_\lambda$	Perron coefficient and transverse norm of u_n , its limit; eigenvalue-product constant	$p_n, q_n^\perp, p_\infty; C_\lambda$; not $p_N(j)$
$\mathcal{K}; \mathbf{z}_n, \varrho_n; \Delta$	$J_\infty/4$; signed errors and forcing; Duhamel window $\lfloor n^{2/3} \rfloor$	$P; z_n, \varrho_n; L$ (renamed as in Part I)
$a_n^{\text{err}}, b_n^{\text{err}}, c_{\text{gap}}$	quantities of Lemma 27.1	a_n, b_n, c_1
$s(t), \sigma_j, h_j; P_j, Q_j, f_j; \mathfrak{F}, \mathfrak{F}^{[M]}$	formal eigenvalue series, its scalars, carrier coefficients; profiles; formal series	s, σ_j, h_j ; same; $F, F^{[M]}$; the host's s_m, h_r are Part I's gauge, with other values; the host's σ is a threshold exponent
$\mathcal{H}_N^{[M]}, E_n^{[M]}, K_E, V_n^{[M]}, Z_n^\Gamma, \Theta_n$	carrier, even-time profile, its constant, normalized profile, gamma conversion, bulk scale	same, $K, U_n^{[M]}, Z_n, B_n^{\text{bulk}}$; not the host's H_N, Z_N
$\mathfrak{B}_\ell$	Airy solution with $\mathfrak{B}_\ell(0) = 0$, $\mathfrak{B}'_\ell(0) = 1$	B_κ
$C_{d,j}, L_{d,j}; E_{d,j}$	multiplicative and logarithmic coefficients of a_n ; of $T_{n,n-1}^{(d)}$	$c_j(d), b_j(d); C_{d,j}$ and $E_{d,j}$ are the host's $C_{d,r}, E_{d,r}$
$a_n; T_{n,k}^{(d)}, T_{n,n-1}^{(d)}, T_n^{(d)}; K_d$	word count (fixed d); network counts; total amplitude	$c_d(n); \text{TC}_{n,k}^{(d)}, M_d(n), T_d(n); K_d$
$\mathbf{p}, \mathbf{r}, \gamma_\star; S(n); \Phi$	factorial power, exponent, amplitude of an inverted scale; that scale; smooth model	$q, \tau, K; A(n); G(x)$ (as in Section 11)

Symbol here	Meaning	Source 104; the false reading to avoid
$y; Y$	logarithm of a target; a threshold value	$Y; y$ (swapped to agree with Sections 11 and 17)
$\hat{x}_{p,r}(y); m_J; k_A, A_0; \Psi$	amplitude-free inverse; Newton count; localization cut $\lceil A_0/\varepsilon \rceil$; digamma	$z_{q,\tau}(Y); K; K, A; \psi$
$\mathfrak{C}_m; \mathfrak{r}(\omega'), \mathfrak{s}(\omega)$	ratio bound of Proposition 28.1; class cardinalities	$C_m; r(\omega'), s(\omega)$
ε, x_k, X	$n^{-1/3}, \varepsilon(k + \frac{1}{2}), \varepsilon(k + \frac{1}{2})$ (two-step time)	same; the host's ε is $N^{-1/3}$

The amplitude γ_d of Part III is the host's γ_d : both are the limit of $a_n/((n!)^{d-1}\Gamma^n e^{Bn^{1/3}} n^\rho)$ for the same sequence a_n , since source 104's $3\rho, \theta$ and g are the host's B, ρ and Γ . *[Write phase: checked symbolically, together with every conversion of a displayed coefficient below.]*

22 The counts and the main results

22.1 Original combinatorial conventions

Fix an integer $d \geq 2$. Throughout this Part, d is fixed as $n \rightarrow \infty$; constants in bounds may depend on d . Use the conventions of Chang, Fuchs, Liu, Wallner, and Yu [3]. A network is a simple directed acyclic graph with one root of indegree zero and outdegree one, n leaves of indegree one and outdegree zero bijectively labelled by $[n]$, and internal vertices of either indegree one and outdegree two (tree vertices), or indegree d and outdegree one (reticulations). There are no vertices of indegree and outdegree one. The tree-child condition requires every nonleaf vertex to have at least one child that is not a reticulation. Networks are counted up to isomorphism preserving leaf labels; no plane ordering is imposed.

Write $T_{n,k}^{(d)}$ for their number with k reticulations. The maximum is $k = n - 1$; the maximal and total counts are $T_{n,n-1}^{(d)}$ and $T_n^{(d)} = \sum_{k=0}^{n-1} T_{n,k}^{(d)}$. These are general tree-child networks, rather than the one-component subclass in which each reticulation is directly followed by a leaf.

The relevant words use n ordered letters $\omega_1, \dots, \omega_n$, each occurring $d + 1$ times. In each prefix, if the count of ω_i exceeds $d - 2$, that count must be at least the count of every ω_j with $j > i$. Let a_n denote the number of these words, and set $a_0 = 1$. Equivalently they are the rectangular $(d + 1) \times n$ Young tableaux with the wall convention in [3, Remark 3.3], with vertical walls in all but the first two rows. Their exact recurrence, from [3, (17)], is (6). *[Write phase: source 104 prints the same display with $b(n, m)$ for $b_{n,m}$ and $c_d(n)$ for a_n ; it is printed once, in Part I.]* The index d on b is suppressed, but never varied within a recurrence. The exact network bijection gives

$$T_{n,n-1}^{(d)} = n! a_{n-1}. \quad (84)$$

It is essential to keep the startup $b_{1,1} = 1$: a convenient rescaling for a two-sided order estimate would change the amplitude here.

22.2 The canonical asymptotic scale

Let $z = z_1 > z_2 > \dots$ be the negative zeros of the Airy function Ai , $z = -2.338107410459767\dots$. With the host's $\Lambda, \Gamma, c, \ell, B, \rho, \alpha$ of Section 2, define further

$$Q_\Gamma = \prod_{i=1}^{d_-} \Gamma(1 + i/d_+), \quad \delta_f = \frac{d_- d}{2d_+}, \quad \zeta_f = \frac{3d_-}{2d_+}, \quad (85)$$

so that $B/3 = z(d_-/d_+)^{2/3}$ and

$$\rho = \delta_f - d_- + \zeta_f - \frac{1}{2} = -\frac{d_-^2 + d_- + 2}{2d_+}, \quad \alpha = \rho - d_- = -\frac{d(3d-1)}{2(d+1)}.$$

Here Γ is the exponential base, not the unknown amplitude. Source 104 chose a separate letter for it to avoid confusing it with the notation $\gamma(d)$ for that base in [3, Theorem 1.8]; Part I's $\Gamma = 4\Lambda$ serves the same purpose.

Theorem 22.1 (Positive amplitude and all orders; source 104). *For each fixed integer $d \geq 2$ there is a canonically determined $\gamma_d \in (0, \infty)$ and a compatible sequence $C_{d,j}$, $C_{d,0} = 1$, such that for every fixed $J \geq 0$,*

$$a_n = \gamma_d (n!)^{d-1} \Gamma^n e^{Bn^{1/3}} n^\rho \left(\sum_{j=0}^J C_{d,j} n^{-j/3} + O(n^{-(J+1)/3}) \right). \quad (86)$$

The first coefficient is

$$C_{d,1} = \frac{z^2(8(d+1)^2 - 27(d-1)^2)}{90(d+1)^{4/3}(d-1)^{2/3}} = \frac{B^2(8(d+1)^2 - 27(d-1)^2)}{810(d-1)^2}. \quad (87)$$

In particular,

$$C_{2,1} = \frac{z^2}{2 \cdot 3^{4/3}} = \frac{B^2}{18}, \quad C_{3,1} = \frac{z^2}{36 \cdot 2^{1/3}} = \frac{B^2}{162}, \quad C_{d,1} < 0 \quad (d \geq 4).$$

All coefficients are effectively obtained by the exact polynomial algorithm in Section 26. They lie in $\mathbb{Q}[z, (d_-/d_+)^{1/3}]$ for fixed d .

This is a Poincaré expansion, meaning an asymptotic statement for each fixed finite depth. Neither convergence of the infinite formal series nor exponentially small terms is part of the theorem.

Remark 22.2 (Relation to Part I; write phase). Theorem 22.1 is Theorem 2.1 for the word counts, proved again by a different route; its $C_{d,j}$ are Part I's $C_{d,r}$. The second form of (87), in the host's $B = 3z(d_-/d_+)^{2/3}$, is a write-phase conversion: $B^2 = 9z^2(d_-/d_+)^{4/3}$, so the two forms agree identically. It reproduces the host's $C_{2,1} = B^2/18$ (Theorem 14.1) and the first coefficient $B^2/162$ of (38). The sign assertion holds because $8(d+1)^2 - 27(d-1)^2$ is 45 at $d = 2$, 20 at $d = 3$, -43 at $d = 4$, and is a downward parabola in d whose larger root is $(35 + 12\sqrt{6})/19 = 3.389\dots$; so $C_{d,1} < 0$ exactly for $d \geq 4$. Source 104's coefficient field $\mathbb{Q}[z, (d_-/d_+)^{1/3}]$ contains B ; Part I's statement $C_{d,r} \in \mathbb{Q}[B]$ (Section 9) is sharper, and (87) is visibly in $\mathbb{Q}[B]$.

Corollary 22.3 (Maximal and total network equivalents; source 104). *For every fixed $d \geq 2$,*

$$T_{n,n-1}^{(d)} \sim \frac{\gamma_d}{\Gamma} (n!)^d \Gamma^n e^{Bn^{1/3}} n^\alpha. \quad (88)$$

Its full Poincaré expansion follows by substituting $n - 1$ in (86) and using (84). Its first correction is again $C_{d,1}n^{-1/3}$. Moreover,

$$T_n^{(d)} \sim K_d(n!)^d \Gamma^n e^{Bn^{1/3}} n^\alpha, \quad K_d = \begin{cases} \sqrt{e} \gamma_2/12, & d = 2, \\ \gamma_d/\Gamma, & d \geq 3. \end{cases} \quad (89)$$

For $d = 3$ specifically, with $B = 3z/2^{2/3}$,

$$T_n^{(3)} = \frac{\gamma_3}{32} (n!)^3 32^n e^{Bn^{1/3}} n^{-3} \left(1 + \frac{z^2}{36 \cdot 2^{1/3}} n^{-1/3} + O(n^{-1/2}) \right), \quad (90)$$

where $z^2/(36 \cdot 2^{1/3}) = B^2/162$.

Proof. The factorial shift is $(n - 1)! = n!/n$, the exponential-base shift is $\Gamma^{n-1} = \Gamma^n/\Gamma$, and the stretched-exponential shift first contributes at order $n^{-2/3}$. The polynomial shift first contributes at order n^{-1} . These prove the maximal claims. For totals, [3, Corollary 1.11] proves $T_n^{(2)}/T_{n,n-1}^{(2)} \rightarrow \sqrt{e}$ and $T_n^{(d)}/T_{n,n-1}^{(d)} \rightarrow 1$ for $d \geq 3$. The quantitative ternary assertion is proved in Section 28.2. $\square$

The prior result [3, Theorem 1.8] already gives the full two-sided Θ scale in (88), both for maximal and total counts. The positive limiting amplitude and the all-order maximal/word expansion are the additional assertions proved here. *[Write phase: “additional” relative to the published literature; in the repository the same assertions are Theorem 2.1 of Part I, proved independently from source 16 and printed first (Section 21.1).]* The published ratio theorem alone would not give an all-order total expansion. For $d \geq 4$ only the leading total equivalent is asserted in this Part.

23 An exact normalization and a positive Jacobi product

23.1 Rotating the recurrence without changing its startup

Put $\tilde{b}(p, q) = b_{p+1, q+1}$ on $0 \leq q \leq p$, so that $\tilde{b}(0, 0) = 1$. Subtract the successive prefix equations in (6). With forbidden inputs zero, the result away from the prescribed initial point is

$$\begin{aligned} \tilde{b}(p, q) &= \mathcal{W}(p, q) \tilde{b}(p - 1, q) + \mathcal{V}(p, q) \tilde{b}(p, q - 1), \\ \mathcal{W}(p, q) &= \frac{1}{d_-!} \prod_{i=1}^{d_-} (dp + q + i), & \mathcal{V}(p, q) &= \frac{dp + q + d_-}{dp + q}. \end{aligned} \quad (91)$$

No expression with the zero denominator at $(0, 0)$ is evaluated. The endpoint identity is

$$a_n = \frac{\tilde{b}(n, n)}{\mathcal{W}(n, n)}, \quad n \geq 0, \quad (92)$$

with the endpoint convention $\mathcal{W}(0, 0) = 1$. For $n \geq 1$, this follows from $b_{n+1, n+1} = \mathcal{W}(n, n) a_n$.

Use the exact diagonal product

$$G_\Gamma(p) = \prod_{l=1}^p \mathcal{W}(l, l) = \Lambda^p \prod_{i=1}^{d_-} (1 + i/d_+)_p, \quad G_\Gamma(0) = 1, \quad (93)$$

where $(x)_p = \Gamma(x+p)/\Gamma(x)$. On the parity-compatible domain $0 \leq h \leq N$, define

$$\mathcal{Y}(N, h) = \frac{\tilde{b}((N+h)/2, (N-h)/2)}{G_\Gamma((N+h)/2)}.$$

Then $\mathcal{Y}(0, 0) = 1$ and

$$\begin{aligned} \mathcal{Y}(N, h) &= \omega_N^-(h) \mathcal{Y}(N-1, h-1) + \omega_N^+(h) \mathcal{Y}(N-1, h+1), \\ \omega_N^-(h) &= \prod_{i=1}^{d_-} \frac{d_+ N + d_- h + 2i}{d_+(N+h) + 2i}, \quad \omega_N^+(h) = \frac{d_+ N + d_- h + 2d_-}{d_+ N + d_- h}. \end{aligned} \quad (94)$$

For instance $\mathcal{Y}(1, 1) = \mathcal{W}(1, 0)/\mathcal{W}(1, 1)$ and $\mathcal{Y}(2, 0) = 1$, exactly.

23.2 The two step matrix and its Gram factor

Let $e_n(k) = \mathcal{Y}(2n, 2k)$, $0 \leq k \leq n$, and $e_0 = \delta_0$. For fixed n, k write $\mathfrak{z} = d_+ n + d_- k$, $\mathfrak{P} = d_+(n+k)$, and define

$$\begin{aligned} A_n(k) &= \prod_{i=1}^{d_-} \frac{\mathfrak{z} + i}{\mathfrak{P} + i}, & B_n(k) &= \frac{\mathfrak{z} + d_-}{\mathfrak{z}}, \\ C_n(k) &= \prod_{i=1}^{d_-} \frac{\mathfrak{z} + i - 1}{\mathfrak{P} + i} = \frac{A_n(k)}{B_n(k)}, & F_n(k) &= \frac{\mathfrak{z} + d_- - 1}{\mathfrak{z} - 1}. \end{aligned} \quad (95)$$

Eliminating odd times in (94) gives $e_n = M_n e_{n-1}$ after zero extension, where the lower, upper and diagonal entries are

$$\begin{aligned} L_n(k) &= A_n(k) C_n(k-1), \\ U_n(k) &= B_n(k) F_n(k), \\ D_n(0) &= B_n(0) C_n(0) = A_n(0) = 1, \\ D_n(k) &= A_n(k) F_n(k-1) + B_n(k) C_n(k) = A_n(k) \left(1 + \frac{\mathfrak{z} - 1}{\mathfrak{z} - d}\right), \quad k \geq 1. \end{aligned} \quad (96)$$

Use this full bulk diagonal also at the new coordinate $k = n$. It multiplies $e_{n-1}(n) = 0$, so it leaves the exact evolution unchanged. It also avoids an artificial order-one jump at the old terminal coordinate when comparing consecutive matrices.

Set $\Sigma_n(0) = 1$ and

$$\left(\frac{\Sigma_n(k)}{\Sigma_n(k-1)} \right)^2 = \frac{L_n(k)}{U_n(k-1)}.$$

Here Σ_n also denotes the diagonal matrix with these entries. The symmetric matrix $J_n = \Sigma_n^{-1} M_n \Sigma_n$ has diagonal D_n and positive off-diagonal entries

$$b_n(k)^2 = L_n(k) U_n(k-1) = A_n(k) A_n(k-1) \frac{\mathfrak{z} - 1}{\mathfrak{z} - d}. \quad (97)$$

It is positive definite: $J_n = K_n K_n^\top$, with the square lower bidiagonal factor

$$K_n(k, k)^2 = A_n(k), \quad K_n(k, k-1)^2 = A_n(k) \frac{\mathfrak{z} - 1}{\mathfrak{z} - d} \quad (k \geq 1). \quad (98)$$

Every displayed nonzero factor is positive. Let λ_n be the largest eigenvalue of J_n , and ψ_n its positive unit eigenvector, extended by zero in $\ell^2(\mathbb{N}_0)$. Put $\psi_0 = \delta_0$.

23.3 An exact contraction at every fixed degree

Write $\varsigma_n(j) = \Sigma_n(j)/\Sigma_n(j-1)$. With $\mathfrak{z} = d_+n + d_-j$, its square factors as

$$\varsigma_n(j)^2 = \prod_{i=1}^{d_-} \frac{\mathfrak{z} + i}{d_+(n+j) + i} \prod_{i=1}^{d_-} \frac{\mathfrak{z} - d + i}{d_+(n+j-1) + i} \cdot \frac{\mathfrak{z} - d_-}{\mathfrak{z}} \frac{\mathfrak{z} - d}{\mathfrak{z} - 1}. \quad (99)$$

For $n \geq j \geq 1$, every factor is in $(0, 1]$. Denominator minus numerator equals, respectively, $2j$, $2j-1$, d_- , and d_- . Each numerator and denominator has the same positive slope d_+ as a function of n , so every factor increases with n . Thus $\Sigma_n(k) \leq 1$, and the zero-extended diagonal operator

$$\mathbf{R}_n(k) = \begin{cases} \Sigma_{n-1}(k)/\Sigma_n(k), & k < n, \\ 0, & k \geq n \end{cases} \quad \text{satisfies} \quad 0 \leq \mathbf{R}_n \leq I. \quad (100)$$

In particular $\mathbf{R}_n(0) = 1$. On $n-1 \leq t \leq n$ and $1 \leq j \leq n-1$, all factors in (99) are bounded below by a positive constant depending only on d , and differentiation gives

$$0 \leq \partial_t \log \varsigma_t(j) \leq C_d(j+1)/n^2.$$

Integrate in t and sum over $j \leq k$. Enlarging the constant to include the new coordinate gives

$$0 \leq 1 - \mathbf{R}_n(k) \leq C_d k(k+1)/n^2. \quad (101)$$

This exact normalization is important at large fixed degrees. If $G_\Gamma(p)$ were replaced by $\Lambda^p(p!)^{d_-}$, its symmetrizer edge would instead satisfy, for fixed j ,

$$\log(\varsigma_n(j)^2) = \frac{d_-(d_- - 4j)}{d_+n} + O(n^{-2}).$$

When $d_- > 4$ and $j = 1$, it decreases as n increases, and the corresponding $\mathbf{R}_n(1)$ exceeds one for all sufficiently large n . Thus that simpler normalization does not prove contraction for $d \geq 6$. Formula (99) resolves the issue by an exact factorwise argument.

Remark 23.1 (The obstruction does not touch Part I; write phase). The obstruction concerns source 104's two-step frame with a factorial gauge. Part I uses a different, one-step frame: its gauge ratio $Q_N = G_{N-1}/G_N$ is built from the weights $p_N(j)$ of (10), each of whose $d-1$ factors has the positive N -derivative displayed after (11), for every d . So Q_N is a contraction at every fixed degree, and Part I is unaffected. Part IV's factorial gauge is the case $d = 2$, where $d_- = 1 < 4$ and no obstruction arises (Lemma 35.1).

Finally let

$$P_n = \prod_{j=1}^n \lambda_j, \quad u_n = P_n^{-1} \Sigma_n^{-1} e_n, \quad \mathbf{T}_n = (J_n/\lambda_n) \mathbf{R}_n. \quad (102)$$

Then $u_n = \mathbf{T}_n u_{n-1}$ and $u_0 = \delta_0$. Each $\mathbf{T}_n$ is entrywise nonnegative and an ℓ^2 contraction. The Gram factor proves positivity of the whole spectrum of J_n/λ_n , not just of its Perron eigenvalue.

24 The Airy limit and quantitative spectral estimates

24.1 Global estimates and form compactness

Put $\varepsilon = n^{-1/3}$, $x_k = \varepsilon(k + 1/2)$, and

$$c_A = (4c)^{1/3} = (8d_-/d_+)^{1/3}, \quad f_A(x) = \frac{\sqrt{c_A} \operatorname{Ai}(z + c_A x)}{\operatorname{Ai}'(z)}. \quad (103)$$

Then $\int_0^\infty f_A^2 dx = 1$, $f_A'(0) = \sqrt{4c}$, and f_A is the ground state of the Dirichlet operator

$$\mathcal{H}_A = -\frac{d^2}{dx^2} + 4cx, \quad \mathcal{H}_A f_A = -c_A^2 z f_A.$$

Let J_∞ have diagonal $1, 2, 2, \dots$ and unit off-diagonals. The exact finite-product coefficients satisfy

$$\begin{aligned} 0 &< c_d \leq \mathbf{b}_n(k) \leq C_d, \\ 0 &\leq J_n \leq (1 + C_d/n) J_\infty \quad \text{entrywise}, \\ 4 - \mathbf{D}_n(k) - \mathbf{b}_n(k) - \mathbf{b}_n(k+1) &\geq c_d k/n - C_d/n \quad (k \geq 1), \\ 4 - \mathbf{D}_n(0) - \mathbf{b}_n(1) &= 2 + O(n^{-1}). \end{aligned} \quad (104)$$

Only existing edges enter these statements; a missing terminal edge is zero. To check them uniformly, put $\vartheta = (d_+ n + d_- k)/(d_+(n+k))$. Then $d/d_+ \leq \vartheta \leq 1$ and

$$\mathbf{A}_n(k) = \vartheta^{d_-} + O(n^{-1}), \quad \mathbf{D}_n(k) = 2\vartheta^{d_-} + O(n^{-1}), \quad \mathbf{b}_n(k), \mathbf{b}_n(k+1) = \vartheta^{d_-} + O(n^{-1})$$

in the bulk. Extract ϑ from every factor; the remaining denominators are bounded below by a constant times n . Moreover

$$1 - \vartheta^{d_-} \geq 1 - \vartheta = \frac{2k}{d_+(n+k)} \geq \frac{k}{d_+ n}.$$

The omitted terminal edge only increases the potential. Uniform positive bounds for the factors prove the edge bounds; $\mathbf{A}_n \leq 1$ and $(\mathfrak{z} - 1)/(\mathfrak{z} - d) = 1 + O(n^{-1})$ prove the entrywise bound. Adjacent bulk coefficients differ by $O(n^{-1})$, also when the explicit expression for $\mathbf{b}_n$ is continued to the fictitious edge $n+1$.

On the common time domain, direct differentiation gives

$$|\mathbf{D}_n(k) - \mathbf{D}_{n-1}(k)| + |\mathbf{b}_n(k) - \mathbf{b}_{n-1}(k)| \leq C_d(k+1)/n^2. \quad (105)$$

Here $\mathbf{D}_n(0) = 1$ is exact, and the new edge is excluded.

The quadratic form of $4I - J_n$ is the weighted edge-gradient sum plus the potential in (104). Its $n^{2/3}$ -scaled low-energy limit is $\mathcal{H}_A$. We give the compactness details because the boundary and increasing dimension matter. For a unit vector of energy at most $C\varepsilon^2$, the form bound implies

$$\sum_k k |u_k|^2 \leq C/\varepsilon, \quad |u_0|^2 \leq C\varepsilon^2.$$

Interpolate $u_k/\sqrt{\varepsilon}$ linearly at x_k , start at zero at $x = 0$, and add a one-mesh ramp to zero after x_n . The first half-cell derivative energy is $2|u_0|^2/\varepsilon^2$, provided by the boundary potential tending to 2. The other local derivative energies are bounded by the edge form. The first moment bounds

the squared L^2 mass beyond R by C/R . Local Rellich compactness together with this tail bound gives strong L^2 compactness, preserving norms and orthogonality. The negative potential correction is only $O(n^{-1})$ and disappears after scaling. Local coefficient convergence gives the gradient and potential lower limits. Sampled compactly supported smooth Dirichlet functions give matching upper bounds. *[Write phase: this is the compactness argument of Remark 5.1 on the half-grid $x_k = \varepsilon(k + \frac{1}{2})$, where the boundary half-cell replaces Part I's fictitious first interval; Part IV prints it at $d = 2$ with a little more detail.]* The min-max principle therefore yields, for each fixed j ,

$$\frac{4 - \lambda_n^{(j)}}{\varepsilon^2} \longrightarrow -c_A^2 z_j. \quad (106)$$

In particular, positive definiteness and separation of the first two Airy levels give

$$0 \leq \lambda_n^{(2)}/\lambda_n \leq 1 - c_d n^{-2/3}. \quad (107)$$

[Write phase: here the Gram factor excludes the negative half of the spectrum, so a full spectral gap holds, whereas Part I's one-step matrix S_N has a parity-symmetric spectrum and needs the parity singular gap of Section 5.]

24.2 The scalar term at order n inverse

On the half-grid, the exact coefficients have expansions

$$\begin{aligned} D_n(k) &= 2 - 2cx\varepsilon^2 + \frac{3}{2}c\varepsilon^3 + O(\varepsilon^4(1+x^2)), \\ b_n(k) &= 1 - cx\varepsilon^2 + \frac{5}{4}c\varepsilon^3 + O(\varepsilon^4(1+x^2)), \\ b_n(k+1) &= 1 - cx\varepsilon^2 + \frac{1}{4}c\varepsilon^3 + O(\varepsilon^4(1+x^2)). \end{aligned} \quad (108)$$

For example, first expand $A_n(k) = 1 - ck/n + O((k^2 + k)/n^2)$ and $(j-1)/(j-d) = 1 + c/(2n) + O((k+1)/n^2)$, then use (97) and (96). *[Write phase: source 104 writes these coefficients as $4r/a$, $3r/a$, $2r/a$, $5r/(2a)$, $r/(2a)$ and $r/(an)$, which are $2c$, $\frac{3}{2}c$, c , $\frac{5}{4}c$, $\frac{1}{4}c$ and $c/(2n)$.]* The expansions are uniform for $k \leq n/2$. Pairing $f_A(x+\varepsilon)$ and $f_A(x-\varepsilon)$ and using the adjacent-coefficient difference bound for odd derivative terms gives

$$J_n f_A = 4f_A + \varepsilon^2 (f_A'' - 4cx f_A) + 3c\varepsilon^3 f_A + O_{\ell^2}(\varepsilon^4),$$

where the samples include their factor $\sqrt{\varepsilon}$. More explicitly, before that sampling factor the pointwise error is bounded by $C\varepsilon^4$ times a fixed polynomial-Airy derivative envelope; its squared lattice sum times ε is bounded. The part $k > n/2$ is exponentially small.

At the first row, $D_n(0) = 1$ and $b_n(1) = 1 - c\varepsilon^3/4 + O(\varepsilon^6)$. The unsampled boundary residual is $O(\varepsilon^4)$: the cubic term in $f_A(3\varepsilon/2) - 3f_A(\varepsilon/2)$ cancels the eigenvalue term by

$$f_A(0) = f_A''(0) = 0, \quad f_A'''(0) = c_A^2 z f_A'(0).$$

Its sampled size is $O(\varepsilon^{9/2})$. The chosen upper terminal diagonal contributes only an exponentially small error.

Let ψ_n^A be the normalized sample $\sqrt{\varepsilon} f_A(x_k)$. Midpoint quadrature gives pre-normalization squared norm $1 + O(\varepsilon^2)$. The residual estimate and isolation from (106) identify the nearby eigenvalue as the largest one, and the gap gives

$$\lambda_n = 4 + c_A^2 z n^{-2/3} + 3c n^{-1} + O(n^{-4/3}), \quad (109)$$

$$\|\psi_n - \psi_n^A\|_2 = O(\varepsilon^2). \quad (110)$$

[Write phase: $c_A^2 z = 4^{2/3} \ell$. This agrees with two steps of Part I's (19): $(2 + \ell N^{-2/3})^2 = 4 + 4^{2/3} \ell n^{-2/3} + O(n^{-1})$ at $N = 2n$; the n^{-1} terms differ because the gauges differ.]

24.3 Localization and changes in dimension

Entrywise domination and positivity of ψ_n give

$$\langle \psi_n, (4I - J_\infty) \psi_n \rangle \leq 4 - \frac{\lambda_n}{1 + C_d/n} \leq C\varepsilon^2.$$

The left side is $\sum_k |\psi_n(k+1) - \psi_n(k)|^2 + 2\psi_n(0)^2$. Discrete Sobolev therefore yields $\|\psi_n\|_\infty \leq C\sqrt{\varepsilon}$. Choose $k_A = \lceil A_0/\varepsilon \rceil$ with A_0 large, and a fixed small $c_0 > 0$. The test function $w(k) = \exp[-c_0\varepsilon(k - k_A)]$ is a strict supersolution $J_n w < \lambda_n w$ for $k > k_A$. Indeed, the full bulk row sum is at most $4 - c_d k/n + C_d/n$. Continue the upper edge first to its fictitious value at $n+1$. The paired exponential shifts change that sum by $O(\varepsilon^2)$; the first-order part uses the adjacent-coefficient difference $O(n^{-1})$. Comparing with (109), choose A_0 large enough, then omit the positive fictitious contribution at $k = n$. The finite maximum principle gives

$$\psi_n(k) \leq C\sqrt{\varepsilon} e^{-c_0\varepsilon(k-k_A)} \quad (k \geq k_A), \quad \|k^p \psi_n\|_2 \leq C_p n^{p/3} \quad (p \geq 0 \text{ fixed}). \quad (111)$$

In particular the old terminal component is exponentially small in $n^{2/3}$. Combining (105) and (111), with the new edge estimated separately, gives

$$\|(J_n - J_{n-1})\psi_{n-1}\|_2 \leq Cn^{-5/3}.$$

Projecting this residual and using the gap proves

$$\|\psi_n - \psi_{n-1}\|_2 \leq C/n, \quad (112)$$

$$\|(I - R_n)\psi_n\|_2 + \|(I - R_n)\psi_{n-1}\|_2 \leq Cn^{-4/3}. \quad (113)$$

The latter estimate is also an immediate consequence of (101) and the second moment in (111).

24.4 Endpoint smoothing

Let $\mathcal{K} = J_\infty/4$. The half-grid sine transform gives

$$\mathcal{K}^m(0, 0) = \frac{2}{\pi} \int_0^\pi \cos^{2m}(u/2) \sin^2(u/2) du \leq C(m+1)^{-3/2}. \quad (114)$$

Hence $\|\delta_0^\top \mathcal{K}^l\|_2 \leq C(l+1)^{-3/4}$. Positivity, $R_j \leq I$, (104), and (109) imply, for $0 \leq l \leq n^{2/3}$,

$$\begin{aligned} 0 &\leq \mathsf{T}_n \cdots \mathsf{T}_{n-l+1} \leq C\mathcal{K}^l, \\ \|\delta_0^\top \mathsf{T}_n \cdots \mathsf{T}_{n-l+1}\|_2 &\leq C(l+1)^{-3/4}. \end{aligned} \quad (115)$$

The accumulated comparison scalar is bounded because its logarithm is $O(\ln n^{-2/3})$. This is an entrywise product comparison and requires no commutativity. The same bound holds for powers of a fixed J_n/λ_n . [Write phase: this is Part I's Catalan bound (26) for the killed kernel $J_\infty/4$ in place of half-line adjacency.]

For $\mathbf{z} = \psi_n - \psi_n^A$, the quasimode estimates give $\mathbf{z} = (J_n/\lambda_n)\mathbf{z} + \varrho$ with $\|\varrho\|_2 = O(\varepsilon^4)$. Iterate $\Delta = \lfloor n^{2/3} \rfloor$ times and apply (115):

$$|\mathbf{z}(0)| \leq C\Delta^{-3/4}\varepsilon^2 + C\varepsilon^4 \sum_{l < \Delta} (l+1)^{-3/4} = O(\varepsilon^{7/2}).$$

Since $\psi_n^A(0) = (f'_A(0)/2)\varepsilon^{3/2}(1 + O(\varepsilon^2))$, this proves the precise endpoint scale

$$\psi_n(0) = \sqrt{c}n^{-1/2} \left(1 + O(n^{-2/3})\right). \quad (116)$$

25 A strictly positive limiting amplitude

25.1 Convergence of the normalized product

Write

$$p_n = \langle \psi_n, u_n \rangle, \quad p_n^\perp = \left\| (I - \psi_n \psi_n^\top) u_n \right\|_2.$$

Contractivity gives $\|u_n\|_2 \leq 1$. The gap, drift, and symmetrizer defect give

$$p_n^\perp \leq (1 - cn^{-2/3})p_{n-1}^\perp + C/n.$$

A power supersolution yields $p_n^\perp = O(n^{-1/3})$. The exact scalar increment is

$$\begin{aligned} p_n - p_{n-1} &= p_{n-1} (\langle \psi_n, \mathbf{R}_n \psi_{n-1} \rangle - 1) \\ &\quad + \langle \mathbf{R}_n \psi_n - \psi_{n-1}, (I - \psi_{n-1} \psi_{n-1}^\top) u_{n-1} \rangle. \end{aligned} \quad (117)$$

The first bracket is $O(n^{-4/3})$: without $\mathbf{R}_n$ it is minus half the squared Perron-vector drift. The second term is $O(n^{-1}p_{n-1}^\perp) = O(n^{-4/3})$. Thus there exists $p_\infty \geq 0$ such that

$$|p_n - p_\infty| + \|u_n - p_\infty \psi_n\|_2 \leq Cn^{-1/3}. \quad (118)$$

Convergence alone does not prove strict positivity. Restart the same exact normalized product at a large N from the vector ψ_N . The preceding estimates hold with constants independent of N above a fixed threshold; its initial perpendicular component is zero and its norm is one. Summing its scalar increments shows that its limiting Perron mass differs from one by at most $CN^{-1/3}$. Choose N so large that this mass exceeds $1/2$. Every coordinate of the actual u_N on $0, \dots, N$ is strictly positive, by the original recurrence. Thus $u_N \geq \theta_0 \psi_N$ for some $\theta_0 > 0$. Positivity of subsequent operators implies $p_\infty \geq \theta_0/2 > 0$.

Set $\mathbf{z}_n = u_n - p_\infty \psi_n$. Its exact evolution has forcing $p_\infty(\mathbf{T}_n \psi_{n-1} - \psi_n)$ of norm $O(n^{-1})$. Iterate through the last $\Delta = \lfloor n^{2/3} \rfloor$ steps, use (118), then (115). The result is

$$|\mathbf{z}_n(0)| \leq C\Delta^{-3/4}n^{-1/3} + \frac{C}{n} \sum_{l < \Delta} (l+1)^{-3/4} = O(n^{-5/6}).$$

Consequently

$$u_n(0) = p_\infty \sqrt{c} n^{-1/2} \left(1 + O(n^{-1/3})\right). \quad (119)$$

[Write phase: this is the restart comparison of source 104, not the contradiction with the lower Θ bound used in Part I (Section 7) and Part II; source 104's positivity proof therefore needs no external enumeration-asymptotic input.]

25.2 The exact gamma normalization

Equation (109) gives

$$\log(\lambda_n/4) = \frac{B}{3}n^{-2/3} + \zeta_f n^{-1} + O(n^{-4/3}).$$

It follows that

$$C_\lambda = \lim_{n \rightarrow \infty} \frac{P_n}{4^n e^{Bn^{1/3}} n^{\zeta_f}} \in (0, \infty), \quad (120)$$

with relative error $O(n^{-1/3})$. A positive convergent representation is

$$C_\lambda = \frac{\lambda_1}{4} e^{-B} \prod_{j=2}^{\infty} \left[\frac{\lambda_j}{4} e^{-B(j^{1/3} - (j-1)^{1/3})} \left(\frac{j-1}{j} \right)^{\zeta_f} \right]. \quad (121)$$

Each factor is $1 + O(j^{-4/3})$. Also, from $\langle \psi_n, \mathsf{T}_n u_{n-1} \rangle = \langle \mathsf{R}_n \psi_n, u_{n-1} \rangle$,

$$p_\infty = 1 + \sum_{n=1}^{\infty} \langle \mathsf{R}_n \psi_n - \psi_{n-1}, u_{n-1} \rangle, \quad (122)$$

and this scalar series is absolutely convergent by (117).

The exact product (93) has the asymptotics

$$\begin{aligned} G_\Gamma(n) &= \Lambda^n (n!)^{d_-} n^{\delta_f} Q_\Gamma^{-1} (1 + O(n^{-1})), \\ \mathcal{W}(n, n) &= \Lambda n^{d_-} (1 + O(n^{-1})), \\ a_n &= \frac{G_\Gamma(n)}{\mathcal{W}(n, n)} e_n(0), \quad e_n(0) = P_n u_n(0). \end{aligned} \quad (123)$$

The equality $\delta_f - d_- + \zeta_f - 1/2 = \rho$ now proves the leading instance of Theorem 22.1, with

$$\boxed{\gamma_d = \frac{\sqrt{c} p_\infty C_\lambda}{\Lambda Q_\Gamma} > 0.} \quad (124)$$

This fixes the amplitude for the exact original startup, including $a_0 = a_1 = 1$. No closed form or certified numerical enclosure of γ_d is asserted. The existence of a convergent defining product should not be confused with convergence of the all-order correction series proved next. *[Write phase: (124) and Part I's (31) are two convergent representations of one number, through different gauges; their factors p_∞ , C_λ and A_∞ , κ_d are not individually comparable.]*

26 The all order coefficient algorithm

26.1 An exact scaled one step equation

Use a different Airy scaling for the one-step equation:

$$t = N^{-1/3}, \quad x = (h+1)t, \quad F(x) = \text{Ai}(z + c^{1/3}x), \quad \eta_f = \zeta_f - \frac{1}{6}. \quad (125)$$

Then $F'' = (cx + \ell)F$, $F(0) = 0$, and $F'(0) = c^{1/3} \text{Ai}'(z) \neq 0$. The exact scaled weights of (94) are

$$\begin{aligned} U(t, x) &= \prod_{i=1}^{d_-} \frac{d_+ + d_-xt^2 + (2i - d_-)t^3}{d_+ + d_+xt^2 + (2i - d_+)t^3}, \\ V(t, x) &= \frac{d_+ + d_-xt^2 + d_-t^3}{d_+ + d_-xt^2 - d_-t^3}, \quad t_1 = t(1 - t^3)^{-1/3}. \end{aligned} \quad (126)$$

Seek formal series

$$\begin{aligned} s(t) &= 2 + \ell t^2 + \sum_{j \geq 3} \sigma_j t^j, \\ \mathfrak{F}(t, x) &= \sum_{j \geq 0} t^j f_j(x), \quad f_j = P_j F + Q_j F', \\ P_0 &= 1, \quad Q_0 = 0, \quad Q_j(0) = 0, \quad P_j(0) + Q_j'(0) = 0 \quad (j \geq 1), \end{aligned} \quad (127)$$

satisfying

$$\begin{aligned} s(t)\mathfrak{F}(t, x) &= U(t, x)\mathfrak{F}(t_1, (x - t)(1 - t^3)^{-1/3}) \\ &\quad + V(t, x)\mathfrak{F}(t_1, (x + t)(1 - t^3)^{-1/3}). \end{aligned} \quad (128)$$

For fixed d , the profiles P_j, Q_j and scalars σ_j have rational polynomial coefficients in ℓ, x or ℓ . The first nontrivial equation is the Airy equation at order t^2 .

Suppose earlier orders are solved. At order t^k , write the known residual as $R_0 F + R_1 F'$. The new profile has index $k - 2$, and it and σ_k must solve

$$\begin{aligned} P'' + 2(cx + \ell)Q' + cQ - \sigma_k + R_0 &= 0, \\ 2P' + Q'' + R_1 &= 0. \end{aligned} \quad (129)$$

Define the polynomial operator

$$\mathcal{T} = D^3 - 4(cx + \ell)D - 2c.$$

On polynomials of degree at most m , this operator is triangular with diagonal $-2c(2j+1)$, $0 \leq j \leq m$. None is zero. Thus the following is an exact finite algorithm:

$$\begin{aligned} \tilde{q} &= \mathcal{T}^{-1}(2R_0 - R_1'), & \sigma_k &= -c\tilde{q}(0), \\ Q &= \tilde{q} + \sigma_k/c, & P(x) &= \int_0^x \frac{-R_1(y) - Q''(y)}{2} dy - Q'(0). \end{aligned} \quad (130)$$

To verify it, eliminate P'' from (129); this gives $\mathcal{T}Q = 2R_0 - R_1' - 2\sigma_k$. Substituting (130) proves both equations and both gauges. Conversely the nonzero triangular diagonal and the gauges give uniqueness. This proves existence and compatibility to every formal depth. It also shows

$$f_j(0) = 0 \quad (j \geq 0), \quad f_j'(0) = 0 \quad (j \geq 1).$$

[Write phase: $\mathcal{T}$ is -2 times the left side of Part I's triangular equation (18); the polynomial inverse is the same.]

For the first correction,

$$R_0 = 2c, \quad R_1 = (c + 2/3)x, \quad \sigma_3 = 3c/2 - 1/3 = 2\eta_f,$$

so $Q_1 = 0$ and $P_1 = -(3c + 2)x^2/12$. The next two scalars are

$$\sigma_4 = \frac{\ell^2(189c^2 - 64)}{540c^2}, \quad (131)$$

$$\sigma_5 = \frac{\ell(351c^3 - 90c^2 + 64)}{324c^2}. \quad (132)$$

They are exact symbolic consequences of the recurrence, not numerical fits.

26.2 Compatible scalar carriers

Define h_j by the formal difference identity

$$\begin{aligned} \log \frac{s(t)}{2} &= \frac{3\ell}{2t} [1 - (1 - t^3)^{1/3}] - \eta_f \log(1 - t^3) \\ &+ \sum_{j \geq 1} h_j t^j [1 - (1 - t^3)^{-j/3}]. \end{aligned} \quad (133)$$

The coefficients at t^2, t^3 are $\ell/2, \eta_f$. At order t^{j+3} , the new unknown h_j has nonzero coefficient $-j/3$. Thus all h_j are uniquely determined, with no arbitrary additive constants. In particular,

$$h_1 = \frac{3\ell^2}{8} - \frac{3\sigma_4}{2} = \frac{\ell^2(32 - 27c^2)}{180c^2}. \quad (134)$$

[Write phase: the next carrier coefficient, used in Section 28 but not printed by source 104, is $h_2 = -\ell(27c^3 - 36c^2 + 16)/(108c^2)$, from (133) and (132) (computed symbolically in writing). At $d = 2$, where $c = 2/3$, $\sigma_4 = \ell^2/12$, $\sigma_5 = 8\ell/9$, $h_1 = \ell^2/4$ and $h_2 = -\ell/6$.] For any fixed finite depth M , put

$$\begin{aligned} \mathcal{H}_N^{[M]} &= 2^N e^{(3\ell/2)N^{1/3}} N^{\eta_f} \exp \left(\sum_{j=1}^M h_j N^{-j/3} \right), \\ \mathfrak{F}^{[M]}(t, x) &= \sum_{j=0}^M t^j f_j(x). \end{aligned} \quad (135)$$

Every carrier has leading multiplicative constant exactly one. Its exact consecutive ratio agrees with the formal s -series through at least order t^{M+3} ; when needed the h_j are computed using profiles beyond the retained profile depth.

At even times, write $N = 2n$, $h = 2k$, $\varepsilon = n^{-1/3}$, and $X = \varepsilon(k + 1/2)$. Then $t = 2^{-1/3}\varepsilon$ and $x = 2^{2/3}X$. Define

$$\begin{aligned} E_n^{[M]}(k) &= K_E \mathcal{H}_{2n}^{[M]} \mathfrak{F}^{[M]}(2^{-1/3}\varepsilon, 2^{2/3}X), \\ K_E &= \frac{\Lambda Q_\Gamma 2^{1/3-\eta_f}}{F'(0)}. \end{aligned} \quad (136)$$

At $k = 0$ the argument is $x = t$, and

$$\mathfrak{F}^{[M]}(t, t) = F'(0)t(1 + O(t^2)).$$

The derivative gauges ensure that higher profiles do not alter this leading endpoint normalization. Consequently every depth shares

$$E_n^{[M]}(0) = \Lambda Q_\Gamma 4^n e^{Bn^{1/3}} n^{\zeta_f - 1/2} (1 + O(\varepsilon)). \quad (137)$$

The exact conversion back to the counted sequence is crucial. Define

$$Z_n^\Gamma = \frac{Q_\Gamma G_\Gamma(n)}{\Lambda^n (n!)^{d_-} n^{\delta_f}} \cdot \frac{\Lambda n^{d_-}}{\mathcal{W}(n, n)}. \quad (138)$$

It tends to one and has a full expansion in integer powers of n^{-1} , from gamma-ratio asymptotics and the finite polynomial $\mathcal{W}(n, n)$. Its first term is

$$Z_n^\Gamma = 1 + \left(\frac{d_- d(2d_- + 1)}{12d_+^2} - \frac{d_- d}{4d_+} \right) n^{-1} + O(n^{-2}). \quad (139)$$

The canonical multiplicative coefficient series is the expansion of

$$\frac{Z_n^\Gamma E_n^{[M]}(0)}{\Lambda Q_\Gamma 4^n e^{Bn^{1/3}} n^{\zeta_f - 1/2}}. \quad (140)$$

The gamma constant Q_Γ and $\text{Ai}'(z)$ cancel here. The coefficients stabilize as the retained depth increases.

In particular, Z_n^Γ has no $n^{-1/3}$ term and the endpoint profile has no term of order t . Hence $C_{d,1} = 2^{-1/3} h_1$, which simplifies to (87). If $\omega^3 = 1$ and the Airy-equation solution is normalized by $\mathfrak{B}_\ell(0) = 0$, $\mathfrak{B}'_\ell(0) = 1$, then

$$\mathfrak{B}_{\omega\ell}(\omega x) = \omega \mathfrak{B}_\ell(x).$$

The weights U, V are invariant under simultaneous $(t, x, \ell) \mapsto (\omega t, \omega x, \omega \ell)$. Formal uniqueness preserves this grading. Thus a scalar monomial $\ell^m t^j$ in the normalized endpoint has $m + j$ divisible by three. On substituting $\ell = (2d_-/d_+)^{2/3} z$, $t = 2^{-1/3} \varepsilon$, the power $(2m - j)/3$ of 2 is an integer. Together with the rational expansion of Z_n^Γ , this proves the coefficient field stated in Theorem 22.1.

27 Transferring arbitrary formal depth to the exact counts

27.1 Residual estimates and the boundary

For each fixed M and C_0 , the exact one-step residual of $\mathcal{H}_N^{[M]} \mathfrak{F}^{[M]}(N^{-1/3}, (h+1)N^{-1/3})$, divided by its scalar carrier, is bounded by

$$C t^{M+3} (1+x)^D e^{-c_0 x^{3/2}}, \quad 0 \leq h \leq C_0 \sqrt{N}, \quad (141)$$

for constants depending on the fixed depth and degree. Here $c_0 > 0$ is a bound constant. In this region $x = O(t^{-1/2})$, so $xt^2 = O(t^{3/2})$, and every denominator in the exact weights stays separated from zero. A finite rational Taylor expansion has a remainder bounded by the stated power of t times a polynomial in x . The shifted arguments differ from x by

$$\pm t + O(xt^3 + t^4) = O(t).$$

All derivatives of the finite polynomial–Airy profiles obey polynomial times $e^{-c_0 x^{3/2}}$ bounds on those intervals. The exact finite-carrier ratio is analytic near $t = 0$ after its constant 2 is removed. Formal cancellation through degree $M + 2$ gives (141). This argument uses only finitely many terms, not convergence of an infinite profile series.

At the lower boundary $h = 0$, the ghost predecessor $h = -1$ has scaled argument zero. Every finite profile vanishes there exactly, before Taylor expansion. The lower boundary is therefore exact. Eliminating the odd step gives precisely the even boundary diagonal $D_n(0) = 1$; the even residual is a bounded linear combination of one-step residuals.

Choose a smooth fixed cutoff χ , equal to one on $[0, 1/2]$ and zero on $[1, \infty)$, and let

$$\tilde{E}_n^{[M]}(k) = \chi(k/\sqrt{n})E_n^{[M]}(k).$$

In the transition region X is comparable to $n^{1/6}$, so the profiles and adjacent values are polynomial times $e^{-c_0 n^{1/4}}$. Every cutoff error is therefore smaller than any fixed inverse power after removing the common scalar carrier. Include also the extra possible residual edge at $k = \lfloor \sqrt{n} \rfloor + 1$. From (99),

$$-\log \Sigma_n(k) = O((k^2 + k)/n) = O(1), \quad k \leq \sqrt{n} + O(1).$$

Thus Σ_n^{-1} is uniformly bounded on the whole residual support.

Let $\Theta_n = 4^n e^{Bn^{1/3}} n^{\eta_f}$. The squared polynomial–Airy envelope has lattice sum $O(\varepsilon^{-1})$, so

$$\left\| \tilde{E}_n^{[M]} - M_n \tilde{E}_{n-1}^{[M]} \right\|_2 = O(\Theta_n \varepsilon^{M+5/2}).$$

Since $\Theta_n/P_n = O(\varepsilon^{1/2})$, we obtain the normalized defect

$$\left\| P_n^{-1} \Sigma_n^{-1} [\tilde{E}_n^{[M]} - M_n \tilde{E}_{n-1}^{[M]}] \right\|_2 = O(\varepsilon^{M+3}) = O(n^{-(M+3)/3}). \quad (142)$$

27.2 The amplitude is the same at every depth

Set $V_n^{[M]} = P_n^{-1} \Sigma_n^{-1} \tilde{E}_n^{[M]}$. The leading coefficient in (136) and the normalized Airy sample give

$$\left\| V_n^{[M]} - \frac{\Lambda Q_\Gamma}{\sqrt{c} C_\lambda} \psi_n \right\|_2 = O(\varepsilon). \quad (143)$$

For the scalar calculation, $K_E 2^{\eta_f} = \Lambda Q_\Gamma (d_+/d_-)^{1/3} / \text{Ai}'(z)$; divide by the Airy normalization $\sqrt{c_A} / \text{Ai}'(z)$ and include the half-power scalar ratio. The finite carrier correction and weighted profile correction are $O(\varepsilon)$. On the cutoff support, $\Sigma_n^{-1} - 1 = O(\varepsilon(1 + X^2))$ in the polynomial–Airy weighted norm; the cutoff discards an exponentially small tail. Finally (110) has error $O(\varepsilon^2)$. These justify (143) globally.

By the exact amplitude formula,

$$\gamma_d \frac{\Lambda Q_\Gamma}{\sqrt{c} C_\lambda} = p_\infty.$$

Hence the signed error $z_n = u_n - \gamma_d V_n^{[M]}$ has norm $O(n^{-1/3})$, limiting Perron projection zero, and exact evolution

$$z_n = T_n z_{n-1} + \varrho_n, \quad \|\varrho_n\|_2 = O(n^{-p}), \quad p = (M + 3)/3. \quad (144)$$

The profiles need not be positive. Positivity was used to prove the actual limiting amplitude; it is not assumed for the signed error.

27.3 A forced product bootstrap

Lemma 27.1. *Suppose (144) holds with $p > 4/3$, $\|z_n\|_2 = O(n^{-1/3})$, and $\langle \psi_n, z_n \rangle \rightarrow 0$. Then*

$$\|z_n\|_2 = O(n^{-p+1}), \quad z_n(0) = O(n^{-p+1/2}).$$

Proof. Put $a_n^{\text{err}} = \langle \psi_n, z_n \rangle$ and $b_n^{\text{err}} = \|(I - \psi_n \psi_n^\top) z_n\|_2$. The gap, drift, and R-defect estimates give

$$\begin{aligned} b_n^{\text{err}} &\leq (1 - c_{\text{gap}} n^{-2/3}) b_{n-1}^{\text{err}} + C n^{-1} |a_{n-1}^{\text{err}}| + C n^{-p}, \\ |a_n^{\text{err}} - a_{n-1}^{\text{err}}| &\leq C n^{-4/3} |a_{n-1}^{\text{err}}| + C n^{-1} b_{n-1}^{\text{err}} + C n^{-p}. \end{aligned} \quad (145)$$

If $|a_n^{\text{err}}| + b_n^{\text{err}} = O(n^{-q})$, the first inequality improves the transverse bound to

$$b_n^{\text{err}} = O(n^{-q-1/3} + n^{-p+2/3}).$$

This follows by a power supersolution for $x_n \leq (1 - c_{\text{gap}} n^{-2/3}) x_{n-1} + O(n^{-s})$, which gives $x_n = O(n^{-s+2/3})$; any finite startup contribution decays as $e^{-c_2 n^{1/3}}$. Substitute the improved transverse bound in the second inequality and sum from $n+1$ to infinity, using $a_\infty^{\text{err}} = 0$. Then

$$a_n^{\text{err}} = O(n^{-q-1/3} + n^{-p+1}).$$

All increments are absolutely summable: initially $q = 1/3$ and $p > 4/3$. Finitely many iterations improve the exponent by $1/3$ until reaching $p-1$, proving the norm estimate.

For the endpoint, iterate over the last $\Delta = \lfloor n^{2/3} \rfloor$ steps and use (115):

$$|z_n(0)| \leq C \Delta^{-3/4} n^{-p+1} + C n^{-p} \sum_{l < \Delta} (l+1)^{-3/4} = O(n^{-p+1/2}).$$

□

[Write phase: Lemma 27.1 is the zero-limit bootstrap (36) of Part I in source 104's frame; Part IV's Lemma is the same statement at $d = 2$.]

The endpoint scale of $E_n^{[M]}$ is $4^n e^{Bn^{1/3}} n^{\zeta_f - 1/2}$, whereas P_n has power ζ_f . Lemma 27.1 therefore gives relative error $O(n^{-p+1}) = O(\varepsilon^M)$ in $e_n(0)$, and after the exact common factor $G_\Gamma(n)/\mathcal{W}(n, n)$, also in a_n . To obtain the expansion through ε^J , choose $M \geq J+1$ and $M \geq 2$. Expand (140) through degree J ; its next term is $O(\varepsilon^{J+1})$ and the transfer error is at most that size. All coefficients come from compatible formal profiles, carriers and gamma factors, so they are independent of the deeper chosen truncation. This completes the proof of Theorem 22.1.

28 Explicit coefficients and total ternary networks

28.1 The second universal coefficient

Writing the normalized logarithmic series as

$$\log \left(1 + \sum_{j \geq 1} C_{d,j} \varepsilon^j \right) = \sum_{j \geq 1} L_{d,j} \varepsilon^j,$$

one has $L_{d,1} = C_{d,1}$ and

$$L_{d,2} = z \left(\frac{d_-}{d_+} \right)^{2/3} \left(\frac{1}{d_+} - \frac{d_+^2}{27d_-^2} \right) = \frac{B}{3} \left(\frac{1}{d+1} - \frac{(d+1)^2}{27(d-1)^2} \right). \quad (146)$$

Consequently

$$C_{d,2} = \frac{C_{d,1}^2}{2} + \frac{B}{3} \left(\frac{1}{d+1} - \frac{(d+1)^2}{27(d-1)^2} \right). \quad (147)$$

To derive it, extract h_2 from (133) using (132). The endpoint contributes $\ell t^2/6$, since $F(t)/(F'(0)t) = 1 + \ell t^2/6 + O(t^3)$. No higher profile or Z_n^Γ contributes before order t^3 . Thus $L_{d,2} = 2^{-2/3}(h_2 + \ell/6)$, which simplifies to (146).

[Write phase: the second forms in (146) and (147) are in the host's B and were checked symbolically against source 104's forms. At $d = 2$, $L_{2,2} = 0$, as in (45); at $d = 3$, $L_{3,2} = 11B/324$, as in (38). For $d = 3, 4, 5, 6, 7, 8$ the bracket in (146) is $11/108, 118/1215, 1/12, 332/4725, 115/1944, 22/441$; it has the sign of $27(d-1)^2 - (d+1)^3$, so it vanishes at $d = 2$, is positive for $3 \leq d \leq 21$ (where $L_{d,2} < 0$, since $B < 0$) and negative for $d \geq 22$ (where $L_{d,2} > 0$; at $d = 21$, $10800 > 10648$, and at $d = 22$, $11907 < 12167$). Numerically $(C_{d,1}, L_{d,2}) \approx (0.12053, -0.15002), (-0.14686, -0.16154)$ and $(-0.31838, -0.14869)$ at $d = 3, 4, 5$.]

The maximal-network expansion has a different second coefficient:

$$\frac{T_{n,n-1}^{(d)}}{(\gamma_d/\Gamma)(n!)^d \Gamma^n e^{Bn^{1/3}} n^\alpha} = 1 + C_{d,1} n^{-1/3} + \left(C_{d,2} - \frac{B}{3} \right) n^{-2/3} + O(n^{-1}). \quad (148)$$

Indeed $B[(n-1)^{1/3} - n^{1/3}] = -(B/3)n^{-2/3} + O(n^{-5/3})$. This shift must not be omitted when using word coefficients for network counts. [Write phase: in Part I's notation (37), $E_{d,1} = C_{d,1}$ and $E_{d,2} = C_{d,2} - B/3$.]

For $d = 3$, $d_- = 2$, $d_+ = 4$, $\Gamma = 32$, $\rho = -1$, $\alpha = -3$, and $B/3 = z2^{-2/3}$. In particular,

$$\begin{aligned} C_{3,1} &= \frac{z^2}{36 \cdot 2^{1/3}} = \frac{B^2}{162}, \\ C_{3,2} &= \frac{z^4}{2592 \cdot 2^{2/3}} + \frac{11z}{108 \cdot 2^{2/3}} = \frac{B^4}{52488} + \frac{11B}{324}. \end{aligned} \quad (149)$$

The exact degree-six reconstruction also gives the next ternary logarithmic coefficient. More explicitly,

$$\begin{aligned} \log \frac{a_n}{\gamma_3(n!)^2 32^n e^{Bn^{1/3}} n^{-1}} &= \frac{z^2}{36 \cdot 2^{1/3}} n^{-1/3} + \frac{11z}{108 \cdot 2^{2/3}} n^{-2/3} \\ &\quad + \left(\frac{z^3}{243} - \frac{3}{16} \right) n^{-1} + O(n^{-4/3}) \quad (d = 3). \end{aligned} \quad (150)$$

The exact gamma factor contributes $-7/(32n)$ to this expression; omitting it would give the wrong cubic logarithmic coefficient. The corresponding multiplicative third coefficient is

$$C_{3,3} = \frac{z^6}{559872} + \frac{43z^3}{7776} - \frac{3}{16} = \frac{B^6}{25509168} + \frac{43B^3}{52488} - \frac{3}{16}.$$

Here a_n in (150) denotes the word count at degree $d = 3$. The exact reconstruction in the source package checks the recurrence, scalar carrier, local Airy solution and gamma conversion before comparing these coefficients.

[Write phase: with $B = 3z/2^{2/3}$ the three coefficients of (150) are $B^2/162$, $11B/324$ and $4B^3/6561 - 3/16$, exactly those of Part I's (38): two independent derivations, by different gauges, agree. The B -forms of $C_{3,2}$ and $C_{3,3}$ are write-phase conversions, checked symbolically; $C_{3,3}$ is new.]

28.2 A quantitative consequence of the published ternary proof

Proposition 28.1 (source 104; conditional on its reading of [3], see Remark 28.2). *For $d = 3$,*

$$T_n^{(3)}/T_{n,n-1}^{(3)} = 1 + O(n^{-1/2}). \quad (151)$$

This rate follows directly from the proof of [3, Lemma 3.24 and Theorem 1.9(ii)].

Proof. The published argument partitions ternary word/network objects into augmentation classes associated with binary objects. In its paired classes, the cardinalities $\mathfrak{r}(\omega')$ and $\mathfrak{s}(\omega)$ have the uniform ratio bound

$$\frac{\mathfrak{r}(\omega')}{\mathfrak{s}(\omega)} \geq \mathfrak{C}_m, \quad \mathfrak{C}_m = \prod_{i=2}^m \frac{4i-3}{4i-5} = \frac{\Gamma(3/4) \Gamma(m+1/4)}{\Gamma(5/4) \Gamma(m-1/4)} \asymp \sqrt{m}.$$

For n leaves the word has exactly $m = n - 1$ letters. The source temporarily shifts its word subscript within this paragraph; using m separates the word size from the leaf count. The binary indexing pairs the classes bijectively. Summing $\mathfrak{r}$ over them gives the ternary maximal count, up to the common permutation factor, whereas summing $\mathfrak{s}$ gives twice the first-deficit count. That factor two is the marked free-edge encoding in the published construction. Hence, explicitly,

$$\frac{T_{n,n-2}^{(3)}}{T_{n,n-1}^{(3)}} \leq \frac{1}{2\mathfrak{C}_{n-1}} = O(n^{-1/2}).$$

The proof of [3, Theorem 1.9(ii)], by iterating its inequality (15), also gives

$$T_{n,n-1-k}^{(3)} \leq \frac{T_{n,n-2}^{(3)}}{2^{k-1}k!}, \quad 1 \leq k \leq n-1.$$

Summing the convergent factorial majorant proves (151). □

Combining the all-order maximal theorem with this ratio proves (90): the first maximal correction is $C_{3,1}n^{-1/3}$, and its next error $O(n^{-2/3})$ is smaller than the ratio error $O(n^{-1/2})$. The second word or maximal coefficient in (148) and (149) does not transfer with this error bound. The displayed rate is a quantitative extraction from the published proof, not a claim to a separate new combinatorial rate theorem. For $d \geq 4$, the published leading ratio suffices for (89); no unproved quantitative extension is used here.

Remark 28.2 (Status of Proposition 28.1; write phase). The proof above reads two quantitative facts out of the *proofs* of [3, Lemma 3.24 and Theorem 1.9(ii)]: the uniform class-ratio bound $\mathfrak{C}_m$ with $m = n - 1$, and the factor two between the summed class sizes and the first-deficit count. Neither is a stated result of that paper (source 104's 104-fixdeg-SOURCES.md says so), the paper is not shipped, and the reading was not checked against it in writing. The algebra after the two facts is correct: $\mathfrak{C}_m = \prod_{i=2}^m (i - \frac{3}{4}) / (i - \frac{5}{4})$ telescopes to the displayed Gamma ratio, which is $\asymp \sqrt{m}$, and $\sum_{k \geq 1} 1/(2^{k-1}k!) = 2(e^{1/2} - 1)$ is finite. Proposition 28.1 and (90) are therefore printed as conditional on that reading, and the verification is listed in Section 32. If the reading holds, the proposition is the first total-count correction known here for any $d \geq 3$ (Section 10).

29 All order inverses and exact threshold rules

29.1 A common Lambert starting point

The factorial power must be included in the inverse. Consider a scale

$$S(n) = \gamma_*(n!)^{\mathfrak{p}} \Gamma^n e^{Bn^{1/3}} n^r (1 + o(1)), \quad \mathfrak{p} > 0.$$

For words use $(\mathfrak{p}, r, \gamma_*) = (d_-, \rho, \gamma_d)$; for maximal networks use $(d, \alpha, \gamma_d/\Gamma)$; for total networks use (d, α, K_d) . With $y > 0$ large, define

$$x_0(y) = \frac{y}{\mathfrak{p} W_0(\Gamma^{1/\mathfrak{p}} y / (\mathfrak{p} e))}, \quad (152)$$

$$L_0(y) = \mathfrak{p} \log(\Gamma^{1/\mathfrak{p}} x_0(y)),$$

$$\hat{x}_{\mathfrak{p},r}(y) = x_0(y) - \frac{Bx_0(y)^{1/3} + (r + \mathfrak{p}/2) \log x_0(y) + (\mathfrak{p}/2) \log(2\pi)}{L_0(y)}. \quad (153)$$

Here W_0 is the positive real principal branch of Lambert's W function. The point x_0 solves

$$\mathfrak{p} x_0 (\log(\Gamma^{1/\mathfrak{p}} x_0) - 1) = y.$$

[Write phase: this is Part I's seed (40), since $\mathfrak{p}x(\log(\Gamma^{1/\mathfrak{p}}x) - 1) = \mathfrak{p}x \log x + \mathfrak{a}x$ with $\mathfrak{a} = \log \Gamma - \mathfrak{p}$.]

Proposition 29.1 (Amplitude free eventual inverse). *Even the known two-sided Θ estimate for any of the three sequences implies*

$$\hat{x}_{\mathfrak{p},r}(\log S(n)) = n + O(1/\log n). \quad (154)$$

Thus nearest-integer rounding eventually recovers the index of an exact sequence value without evaluating the amplitude.

Proof. By Stirling's formula, the two-sided scale gives $\log S(n) = \Phi(n) + O(1)$, where

$$\Phi(x) = \mathfrak{p}x(\log(\Gamma^{1/\mathfrak{p}}x) - 1) + Bx^{1/3} + (r + \mathfrak{p}/2) \log x + (\mathfrak{p}/2) \log(2\pi).$$

Write $\Phi = \Phi_0 + \varphi_1$. At x_0 , $\Phi_0(x_0) = y$ and $\Phi'_0(x_0) = L_0$. For $h = -\varphi_1(x_0)/L_0 = O(x_0^{1/3}/\log x_0)$,

$$\Phi(x_0 + h) - y = \varphi'_1(x_0)h + O(h^2/x_0) = O(x_0^{-1/3}/\log x_0).$$

Since $\Phi'(x) \sim \mathfrak{p} \log x$, the mean value theorem proves (154). The omitted $\log \gamma_*$ is a bounded constant, so it is consistent with this accuracy. $\square$

The statement is eventual and has no effective numerical cutoff. It does not assert that a floating-point implementation rounds correctly from the first few terms.

29.2 Arbitrary order word and maximal inverses

Let $L_{d,j}$ be the logarithmic coefficients defined in Section 28. With the true canonical amplitude, set

$$\mathcal{F}_J(x) = d_- \log \Gamma(x+1) + x \log \Gamma + Bx^{1/3} + \rho \log x + \log \gamma_d + \sum_{j=1}^J L_{d,j} x^{-j/3}, \quad (155)$$

where $\Gamma(x+1)$ is Euler's Gamma function and $\log \Gamma$ the logarithm of the base. For sufficiently large x , its derivative is positive and asymptotic to $d_- \log x$. If $x_J(y)$ is its unique sufficiently large inverse, Theorem 22.1 and the mean value theorem give

$$x_J(\log a_n) = n + O\left(\frac{n^{-(J+1)/3}}{\log n}\right). \quad (156)$$

The exact shifted maximal model is

$$\mathcal{M}_J(x) = \log \Gamma(x+1) + \mathcal{F}_J(x-1). \quad (157)$$

Its derivative is asymptotic to $d \log x$, and its inverse at $\log T_{n,n-1}^{(d)}$ has the same error order as (156). This shifted model automatically includes every index correction, including the $-B/3$ in (148).

For either model $\mathcal{A}_J$, a prescribed finite number of Newton steps suffices asymptotically. Start at (152), with $\mathfrak{p} = d_-$ for $\mathcal{F}_J$ or $\mathfrak{p} = d$ for $\mathcal{M}_J$, and iterate

$$x_{k+1} = x_k - \frac{\mathcal{A}_J(x_k) - y}{\mathcal{A}'_J(x_k)}, \quad m_J = \lceil \log_2(J+4) \rceil - 1. \quad (158)$$

The m_J th iterate already has the inverse error in (156). Indeed, if x_* is the exact model root and $L = \log x_0$, then

$$x_* - x_0 = O(x_0^{1/3}/L), \quad \mathcal{A}'_J(x) \asymp L, \quad \mathcal{A}''_J(x) = O(1/x_0)$$

in the relevant neighborhood. Taylor's theorem gives

$$|x_{k+1} - x_*| \leq \frac{C}{x_0 L} |x_k - x_*|^2, \quad x_k - x_* = O\left(\frac{x_0^{1-2^{k+1}/3}}{L^{2^{k+1}-1}}\right).$$

The iterates stay in that neighborhood. Since $2^{m_J+1} \geq J+4$, the numerical-model error is at most the asymptotic-model error. All statements here refer to exact mathematical evaluations; uncertified amplitude digits do not provide a certified numerical inverse. *[Write phase: the same quadratic estimate is Part II's (82).]*

For reference,

$$\mathcal{F}'_J(x) = d_- \Psi(x+1) + \log \Gamma + \frac{B}{3} x^{-2/3} + \frac{\rho}{x} - \sum_{j=1}^J \frac{j}{3} L_{d,j} x^{-j/3-1},$$

where Ψ is the digamma function. If only elementary functions and W_0 are desired, replace each $\log \Gamma(x+1)$ by

$$(x + \tfrac{1}{2}) \log x - x + \tfrac{1}{2} \log(2\pi) + \sum_{m=1}^R \frac{B_{2m}}{2m(2m-1)x^{2m-1}}, \quad R = \max\left(0, \left\lceil \frac{J-2}{6} \right\rceil\right), \quad (159)$$

with the Bernoulli numbers B_{2m} . Its remainder is $O(x^{-2R-1})$ and $2R+1 \geq (J+1)/3$. Use the corresponding expression at $x-1$ in the shifted model and differentiate the replacement consistently. The same inverse order and finite Newton count follow.

29.3 The information available for total counts

For totals at any fixed degree, the leading model

$$\mathcal{T}_{d,0}(x) = d \log \Gamma(x+1) + x \log \Gamma + Bx^{1/3} + \alpha \log x + \log K_d \quad (160)$$

has eventual inverse satisfying

$$\mathcal{T}_{d,0}^{-1}(\log T_n^{(d)}) = n + o(1/\log n).$$

This uses just the leading equivalent. For $d \geq 4$, no power-law remainder for this inverse is inferred from the published ratio limit. For $d = 3$, the source-derived refinement gives the stronger model

$$\mathcal{T}_{3,1}(x) = 3 \log \Gamma(x+1) + x \log 32 + Bx^{1/3} - 3 \log x + \log(\gamma_3/32) + \frac{z^2}{36 \cdot 2^{1/3}} x^{-1/3}, \quad (161)$$

and

$$\mathcal{T}_{3,1}^{-1}(\log T_n^{(3)}) = n + O\left(\frac{n^{-1/2}}{\log n}\right). \quad (162)$$

Starting from (152) with $\mathbf{p} = 3$, $\Gamma = 32$, two Newton steps suffice for this accuracy: their root error is $O(x_0^{-5/3}/\log^7 x_0)$, smaller than the model error. This is not an all-order total-count inverse. *[Write phase: (161) and (162) inherit the conditional status of Proposition 28.1. At $d = 2$ Part II's (77) and (78) are stronger.]*

29.4 Exact floor and ceiling thresholds

The asymptotic inverse of an exact sequence value and a threshold query are different tasks. A threshold can be arbitrarily close to a count. For words, (6) gives, for $n \geq 2$,

$$a_n \geq b_{n,n} = \binom{d_+ n - 2}{d_-} a_{n-1} > a_{n-1}.$$

The initial equality $a_0 = a_1 = 1$ is a finite exception. The maximal sequence is strictly increasing eventually by its exact identity; the total equivalent also gives $T_{n+1}^{(d)}/T_n^{(d)} \sim \Gamma n^d$, hence eventual strict monotonicity. The amplitude-free approximation satisfies $\hat{x}'_{\mathbf{p},r}(y) \sim 1/(\mathbf{p} \log x_0) > 0$.

Let $S(n)$ be any of these sequences and let Y be a sufficiently large positive threshold. Define

$$k_{\leq}(Y) = \max\{n : S(n) \leq Y\}, \quad j_0 = \text{round}(\hat{x}_{\mathbf{p},r}(\log Y)).$$

Since the monotone approximation is $n + o(1)$ at each exact count, $j_0 \in \{k_{\leq}(Y), k_{\leq}(Y) + 1\}$. Therefore one exact or rigorously certified comparison gives

$$k_{\leq}(Y) = \begin{cases} j_0, & S(j_0) \leq Y, \\ j_0 - 1, & S(j_0) > Y. \end{cases} \quad (163)$$

For the least index $k_{\geq}(Y) = \min\{n : S(n) \geq Y\}$,

$$k_{\geq}(Y) = \begin{cases} j_0, & S(j_0) \geq Y, \\ j_0 + 1, & S(j_0) < Y. \end{cases} \quad (164)$$

Equality belongs to the first branch in both formulas. To see the ceiling assertion at equality, note that an exact sufficiently large count rounds to its exact index; strictly between counts the only candidates are the adjacent two. Any deterministic tie rule for nearest-integer rounding works eventually. No unverified floating-point comparison arbitrarily near equality is enough. No effective cutoff or certified numerical precision prescription is supplied here; a practical implementation must certify the comparison and account for the finite initial region. *[Write phase: $k_{\geq}(Y)$ is Part II's $\mathcal{N}(Y)$; the bracket (83) and this one-comparison rule are two forms of the separation condition of Remark 11.1.]*

30 Sequence identification and the scope of the literature check

The $d = 2$ word sequence is OEIS A213863 [6]. For $d = 3, 4$, the original recurrence gives

d	a_0	a_1	a_2	a_3
2	1	1	7	106
3	1	1	25	2305
4	1	1	91	51821

The corresponding maximal-network prefixes, starting at $n = 1$, are

$$\begin{aligned} d = 3 : & \quad 1, 2, 150, 55320, 57939000, 132120450000, \dots, \\ d = 4 : & \quad 1, 2, 546, 1243704, 11351644920, 291451508298720, \dots \end{aligned}$$

They agree with the maximal entries in [3, Appendix A, Tables 3–4]. No OEIS identifier was established for the exact $d = 3$ or $d = 4$ word or unranked-network sequence in the focused search. This means no entry was located, not that the database contains none. *[Write phase: the $d = 3$ row continues Part I's diagonal in Section 3; the $d = 4$ prefixes and both maximal prefixes were recomputed from (6) in writing and agree.]*

A nearby identifier is an important trap: A213864 begins $1, 1, 19, 1075, \dots$ and is not the $d = 3$ sequence. Although it also uses four copies of each letter, it imposes its prefix ordering once a letter has appeared. The $d = 3$ condition permits one occurrence before requiring the comparison. Varying multiplicity while keeping the zero-occurrence exception fixed, as in A213275, changes the family. Ranked tree-child counts, including A167484, also count different objects [14, 15, 16].

The main baseline is [3], whose Theorem 1.8 gives the full Airy Θ scale, whose Corollary 1.11 gives the ratio limit, and whose Remark 1.12 explicitly distinguishes those facts from a first-order equivalent. Its one-component equivalents concern a different subclass. The positive amplitude above does not rediscover the already published exponent, stretched exponential, or exponential base.

A focused primary-source check through 2 October 2026 located no prior positive-amplitude equivalent or all-order expansion in the fixed- $d \geq 3$ maximal-reticulation regime. The relevant newer binary work has different scope: the Young-tableau and Pons–Batle results in [4, 8, 11] concern exact identities and fixed or bounded reticulation regimes; [9, 10] study sparse binary regimes; and the trident-tableau work [12] concerns binary networks with exactly one reticulation. These scope distinctions do not constitute an exhaustive priority certificate. The article's mathematical claims rest on the explicit proof above, not on absence of a search result. *[Write phase: in the repository Part I (source 16) proves the same positive amplitude and all-order expansion for every fixed d ; it*

was written independently of source 104 and printed first (Section 21.1). The sentence is accurate for the literature source 104 inspected.]

31 Reproducibility and remaining boundaries

The accompanying package contains the standalone TeX source, this PDF, build instructions, exact finite certificates and a fail-closed checker. It separates finite algebraic verification from the analytic proof.

The rational checks compare the original recurrence with the rotated one-step recurrence, the two-step evolution, the exact endpoint normalization, the symmetrizer edges and the Gram factor. They also test the repaired contraction at degrees including $d \geq 6$, where the naive normalization obstruction matters. The polynomial–Airy checks clear the exact recurrence denominators and check both Airy coordinates, value and derivative gauges, and the modulo-three grading. The endpoint reconstruction uses the normalized Airy differential equation, integrates the scalar ratio, and includes the exact gamma conversion Z_n^Γ . These tests identify the displayed coefficients and detect algebraic or startup mistakes; they are not machine proofs of the global spectral or all-order analytic estimates.

All verification gates use explicit exceptions, so they remain active under `python -O`. The release suite is run in ordinary and optimized modes and requires deliberately corrupted inputs to be rejected. The source archive has been rebuilt and replayed from a fresh extraction; every page of the final PDF is visually reviewed after rendering. The shipped provenance identifies the primary sources and relevant versions without redistributing third-party papers or development notes.

[Write phase: in the repository the checker, runner and build script are code/104-fixdeg-verify_exact.py, code/104-fixdeg-run_checks.py and code/104-fixdeg-build.sh; the inputs, results and provenance are data/104-fixdeg-; the package documentation and review are 104-fixdeg-checks-README.md, 104-fixdeg-checks-RESULTS.md, 104-fixdeg-SOURCES.md and 104-fixdeg-qa-REVIEW.md. The manuscript, its PDF, the release verifier and the release manifest are not shipped; they survive in the arrival commit. The intake reran the suite on copies of the delivered layout on 5 October 2026. A first run failed after 34 of 50 rejections, during a machine-wide resource shortage, with “verifier did not produce result JSON”; that case rejects correctly in isolation. A rerun on a fresh copy passed: both baselines (5,655 gates each) and 50 of 50 rejections, in 475 seconds on a loaded laptop. The regenerated result files equal the delivered ones apart from timestamps, durations and the Python version. The release verifier verify_release.py fails on Windows (“Unsafe inventory path”: it requires `str(Path(name)) == name`, which backslashes break); this is a portability defect, not an integrity failure, and the delivered manifest was verified (25 of 25) at placement. The README of this report gives the rerun route.]*

Amplitude digits. The convergent representations (121), (122), and (124) define γ_d exactly. Certified decimal enclosures would additionally require effective bounds for the spectral, drift, projection and smoothing tails, together with rigorously controlled finite computations. None is claimed here.

[Write phase: uncertified diagnostics, computed at the intake (5 October 2026; exact integer recurrence for $n \leq 900$, 700, 600, then least-squares fits of the logarithmic remainder with $C_{d,1}$ fixed by (87); the script is not shipped) give $\gamma_3 \approx 3.784368$, $\gamma_4 \approx 3.478351$ and $\gamma_5 \approx 2.384755$. The fitted second logarithmic coefficients, -0.15001 , -0.16151 and -0.14867 , agree with (146) to four or five digits.

Part I's shipped diagnostics (data/16-dcomb-numerical-diagnostics.json) give 3.7843665 for $d = 3$ at $n = 6000$, corrected through n^{-1} , and, for $d = 4$, uncorrected ratios that increase from 3.4036 at $n = 500$ to 3.4485 at $n = 6000$, as a negative $C_{4,1}$ requires. None of these digits is certified.]

Total count corrections. The $d = 3$ augmentation argument transfers the first coefficient with an $O(n^{-1/2})$ relative remainder. It does not determine the next total coefficient. Sharper deficit-sector asymptotics would be needed for an all-order total expansion; for $d \geq 4$ this Part uses only the leading published ratio.

Uniformity and exponentially small terms. Every estimate allows dependence on the fixed integer d , and on fixed truncation depth where applicable. There is no uniform limit with $d \rightarrow \infty$. Formal solvability at every finite depth and a Poincaré remainder do not establish a convergent infinite series or a transseries. Separate exponentially small amplitudes and their interaction would require further analysis.

What makes the argument apply at every fixed degree. The exact gamma-product normalization, full new-coordinate diagonal, factorwise contraction, form compactness at the half-grid boundary, localization across changing dimension, boundary kernel, strictly positive product comparison, finite polynomial inverse, and signed forced-product bootstrap are all proved here. No unproved analogy with the binary case is needed.

32 Further questions and research (Part III)

Written in the write phase (5 October 2026), under the repository's rule that unproved claims are recorded, not dropped. No claim of source 104 was found to be wrong: every checked constant agrees with Parts I–II or with an independent computation.

1. *Verify the ternary reading.* Proposition 28.1, hence (90) and (162), rests on source 104's reading of the proofs of [3, Lemma 3.24 and Theorem 1.9(ii)]: the uniform class-ratio bound $\mathfrak{C}_m \asymp \sqrt{m}$ with $m = n - 1$, and the factor two from the marked free-edge encoding (sketch in the proof above). What is missing is a check of both facts against the paper, or a self-contained proof of $T_{n,n-2}^{(3)}/T_{n,n-1}^{(3)} = O(n^{-1/2})$ from the recurrence. (Source 104.)
2. *Further total-count corrections for $d \geq 3$.* The second total coefficient at $d = 3$, and any total correction for $d \geq 4$, need sharper deficit-sector asymptotics; the published ratio limit gives no rate for $d \geq 4$. This is Part I's question 1 of Section 12, now re-scoped. At $d = 2$ Part II settles it through the binary identity (T1), which has no known analogue for $d \geq 3$. (Sources 104 and 16.)
3. *Certified amplitudes.* Enclosures of γ_d through (121), (122) need effective constants for the spectral, drift, projection and smoothing tails. The intake values $\gamma_3 \approx 3.784368$, $\gamma_4 \approx 3.478351$, $\gamma_5 \approx 2.384755$ (Section 31) are uncertified. Part I's question 2. (Sources 104 and 16.)
4. *Uniformity in d .* All constants depend on d ; a joint regime with d growing with n is open. The closed forms (87), (146) have limits as $d \rightarrow \infty$ (for instance $C_{d,1} \rightarrow -19z^2/90$, since $B \rightarrow 3z$), but this proves nothing uniform about the remainders. Part I's question 3. (Sources 104 and 16.)

5. *Exponentially small terms and convergence.* Neither a transseries nor convergence or divergence of the correction series is established. Part I’s question 4. (Sources 104 and 16.)
6. *Closed forms beyond two coefficients.* (87) and (146) are universal in d ; whether every $L_{d,j}$ is a polynomial in B with coefficients rational in d of a recognizable shape, and what the next one is, is not computed. The algorithm of Section 26 gives each one for fixed d . (Write phase.)
7. *OEIS.* No accession was established for $d = 3, 4$ (Section 30); none is claimed or submitted. (Source 104.)

Part IV

The binary word count in the factorial gauge: six logarithmic terms and an identity-free ratio rate (source 102)

Source 102 (bundle Report 102, archive A213863_Tableaux_and_Tree_Child_Networks_Source), printed with its binary-specific material in full and its generic steps as pointers to Part III at $d = 2$; its title was “Positive Airy amplitudes for Young tableaux and tree child networks: a self contained Jacobi product proof for A213863”. Its abstract reads (symbols renamed as in Section 33.3): For the Young-tableau and constrained-word sequence a_n recorded as OEIS A213863, we prove the existence of a strictly positive finite amplitude γ_2 in

$$a_n = \gamma_2 n! 12^n e^{z(3n)^{1/3}} n^{-2/3} (1 + O(n^{-1/3})),$$

where z is the largest zero of the Airy function. The exact recurrence is converted into a product of positive contractive operators obtained from finite Jacobi matrices. A Dirichlet Airy limit, quantitative eigenvector drift, and a killed-walk boundary estimate yield a convergent amplitude representation. We then construct compatible polynomial–Airy profiles to arbitrary depth and transfer them to genuine Poincaré expansions with the same amplitude. The known exact normalization for maximally reticulated binary tree-child networks and the proved total-to-maximal ratio $\sqrt{e}$ give a first-order equivalent for the total network count. A direct quantitative consequence of published lemmas gives an $O(n^{-2/3})$ ratio error and transfers the first correction to the total count; a full all-order total-network expansion is not claimed. We give exact coefficient algorithms, inverse asymptotics, reproducible finite checks, and precise proof boundaries. No certified decimal value of γ_2 , exponentially small sector, or exhaustive literature-priority claim is made.

33 Reading Part IV: provenance, credit and notation

This section was written when source 102 was added to the report (batch 102, 5 October 2026). In Part IV, text added in writing is marked “[Write phase: ...]”.

33.1 Provenance and credit

Source 102 is bundle Report 102, archive A213863_Tableaux_and_Tree_Child_Networks_Source.zip (543,991 bytes), main file `tree_child_amplitude_sources/tree_child_amplitude.tex` (903

lines, 21-page PDF), dated 2 October 2026. Its title page and its PDF metadata give the author as “Research prepared for private review”; that wording is the delivery’s and is disclosed here, as in the placement commit. It arrived in commit `60f54ea06` and was placed in this directory by `6ab1f1979` (batch 102), with the file prefix `102-tableaux-`. It names no repository commit, so it has no pin.

Credit order. The archive of source 102 was closed on 1 October 2026 at 22:36 Pacific time (05:36 UTC on 2 October), before the batch-77 archives of sources 16 and 69 reached the repository (`096ee7b87`, 2 October, 11:05 Pacific). Source 102 thus proves Theorem 14.1 (and Theorem 2.1 at $d = 2$) independently of Parts I–II. It does not cite source 104 (Part III), which belongs to the same delivery series, was closed 42 minutes later and shares its architecture. In the repository the theorem was printed first, from sources 16 and 69; source 102 is a further independent route at $d = 2$ and is credited as such. Source 104 proves the same theorem for every d with the same architecture in another gauge, so Part IV is printed after Part III and points to it where the two arguments coincide at $d = 2$.

33.2 What Part IV proves again and what it adds

Source 102	Relation to Parts I–III
Theorems 34.1, 37.1, 38.1	second proofs of Theorem 14.1 at $d = 2$ (the fourth route, counting Parts I–III), in the factorial gauge $3^p p!$; same architecture as Part III
$L_1, \dots, L_4, c_1, c_2, c_3$ in (217), (218) L_5, L_6 in (217)	second derivation of (45), identical values new; confirmed in writing by Part II’s own generator at order nine (Remark 39.1)
Ratio expansion (220) through n^{-3} Proposition 34.3 ($O(n^{-2/3})$ ratio and total variation)	new weaker than Theorems 16.2 and 18.2, but by a route that does not use the identity (T1); a second route (Remark 34.4)
Total count with first correction (219) Eigenvalue (181), amplitude formula (205), gauge-specific constants Inverses (Section 40)	weaker re-proof of (46) new formulas of the factorial gauge (a further convergent representation of γ_2) second route of Section 17 and Part III’s Section 29 at $d = 2$; no novelty claimed
Numerical table (Section 39.3)	uncertified; consistent with Section 19

33.3 Notation of Part IV

Part IV uses Part II’s letters for every notion Part II names, and for its proof apparatus the letters of Part III, which play the same roles at $d = 2$ in the factorial gauge $3^p p!$ instead of the gamma gauge $3^p(4/3)_p$. Values therefore differ from Part III’s (for example the n^{-1} term of the eigenvalue is $10/3$ here and $3c = 2$ there). The renamings of source 102’s letters are listed below; no formula changed otherwise. A letter not listed has its Part III meaning at $d = 2$ (Section 21.3) or, failing that, the host’s.

Symbol here	Meaning	Source 102; the false reading to avoid
$z, z_j; B$	Airy zeros; $3^{1/3}z$ (Part II’s)	α, α_j ; α is the network exponent here
$B/3$	$3^{-2/3}z$	ρ ; not the exponent $\rho = -2/3$

Symbol here	Meaning	Source 102; the false reading to avoid
$\gamma_2; \gamma_T$	the amplitude (γ of Part II); $\sqrt{e}\gamma_2/12$	$\gamma_w; K_T$
$b_{n,m}; \tilde{b}(p,q)$	the array (6) at $d=2$; $b_{p+1,q+1}$	$b(n,m); B(p,q)$
$\mathcal{Y}(N,h), \omega_N^\pm(h), e_n(k)$	factorial-gauge array, weights, even-time vector	$d(N,h), u_N(h), v_N(h), e_n(k)$; not Part I's $D_N(j)$, which uses $3^n n!$ with the other index pair
$M_n; L_n, U_n, D_n; b_n(k); K_n$	two-step matrix; its entries; Jacobi off-diagonal; bidiagonal factor	$A_n; L_n, U_n, D_n; \beta_n(k); C_n$; not $A_{n,k}$, Part II's $\beta_n(m)$ or $C_r(m)$
$\Sigma_n, \varsigma_n(j), R_n; f_1, f_2$	symmetrizer, edge ratio, inter-time ratio; factors of $\varsigma_n(j)^2$	$S_n, r_n(j), R_n; F_1, F_2$
$P_n, u_n, T_n, J_n/\lambda_n$	as in Part III	Λ_n, v_n, T_n, H_n
$\psi_n, \psi_n^A, f_A, \mathcal{H}_A, c_A$	Perron vector, Airy sample, ground state, operator, $2 \cdot 3^{-1/3}$	$\phi_n, q_n, f, \mathcal{H}, c_A$
$V_n(k); \vartheta_n(k)$	potential of the form; $(3n+k)/(3(n+k))$	$V_n(k); t_n(k)$
$p_n, p_n^\perp, p_\infty; z, \varrho, \varrho_n$	Perron coefficient, transverse norm, limit; errors and forcings	$c_n, d_n, c_\infty; z, \rho, \eta_n$
$c, \ell, F; h_j; \mathfrak{B}_\ell$	$2/3, c^{2/3}z, \text{Ai}(z+c^{1/3}x)$ (Part II's); carrier coefficients; normalized Airy solution	$c, \kappa, \mathcal{A}; \ell_j; \mathcal{B}(\kappa, \cdot)$
$\mathcal{H}_N^{[M]}, \mathfrak{F}, E_n^{[M]}, V_n^{[M]}, \Theta_n$	carrier, formal series, even-time profile, normalized profile, bulk scale	$H_N^{[M]}, F, E_n^{[M]}, U_n^{[M]}, B_n^{\text{bulk}}$
$L_j; c_j$	logarithmic and multiplicative coefficients of a_n (Part II's L_r, c_r)	$b_j; c_j$
$\text{TC}_{n,k}, \text{TC}_n$	binary counts with n leaves ($T_{n,k}^{(2)}, T_n$)	same
$\theta_k, \varpi_k, f_k; \mathcal{R}_{n-1}$	$\text{TC}_{n,n-1-k} / \text{TC}_{n,n-1}, 1/(2^k k!)$, free-root-child ratio; $\text{TC}_n / \text{TC}_{n,n-1}$	$t_k, w_k, f_k; T; W = \sqrt{e}$; the counts $F_{n,r}$ of [3] keep their letter
$x_0(y), L_0(y), \hat{x}(y), \hat{x}_T(y); \Phi; y, Y; m_J; \Psi$	inverse quantities as in Part III; the target logarithm and threshold; Newton count; digamma	$x_0(Y), L_0(Y), z(Y), z_T(Y); G; Y, y; K; \psi$

34 The result and its combinatorial meaning

Let $b_{1,1} = 1$, let $b_{n,m} = 0$ outside $1 \leq m \leq n$, and for $n \geq 2$ let $b_{n,m} = (2n+m-2) \sum_{j \leq m} b_{n-1,j}$; this is (6) at $d=2$. Set $a_0 = 1$ and $a_n = \sum_{m=1}^n b_{n,m}$ for $n \geq 1$. The first values are

$$1, 1, 7, 106, 2575, 87595, 3864040, 210455470.$$

This is OEIS A213863. Its word interpretation uses three copies of each of n letters: in every prefix, whenever a smaller letter has appeared, its count is at least the count of any larger letter. It also counts the three-row Young-tableau family with walls considered in [2]. The recurrence and the network bijection are established in [1].

Let $z = z_1 > z_2 > \dots$ be the negative zeros of Ai , the real solution of $\text{Ai}''(x) = x \text{Ai}(x)$ decaying at positive infinity. Thus $z = -2.338107410459767\dots$; this is a standard Airy constant, distinct from the uncomputed amplitude below.

Theorem 34.1 (Positive amplitude; source 102). *There exists $\gamma_2 \in (0, \infty)$ such that*

$$a_n = \gamma_2 n! 12^n e^{z(3n)^{1/3}} n^{-2/3} \left(1 + O(n^{-1/3})\right). \quad (165)$$

The positive convergent representation in Theorem 37.1 fixes γ_2 canonically.

The already known result for a_n is the two-sided bound (43) with the same full scale, proved in the network analysis of [1] and stated for the identical tableau sequence in [2, Theorem 4.1]. The existence of a limiting multiplicative constant is the additional assertion here. In particular, the stretched exponential is $z(3n)^{1/3} = Bn^{1/3}$, with no additional factor 3 before z . *[Write phase: Theorem 34.1 is the leading case of Theorem 14.1, and γ_2 is Part II's γ , Part I's and Part III's γ_2 : all are the limit of $a_n/(n!12^n e^{Bn^{1/3}} n^{-2/3})$. "Additional" refers to the published literature; in the repository Parts I–II hold the assertion (Section 33.1).]*

34.1 Network conventions and the consequence for total counts

Use the convention of [1, Definition 1]: a rooted, directed acyclic binary phylogenetic network has root indegree zero and outdegree one, n leaves of indegree one and outdegree zero bijectively labelled by $[n]$, tree nodes of indegree one and outdegree two, and reticulation nodes of indegree two and outdegree one. The tree-child condition says that every nonleaf vertex has a child that is a tree node or leaf. Networks are counted up to isomorphism preserving the leaf labels. Write $\text{TC}_{n,k}$ for the count with k reticulations and TC_n for the total. The maximal possible k is $n - 1$.

The exact identity and the later ratio theorem are

$$\text{TC}_{n,n-1} = n!a_{n-1}, \quad \text{TC}_n \sim \sqrt{e} \text{TC}_{n,n-1}. \quad (166)$$

The first is [1, Proposition 2]. The second is [3, Corollary 1.11(i)], at $d = 2$, under the same root, leaf, and binary-degree convention. Its class TC, rather than its one-component subclass OTC, is used here.

Corollary 34.2 (Total binary tree-child networks; source 102). *Under these conventions,*

$$\begin{aligned} \text{TC}_n &\sim \frac{\gamma_2 \sqrt{e}}{12} (n!)^2 12^n e^{z(3n)^{1/3}} n^{-5/3} \\ &\sim \frac{\gamma_2 \pi \sqrt{e}}{6} n^{-2/3} \left(\frac{12}{e^2}\right)^n n^{2n} e^{z(3n)^{1/3}}. \end{aligned} \quad (167)$$

Proof. Substitute (165) at $n - 1$ into (166). Use $(n - 1)! = n!/n$, $12^{n-1} = 12^n/12$, and $e^{z(3(n-1))^{1/3}}/e^{z(3n)^{1/3}} = 1 + O(n^{-2/3})$. The power changes by the extra $1/n$ to $n^{-5/3}$. Stirling's formula gives the second line. $\square$

[Write phase: (167) is the leading term of (46) and of (T24).] The ratio limit alone would not transfer an error rate. The following quantitative consequence of the same paper's lemmas does so. It is a direct deduction from published estimates, not a separate literature-priority claim.

Proposition 34.3 (A quantitative ratio from published lemmas; source 102). *As $n \rightarrow \infty$,*

$$\frac{\text{TC}_n}{\text{TC}_{n,n-1}} = \sqrt{e} \left(1 + O(n^{-2/3})\right). \quad (168)$$

Moreover, the reticulation deficit of a uniformly selected binary tree-child network with n leaves has total-variation distance $O(n^{-2/3})$ from Poisson(1/2).

Proof. For fixed $n \geq 2$ put $M = \text{TC}_{n,n-1}$, $\theta_k = \text{TC}_{n,n-1-k}/M$, and $\varpi_k = 1/(2^k k!)$, for $0 \leq k \leq n-1$. In the notation of [3], $F_{n,r}$ counts networks whose root's child is a free tree node, meaning that neither of that node's children is a reticulation node. Set $\mathfrak{f}_k = F_{n,n-1-k}/M$. Lemma 3.17 and Remark 3.18 of [3] give

$$\theta_k = 2(k+1)\theta_{k+1} + \mathfrak{f}_k \quad (0 \leq k \leq n-2), \quad \theta_0 = 1, \quad \mathfrak{f}_0 = 0.$$

Lemmas 3.19 and 3.20 give, on the full finite range,

$$0 \leq \mathfrak{f}_k \leq \frac{\mathfrak{f}_1}{2^{k-1}(k-1)!} \quad (1 \leq k \leq n-1), \quad \mathfrak{f}_1 = O(n^{-2/3}).$$

Since $\varpi_k = 2(k+1)\varpi_{k+1}$, induction yields

$$0 \leq \frac{\varpi_k - \theta_k}{\varpi_k} = \sum_{j=0}^{k-1} \frac{\mathfrak{f}_j}{\varpi_j} \leq 2\mathfrak{f}_1 \sum_{j=1}^{k-1} j = \mathfrak{f}_1 k(k-1).$$

Write $\sum_{k \geq 0} \varpi_k = \sqrt{e}$ and $\mathcal{R}_{n-1} = \sum_{k=0}^{n-1} \theta_k = \text{TC}_n/M$. The factorial second moment satisfies $\sum_{k \geq 0} k(k-1)\varpi_k = \sqrt{e}/4$. Therefore the exact finite bound is

$$0 \leq \sqrt{e} - \mathcal{R}_{n-1} \leq \frac{\sqrt{e}}{4} \mathfrak{f}_1 + \sum_{k=n}^{\infty} \frac{1}{2^k k!}. \quad (169)$$

The tail is at most $[2^n n!(1 - 1/(2(n+1)))]^{-1}$, hence smaller than every negative power of n , proving (168). Extend $\theta_k = 0$ for $k \geq n$. The actual deficit law is $\theta_k/\mathcal{R}_{n-1}$ and the Poisson law is $\varpi_k/\sqrt{e}$. Since $0 \leq \theta_k \leq \varpi_k$,

$$\frac{1}{2} \sum_{k \geq 0} \left| \frac{\theta_k}{\mathcal{R}_{n-1}} - \frac{\varpi_k}{\sqrt{e}} \right| \leq \frac{1}{2} \left(\frac{\sqrt{e} - \mathcal{R}_{n-1}}{\sqrt{e}} + \frac{\sqrt{e} - \mathcal{R}_{n-1}}{\sqrt{e}} \right) = \frac{\sqrt{e} - \mathcal{R}_{n-1}}{\sqrt{e}} = O(n^{-2/3}).$$

This proves the total-variation assertion. $\square$

Remark 34.4 (Relation to Part II; write phase). Proposition 34.3 is weaker than Part II, which has every finite order of $\mathcal{R}_n$ (Theorem 16.2) and the exact eventual identity $d_{\text{TV}} = \frac{3}{2}(1/\mathcal{R}_n - e^{-1/2})$ with its expansion (Theorem 18.2). It is kept as a second route because it uses only Lemmas 3.17–3.20 and Remark 3.18 of [3], not the Pons–Batle identity (T1) proved in [4], on which Part II's transfer rests. The two are consistent. Part II's identity can be written $d_{\text{TV}} = \frac{3}{2\mathcal{R}_n} \cdot \frac{\sqrt{e} - \mathcal{R}_n}{\sqrt{e}}$; since $\mathcal{R}_n \rightarrow \sqrt{e} > 3/2$, the prefactor tends to $3/(2\sqrt{e}) = 0.9098\dots < 1$, so the exact value lies below source 102's bound $(\sqrt{e} - \mathcal{R}_n)/\sqrt{e}$ for all large n , and both are of exact order $n^{-2/3}$ because $\mathcal{R}_n = \sqrt{e}(1 + \frac{B}{72}n^{-2/3} + O(n^{-1}))$ with $B \neq 0$ ((D8)). Source 102 counts n leaves and Part II $n+1$; the shift does not affect these orders. Source 102 cites [4] but does not use it.

In particular (167) has relative error $O(n^{-1/3})$ by Theorem 34.1. Section 39 sharpens it by one explicit correction. The all-order theorem below applies to a_n and, by exact shifting, to $\text{TC}_{n,n-1}$. The quantitative ratio bound alone does not yield every correction for TC_n .

34.2 Position relative to the inspected literature

The final discussion of [1] asks for a first-order equivalent and identifies two obstacles: the amplitude of a_n , and the total/maximal ratio. The latter was resolved by [3], whose Remark 1.12 still points

to the former. The result here supplies the amplitude input. *[Write phase: so do Parts I–II and Part III, independently; in the repository Part II, from source 69, was printed first and goes further (Section 33.1).]* The September 2026 version of [4] proves an enumerative identity, while its asymptotics fix the number of reticulations. The 2026 works [8, 9, 10] concern bounded or sparse reticulation regimes, not $k = n - 1$. The September paper [11] supplies a short proof of the exact Pons–Batle enumerative identity and recurrence, rather than a diagonal asymptotic theorem. The April 2026 version of [13] discusses tableau and phylogenetic applications in its Section 10.1. These inspected sources locate the particular gap addressed here. They do not constitute an exhaustive worldwide priority certificate. Formal Airy expansions and numerical extrapolation are established techniques; the analytic convergence and transfer arguments, rather than the ansatz alone, carry the theorem.

35 The exact finite Jacobi product in the factorial gauge

35.1 Safe startup and rotated coordinates

The prefix sum in the recurrence is constant above the preceding diagonal, not zero. Subtracting neighboring prefix identities gives, away from the separately initialized first entry,

$$b_{n,m} = (2n + m - 2)b_{n-1,m} + \frac{2n + m - 2}{2n + m - 3}b_{n,m-1}.$$

Put $\tilde{b}(p, q) = b_{p+1, q+1}$ for $0 \leq q \leq p$, and set it to zero elsewhere. Then $\tilde{b}(0, 0) = 1$ and, at noninitial allowed indices,

$$\tilde{b}(p, q) = (2p + q + 1)\tilde{b}(p - 1, q) + \frac{2p + q + 1}{2p + q}\tilde{b}(p, q - 1). \quad (170)$$

The only zero denominator would be at the excluded initial entry. The zero term with $q = -1$ causes no difficulty at any other index. Moreover,

$$a_n = \frac{\tilde{b}(n, n)}{3n + 1}, \quad n \geq 0. \quad (171)$$

[Write phase: (170) and (171) are (91) and (92) at $d = 2$, where $\mathcal{W}(p, q) = 2p + q + 1$ and $\mathcal{W}(n, n) = 3n + 1$.] For integers $N \geq h \geq 0$ of the same parity, define

$$\mathcal{Y}(N, h) = \frac{\tilde{b}((N + h)/2, (N - h)/2)}{3^{(N+h)/2}((N + h)/2)!}.$$

Outside this wedge or off parity let $\mathcal{Y} = 0$, and let $\mathcal{Y}(0, 0) = 1$. Equation (170) yields

$$\begin{aligned} \mathcal{Y}(N, h) &= \omega_N^-(h)\mathcal{Y}(N - 1, h - 1) + \omega_N^+(h)\mathcal{Y}(N - 1, h + 1), \\ \omega_N^-(h) &= \frac{3N + h + 2}{3(N + h)}, \quad \omega_N^+(h) = \frac{3N + h + 2}{3N + h}. \end{aligned} \quad (172)$$

This is evaluated only at noninitial allowed indices. Its lower ghost coordinate is $\mathcal{Y}(N, -1) = 0$. *[Write phase: Part III divides by $G_\Gamma(p) = 3^p(4/3)_p$ at $d = 2$ instead of $3^p p!$; the two gauges differ by $(4/3)_p/p! \sim p^{1/3}/\Gamma(4/3)$, which accounts for the different powers and n^{-1} terms below.]*

Define $e_n(k) = \mathcal{Y}(2n, 2k)$ for $0 \leq k \leq n$, and zero-extend vectors to $\ell^2(\mathbb{N}_0)$. Eliminating the odd step gives $e_0 = \delta_0$ and $e_n = \mathbf{M}_n e_{n-1}$, where

$$(\mathbf{M}_n u)(k) = \mathbf{L}_n(k)u(k - 1) + \mathbf{D}_n(k)u(k) + \mathbf{U}_n(k)u(k + 1) \quad (173)$$

with

$$\begin{aligned}
L_n(k) &= \frac{(3n+k+1)(3n+k-1)}{9(n+k)(n+k-1)}, & 1 \leq k \leq n, \\
U_n(k) &= \frac{3n+k+1}{3n+k-1}, & 0 \leq k < n, \\
D_n(k) &= \frac{(3n+k+1)(6n+2k-3)}{3(n+k)(3n+k-2)}, & 1 \leq k \leq n, \\
D_n(0) &= \frac{3n+1}{3n}.
\end{aligned} \tag{174}$$

Missing off-diagonal terms at 0 and n are omitted. We deliberately keep the full bulk diagonal at $k = n$: its extra down-up contribution multiplies $e_{n-1}(n) = 0$, so the actual evolution is unchanged. This choice removes an exceptional old terminal diagonal from the later drift estimate.

35.2 Symmetrization and positivity

Let $(x)_k = x(x+1) \cdots (x+k-1)$, with $(x)_0 = 1$. Define $\Sigma_n(0) = 1$ and

$$\Sigma_n(k)^2 = \frac{(3n-1)_k(3n)_k(3n+2)_k}{9^k(n)_k(n+1)_k(3n+1)_k}, \quad 1 \leq k \leq n. \tag{175}$$

Also $\Sigma_0(0) = 1$. Since $\Sigma_n(k)/\Sigma_n(k-1) = \sqrt{L_n(k)/U_n(k-1)}$, the matrix $J_n = \Sigma_n^{-1}M_n\Sigma_n$ is symmetric, with diagonal D_n and off-diagonal

$$b_n(k) = \sqrt{\frac{(3n+k+1)(3n+k-1)(3n+k)}{9(n+k)(n+k-1)(3n+k-2)}}, \quad 1 \leq k \leq n. \tag{176}$$

It factors as $J_n = K_n K_n^T$, where K_n is the square lower bidiagonal matrix of order $n+1$ whose nonzero entries are positive and obey

$$\begin{aligned}
K_n(k, k)^2 &= \frac{3n+k+1}{3(n+k)}, & 0 \leq k \leq n, \\
K_n(k, k-1)^2 &= \frac{(3n+k+1)(3n+k-1)}{3(n+k)(3n+k-2)}, & 1 \leq k \leq n.
\end{aligned} \tag{177}$$

The adjacent squared products give $b_n(k)^2$, and the diagonal sums give precisely (174), including $D_n(0)$. Thus J_n is positive definite on its finite support and positive semidefinite after zero extension. Let λ_n be its largest eigenvalue and ψ_n its strictly positive unit eigenvector. Set $\psi_0 = \delta_0$.

35.3 A contractive change of gauge between times

For $k < n$, put $R_n(k) = \Sigma_{n-1}(k)/\Sigma_n(k)$; put $R_n(k) = 0$ for $k \geq n$, with $R_1(0) = 1$.

Lemma 35.1 (source 102). *For every $n \geq 1$,*

$$0 \leq R_n \leq I, \quad 0 \leq 1 - R_n(k) \leq Ck^2/n^2 \quad (0 \leq k \leq n). \tag{178}$$

Proof. Write $\varsigma_n(j) = \Sigma_n(j)/\Sigma_n(j-1)$. Its square factors as $f_1 f_2$, where

$$f_1 = \frac{3n+j-1}{3(n+j-1)}, \quad f_2 = \frac{(3n+j-2)(3n+j+1)}{3(n+j)(3n+j)}.$$

Both lie in $(0, 1]$ and increase with real $n \geq 1$. For $\mathfrak{f}_1$ this follows from $\mathfrak{f}_1 = 1 - 2(j-1)/(3(n+j-1))$. For $\mathfrak{f}_2$, set $\mathfrak{z} = 3n + j$, $\mathfrak{n} = n + j$; then

$$\mathfrak{f}_2 = \frac{\mathfrak{z} - 1 - 2/\mathfrak{z}}{3\mathfrak{n}}, \quad \partial_n \mathfrak{f}_2 = \frac{2j + 1 + 6\mathfrak{n}/\mathfrak{z}^2 + 2/\mathfrak{z}}{3\mathfrak{n}^2} > 0.$$

Thus $R_n(k) = \prod_{j=1}^k \varsigma_{n-1}(j)/\varsigma_n(j) \leq 1$ on the common support. For $n-1 \leq t \leq n$ and $j \leq n-1$, the factors are uniformly bounded below and $|\partial_t \log \varsigma_t(j)| \leq C(j+1)/n^2$. Integration and summation give $0 \leq -\log R_n(k) \leq Ck(k+1)/n^2$. For integral $k \geq 1$ this implies (178); $k = 0$ is exact, and $k = n$ is covered by increasing C . $\square$

Set

$$P_n = \prod_{j=1}^n \lambda_j, \quad P_0 = 1, \quad u_n = P_n^{-1} \Sigma_n^{-1} e_n, \quad \mathsf{T}_n = (J_n/\lambda_n) R_n. \quad (179)$$

Then $u_n = \mathsf{T}_n u_{n-1}$ exactly. Positive semidefiniteness makes J_n/λ_n a Hilbert-space contraction; Lemma 35.1 does the same for R_n . All matrices are entrywise nonnegative. Hence

$$\|\mathsf{T}_n\|_{2 \rightarrow 2} \leq 1, \quad \|u_n\|_2 \leq 1. \quad (180)$$

36 A quantitative frozen Airy limit, localization and smoothing

We prove all the spectral estimates needed in the product argument. Let $\lambda_n^{(2)}$ be the second largest eigenvalue, with the spectrum sorted in decreasing order.

Proposition 36.1 (source 102). *As $n \rightarrow \infty$,*

$$\lambda_n = 4 + 4 \cdot 3^{-2/3} z n^{-2/3} + \frac{10}{3} n^{-1} + O(n^{-4/3}), \quad (181)$$

$$0 \leq \lambda_n^{(2)}/\lambda_n \leq 1 - cn^{-2/3}, \quad (182)$$

$$\|k^p \psi_n\|_2 \leq C_p n^{p/3} \quad (p \geq 0 \text{ fixed}), \quad (183)$$

$$\|\psi_n - \psi_{n-1}\|_2 \leq C/n, \quad (184)$$

$$\psi_n(0) = \sqrt{2/3} n^{-1/2} (1 + O(n^{-2/3})). \quad (185)$$

[Write phase: in (182) c is a positive constant, not $2/3$. The terms $4 \cdot 3^{-2/3} z = c_A^2 z$ and $\sqrt{2/3} = \sqrt{c}$ agree with Part III's (109) and (116) at $d = 2$; the n^{-1} term is $10/3$ here and $3c = 2$ there, because the gauges differ.]

36.1 Form convergence and the spectral gap

Put $\varepsilon = n^{-1/3}$ and $x_k = \varepsilon(k + 1/2)$. The continuum operator is

$$\mathcal{H}_A = -\frac{d^2}{dx^2} + \frac{8}{3}x \quad \text{on } L^2(0, \infty), \quad f(0) = 0.$$

Write $c_A = 2 \cdot 3^{-1/3}$. Its eigenvalues are $-c_A^2 z_j$, and its positive unit ground state is

$$f_A(x) = \frac{\sqrt{c_A} \operatorname{Ai}(c_A x + z)}{\operatorname{Ai}'(z)}. \quad (186)$$

The Airy equation and the integral of $[\text{Ai}'(x)^2 - x \text{Ai}(x)^2]' = -\text{Ai}(x)^2$ give $\int_0^\infty f_A^2 = 1$ and $f_A'(0) = c_A^{3/2} = \sqrt{8/3}$.

For this paragraph set $\mathbf{b}_n(0) = \mathbf{b}_n(n+1) = 0$ and $\mathbf{V}_n(k) = 4 - \mathbf{D}_n(k) - \mathbf{b}_n(k) - \mathbf{b}_n(k+1)$. The exact quadratic form is

$$\langle u, (4I - J_n)u \rangle = \sum_{k=0}^{n-1} \mathbf{b}_n(k+1) |u(k+1) - u(k)|^2 + \sum_{k=0}^n \mathbf{V}_n(k) |u(k)|^2. \quad (187)$$

For $1 \leq k \leq n$, let $\vartheta_n(k) = (3n+k)/(3(n+k))$. Factoring the explicit rational coefficients gives, uniformly on existing bulk edges,

$$\mathbf{D}_n(k) = 2\vartheta_n(k) + O(n^{-1}), \quad \mathbf{b}_n(k) = \vartheta_n(k) + O(n^{-1}), \quad \mathbf{b}_n(k+1) = \vartheta_n(k) + O(n^{-1}) \quad (k < n).$$

For example, with $\mathfrak{z} = 3n+k$, $\mathfrak{n} = n+k$,

$$\mathbf{D}_n(k) = \frac{\mathfrak{z}}{3\mathfrak{n}} \frac{(1+1/\mathfrak{z})(2-3/\mathfrak{z})}{1-2/\mathfrak{z}}, \quad \mathbf{b}_n(k)^2 = \left(\frac{\mathfrak{z}}{3\mathfrak{n}}\right)^2 \frac{1-1/\mathfrak{z}^2}{(1-1/\mathfrak{n})(1-2/\mathfrak{z})}.$$

All denominators are bounded away from zero for large n . In particular,

$$\mathbf{V}_n(k) \geq \frac{4k}{3n} - \frac{C}{n} \quad (k \geq 1), \quad \mathbf{V}_n(0) = 2 + O(n^{-1}), \quad (188)$$

and existing $\mathbf{b}_n(k)$ are bounded above and below by positive constants. The omitted edge at n only increases the potential. On fixed scaled compact intervals away from zero,

$$\mathbf{b}_n(k) \longrightarrow 1, \quad \mathbf{V}_n(k)/\varepsilon^2 \longrightarrow \frac{8}{3}x_k$$

uniformly. Adding C/n to the quadratic form makes its potential nonnegative; the scaled added energy is $O(\varepsilon)$ and disappears in the limit.

Here are details of the compactness and boundary argument. A unit vector with form at most $C\varepsilon^2$ satisfies

$$\sum k |u(k)|^2 \leq C/\varepsilon, \quad |u(0)|^2 \leq C\varepsilon^2, \quad \sum |u(k+1) - u(k)|^2 \leq C\varepsilon^2.$$

Interpolate $u(k)/\sqrt{\varepsilon}$ linearly at x_k , from zero at $x=0$ to the first node, and from the last node to zero one mesh interval later; set the function to zero thereafter. The first half-cell has derivative energy $2|u(0)|^2/\varepsilon^2$, supplied by $\mathbf{V}_n(0) \rightarrow 2$. The other cells give a bounded H^1 seminorm on every fixed compact interval. The moving final ramp eventually lies outside each such interval. Its mass, and all mass beyond $x=R$, are $O(1/R)$ by the first-moment estimate.

Local Rellich compactness and this tail bound yield strong global L^2 compactness. Locally the difference from the piecewise-constant embedding is bounded by $C\varepsilon$ times the H^1 seminorm, while both embeddings have small tails. Norms and inner products, hence orthogonality, are preserved in the limit. Weak lower semicontinuity gives the gradient liminf including the boundary half-cell. For the potential liminf, first restrict away from zero to a fixed compact interval, use uniform convergence, then enlarge the interval. The negative $O(n^{-1})$ part vanishes after scaling. Smooth compactly supported Dirichlet trial functions sampled on the grid supply matching upper bounds. The min-max principle on each fixed-dimensional trial or low-energy subspace now gives

$$(4 - \lambda_n^{(j)})/\varepsilon^2 \longrightarrow -c_A^2 z_j \quad \text{for every fixed } j. \quad (189)$$

This identifies the top eigenvalue and proves (182); positive semidefiniteness controls the opposite end of the spectrum. *[Write phase: this is the argument of Part III, Section 24, at $d=2$, written out in more detail; compare Remark 5.1.]*

36.2 A global quasimode and the coefficient ten thirds

Sample $\sqrt{\varepsilon}f_A(x_k)$ for $0 \leq k \leq n$ and normalize it to a unit vector ψ_n^A . Midpoint quadrature and Airy decay give the pre-normalization norm $1 + O(\varepsilon^2)$. In the bulk, direct expansion gives

$$\begin{aligned} D_n(k) &= 2 - \frac{4}{3}x_k\varepsilon^2 + \frac{5}{3}\varepsilon^3 + O(\varepsilon^4(1+x_k^2)), \\ \mathbf{b}_n(k) &= 1 - \frac{2}{3}x_k\varepsilon^2 + \frac{7}{6}\varepsilon^3 + O(\varepsilon^4(1+x_k^2)), \\ \mathbf{b}_n(k+1) &= 1 - \frac{2}{3}x_k\varepsilon^2 + \frac{1}{2}\varepsilon^3 + O(\varepsilon^4(1+x_k^2)). \end{aligned} \tag{190}$$

The remainders are uniform for $k \leq n/2$, since the rescaled denominators stay separated from zero. Also $|\mathbf{b}_n(k+1) - \mathbf{b}_n(k)| \leq C/n$ on the entire bulk. Pair the Taylor expansions of $f_A(x+\varepsilon)$ and $f_A(x-\varepsilon)$. The operator then acts as

$$J_n f_A = 4f_A + \varepsilon^2(f_A'' - \frac{8}{3}xf_A) + \frac{10}{3}\varepsilon^3 f_A + O_{\ell^2}(\varepsilon^4).$$

To justify the global norm, the unsampled remainder is bounded by $C\varepsilon^4$ times a fixed polynomial in $1+x$ times the maximum of $|f_A^{(j)}(y)|$, $j \leq 4$, $|y-x| \leq \varepsilon$, $y \geq 0$. The odd derivative terms carry the coefficient differences $\varepsilon(\mathbf{b}_n(k+1) - \mathbf{b}_n(k))$ and $\varepsilon^3(\mathbf{b}_n(k+1) - \mathbf{b}_n(k))/6$, and fit this bound. The Airy equation and decay bound the envelope by $C(1+x)^{M'}e^{-c'x^{3/2}}$, whose squared sampled norm $\varepsilon \sum$ is uniformly bounded. The region $k > n/2$ contributes exponentially little in n .

At zero, $D_n(0) = 1 + \varepsilon^3/3$ and $\mathbf{b}_n(1) = 1 + \varepsilon^3/6 + O(\varepsilon^6)$. Since $f_A(0) = f_A''(0) = 0$ and $f_A'''(0) = c_A^2 z f_A'(0)$, the cubic term in $f_A(3\varepsilon/2) - 3f_A(\varepsilon/2)$ cancels $c_A^2 z \varepsilon^2 f_A(\varepsilon/2)$. The remaining unsampled boundary residual is $O(\varepsilon^4)$, hence the sampled one is $O(\varepsilon^{9/2})$. The terminal diagonal affects only an exponentially small term. Therefore

$$\left\| (J_n - [4 + c_A^2 z \varepsilon^2 + \frac{10}{3}\varepsilon^3]) \psi_n^A \right\|_2 \leq C\varepsilon^4. \tag{191}$$

For a self-adjoint matrix, expansion in an orthonormal eigenbasis puts an eigenvalue within the residual norm of the proposed value. The gap from (189) identifies it as λ_n and proves (181). The same expansion, projected off the Perron vector, gives

$$\left\| \psi_n - \psi_n^A \right\|_2 \leq C\varepsilon^2. \tag{192}$$

36.3 Uniform moments and the moving upper edge

Let J_∞ have diagonal 1 at zero, diagonal 2 elsewhere, and off-diagonals 1. The coefficient formulas give the entrywise comparison

$$0 \leq J_n \leq (1 + C/n)J_\infty. \tag{193}$$

The factor $1 + C/n$ is necessary: some downward weights exceed one and $D_n(0) = 1 + 1/(3n) > 1$. For the nonnegative unit Perron vector,

$$\langle \psi_n, (4I - J_\infty)\psi_n \rangle \leq 4 - \lambda_n/(1 + C/n) \leq C\varepsilon^2.$$

The left side is $\sum_{k \geq 0} |\psi_n(k+1) - \psi_n(k)|^2 + 2\psi_n(0)^2$. Telescoping $u(k)^2$ from infinity and applying Cauchy-Schwarz bounds $\|u\|_\infty^2$ by $2\|u\|_2 \|\nabla u\|_2$, with ∇ the forward difference. Thus $\|\psi_n\|_\infty \leq C\sqrt{\varepsilon}$.

Let $k_A = \lceil A_0/\varepsilon \rceil$, with fixed A_0 sufficiently large, and choose a small fixed $c_0 > 0$. On $k > k_A$ take $w(k) = e^{-c_0\varepsilon(k-k_A)}$. For this estimate only, continue the explicit off-diagonal formula to the fictitious

positive edge $\mathbf{b}_n(n+1)$. The full row sum is at most $4 - c'k/n + C/n$, while adjacent coefficients differ by $O(n^{-1})$. Replacing neighboring values by $e^{\pm c_0\varepsilon}w(k)$ changes the full row sum by $O(\varepsilon^2)$: the symmetric part is $O(\varepsilon^2)$ and the antisymmetric part is $O(\varepsilon/n)$. Choose A_0 large enough and use (181); then $(J_n w)(k) < \lambda_n w(k)$ for $k > k_A$. Omitting the fictitious positive upper-edge contribution at $k = n$ only improves this inequality.

The finite maximum principle applied to $\psi_n(k)/w(k)$, with its value at k_A as boundary, gives

$$\psi_n(k) \leq C\sqrt{\varepsilon} e^{-c_0\varepsilon(k-k_A)}, \quad k \geq k_A. \quad (194)$$

Indeed a strictly larger interior maximum would contradict the positive supersolution inequality. Together with the supremum bound below k_A , this proves every moment estimate (183). In particular the mass beyond $n/2$ is exponentially small in $n^{2/3}$.

On common coordinates, differentiating the explicit rational factors yields

$$|D_n(k) - D_{n-1}(k)| + |\mathbf{b}_n(k) - \mathbf{b}_{n-1}(k)| \leq C(k+1)/n^2. \quad (195)$$

For instance, $\partial_n \vartheta_n(k) = 2k/(3(n+k)^2)$, while derivatives of its correction factors are $O(n^{-2})$. There is no old exceptional terminal diagonal, by our full-bulk choice. The new edge $\mathbf{b}_n(n)$ is order one, but acts on $\psi_{n-1}(n-1)$, exponentially small by (194); the new diagonal acts on the zero extension. Hence

$$\|(J_n - J_{n-1})\psi_{n-1}\|_2 \leq Cn^{-2} \|(k+1)\psi_{n-1}\|_2 + O(e^{-cn^{2/3}}) \leq Cn^{-5/3}.$$

The continuum spectral limit separates λ_{n-1} from every non-Perron eigenvalue of J_n by $cn^{-2/3}$. Orthogonal projection of this residual proves (184), with the positive signs of both eigenvectors fixing the phase. Lemma 35.1 and the second moment estimate also give

$$\|(I - R_n)\psi_n\|_2 + \|(I - R_n)\psi_{n-1}\|_2 \leq Cn^{-4/3}. \quad (196)$$

36.4 The killed half-line kernel

Put $\mathcal{K} = J_\infty/4$. The functions $\sin((k+1/2)\theta)$ diagonalize J_∞ , with eigenvalue $2 + 2\cos\theta$ and squared integral $\pi/2$ on $(0, \pi)$. Therefore

$$\mathcal{K}^m(0,0) = \frac{2}{\pi} \int_0^\pi \cos^{2m}(\theta/2) \sin^2(\theta/2) d\theta \leq C(m+1)^{-3/2}.$$

The bound follows by splitting away from zero and using Gaussian decay near zero, where the weight is $O(\theta^2)$. By symmetry, $\|\delta_0^\top \mathcal{K}^l\|_2^2 = \mathcal{K}^{2l}(0,0)$ and $\|\delta_0^\top \mathcal{K}^l\|_2 \leq C(l+1)^{-3/4}$. For $0 \leq l \leq n^{2/3}$, positivity, (193), and $R_j \leq I$ imply

$$0 \leq T_n \cdots T_{n-l+1} \leq \left(\prod_{j=n-l+1}^n \frac{4(1+C/j)}{\lambda_j} \right) \mathcal{K}^l \leq C' \mathcal{K}^l.$$

The logarithm of the scalar product is $O(\ln n^{-2/3}) = O(1)$ by (181). This entrywise argument does not assume commutation of the varying matrices. We conclude

$$\|\delta_0^\top T_n \cdots T_{n-l+1}\|_2 \leq C(l+1)^{-3/4}, \quad 0 \leq l \leq n^{2/3}. \quad (197)$$

The identical proof bounds $\delta_0^\top (J_n/\lambda_n)^l$ on that window. *[Write phase: this is (114)–(115) of Part III at $d = 2$, with the sine basis written out.]*

Let $\mathbf{z} = \psi_n - \psi_n^A$. Equations (181) and (191) give $\mathbf{z} = (J_n/\lambda_n)\mathbf{z} + \varrho$, with $\|\varrho\|_2 \leq C\varepsilon^4$ and $\|\mathbf{z}\|_2 \leq C\varepsilon^2$. Iterate $\Delta = \lfloor n^{2/3} \rfloor$ times and use the fixed-matrix version of (197):

$$|\mathbf{z}(0)| \leq C(\Delta + 1)^{-3/4}\varepsilon^2 + C\varepsilon^4 \sum_{l=0}^{\Delta-1} (l+1)^{-3/4} = O(\varepsilon^{7/2}).$$

Since $\psi_n^A(0) = \frac{1}{2}f'_A(0)\varepsilon^{3/2}(1 + O(\varepsilon^2))$, this proves (185) and completes Proposition 36.1.

37 Convergence and strict positivity of the amplitude

Let

$$p_n = \langle \psi_n, u_n \rangle, \quad p_n^\perp = \left\| (I - \psi_n \psi_n^\top) u_n \right\|_2.$$

Project the exact recurrence $u_n = (J_n/\lambda_n)\mathbf{R}_n u_{n-1}$ onto the complement of ψ_n . Use the spectral gap on that complement, contractivity of $\mathbf{R}_n$, the $O(n^{-1})$ drift, and (196). Since $\|u_n\|_2 \leq 1$,

$$p_n^\perp \leq (1 - cn^{-2/3})p_{n-1}^\perp + C/n. \quad (198)$$

A sufficiently large multiple of $n^{-1/3}$ is a supersolution for large n ; absorb the finite initial segment into its constant. Thus $p_n^\perp = O(n^{-1/3})$.

Because $(J_n/\lambda_n)\psi_n = \psi_n$, the scalar increment is exactly

$$\begin{aligned} p_n - p_{n-1} &= p_{n-1} [\langle \psi_n, \mathbf{R}_n \psi_{n-1} \rangle - 1] \\ &\quad + \langle \mathbf{R}_n \psi_n - \psi_{n-1}, (I - \psi_{n-1} \psi_{n-1}^\top) u_{n-1} \rangle. \end{aligned} \quad (199)$$

Without $\mathbf{R}_n$, the first bracket is $-\|\psi_n - \psi_{n-1}\|_2^2/2 = O(n^{-2})$; its $\mathbf{R}_n$ correction is $O(n^{-4/3})$. The second line is $O(n^{-1}p_{n-1}^\perp) = O(n^{-4/3})$. Consequently the scalar increments are absolutely summable. There is $p_\infty \geq 0$ such that

$$|p_n - p_\infty| + \|u_n - p_\infty \psi_n\|_2 \leq Cn^{-1/3}. \quad (200)$$

A separate comparison proves strict positivity. Restart the normalized evolution at a sufficiently large N with initial vector ψ_N . Its norm is one and its perpendicular component initially vanishes. The same supersolution and increment bounds have constants independent of N once N is above a fixed threshold. Summing the increments of this restarted product shows that its limiting Perron coefficient p'_∞ satisfies $|p'_\infty - 1| \leq CN^{-1/3}$. Choose N so this is less than $1/2$. Every coordinate of the actual u_N and ψ_N on $0, \dots, N$ is strictly positive. Their finite support therefore allows $u_N \geq \theta_0 \psi_N$ coordinatewise for some $\theta_0 > 0$. Positive subsequent products preserve the inequality, so $p_\infty \geq \theta_0 p'_\infty > 0$.

To reach the counting coordinate, set $\mathbf{z}_n = u_n - p_\infty \psi_n$. Its recurrence is

$$\mathbf{z}_n = \mathbf{T}_n \mathbf{z}_{n-1} + p_\infty \varrho_n, \quad \varrho_n = \mathbf{T}_n \psi_{n-1} - \psi_n, \quad \|\varrho_n\|_2 \leq C/n.$$

Iterate through the last $\Delta = \lfloor n^{2/3} \rfloor$ steps. Equations (200) and (197) give

$$|\mathbf{z}_n(0)| \leq C\Delta^{-3/4}n^{-1/3} + \frac{C}{n} \sum_{l=0}^{\Delta-1} (l+1)^{-3/4} = O(n^{-5/6}).$$

Together with (185) and $p_\infty > 0$, this yields

$$u_n(0) = p_\infty \sqrt{2/3} n^{-1/2} (1 + O(n^{-1/3})). \quad (201)$$

37.1 The convergent scalar product

The eigenvalue expansion gives

$$\log(\lambda_n/4) = \frac{B}{3}n^{-2/3} + \frac{5}{6}n^{-1} + O(n^{-4/3}),$$

since $3^{-2/3}z = B/3$. The summable remainder proves that

$$C_\lambda = \lim_{n \rightarrow \infty} \frac{P_n}{4^n e^{Bn^{1/3}} n^{5/6}} \in (0, \infty), \quad (202)$$

with relative error $O(n^{-1/3})$. An explicit product is

$$C_\lambda = \frac{\lambda_1}{4} e^{-B} \prod_{j=2}^{\infty} \left[\frac{\lambda_j}{4} e^{-B(j^{1/3} - (j-1)^{1/3})} \left(\frac{j-1}{j} \right)^{5/6} \right]. \quad (203)$$

Each factor is $1 + O(j^{-4/3})$, and finite products telescope to the quotient in (202). The scalar projection has the absolutely convergent representation

$$p_\infty = 1 + \sum_{n=1}^{\infty} \langle R_n \psi_n - \psi_{n-1}, u_{n-1} \rangle, \quad (204)$$

whose summands are $O(n^{-4/3})$. Its positivity follows from the preceding restart argument, not merely from this signed series.

Theorem 37.1 (Canonical amplitude; source 102). *With the finite matrices, eigenvectors, and initial state specified in Section 35,*

$$\gamma_2 = \sqrt{2/27} p_\infty C_\lambda \in (0, \infty), \quad (205)$$

where (203) and (204) converge as stated.

Proof of Theorems 34.1 and 37.1. The exact diagonal identity is

$$a_n = \frac{3^n n!}{3n+1} e_n(0) = \frac{3^n n!}{3n+1} P_n u_n(0),$$

since $\Sigma_n(0) = 1$. Combine (202) and (201). The exponential is $Bn^{1/3} = z(3n)^{1/3}$ and the power is $5/6 - 1/2 - 1 = -2/3$. The leading constant is $\sqrt{2/3} p_\infty C_\lambda/3$, which is (205). $\square$

[Write phase: (205), Part II's (67) and Part III's (124) at $d = 2$ are three convergent representations of one number, through three gauges; none is evaluated with certified digits. Source 102's restart comparison, like Part III's, needs no lower Θ bound.]

38 The all order construction and its transfer

The next results establish genuine arbitrary-order asymptotics, not just finitely verified identities. The infinite series need not converge.

Theorem 38.1 (All-order expansion; source 102). *There are compatible real coefficients c_j , $c_0 = 1$, such that for every fixed $J \geq 0$,*

$$\frac{a_n}{\gamma_2 n! 12^n e^{Bn^{1/3}} n^{-2/3}} = \sum_{j=0}^J c_j n^{-j/3} + O(n^{-(J+1)/3}). \quad (206)$$

Equivalently, compatible L_j satisfy

$$\log \frac{a_n}{\gamma_2 n! 12^n e^{Bn^{1/3}} n^{-2/3}} = \sum_{j=1}^J L_j n^{-j/3} + O(n^{-(J+1)/3}). \quad (207)$$

The coefficients lie in $\mathbb{Q}[z, 3^{1/3}]$ and are generated by the polynomial recursion below. The amplitude is the same γ_2 at every depth.

[Write phase: this is Theorem 14.1; Part II's coefficient field $\mathbb{Q}[B]$, with $B = 3^{1/3}z$, is contained in $\mathbb{Q}[z, 3^{1/3}]$ and is the sharper statement.]

38.1 An invertible polynomial equation at each depth

Return to the one-step recurrence (172). Put

$$t = N^{-1/3}, \quad x = (h+1)t, \quad c = 2/3, \quad \ell = c^{2/3}z, \quad F(x) = \text{Ai}(z + c^{1/3}x).$$

Then $F'' = (cx + \ell)F$, $F(0) = 0$, and $F'(0) \neq 0$. Primes here and below mean derivatives in x . Let $t_1 = t(1 - t^3)^{-1/3}$. The exact weights become

$$U(t, x) = \frac{3 + xt^2 + t^3}{3 + 3xt^2 - 3t^3}, \quad V(t, x) = \frac{3 + xt^2 + t^3}{3 + xt^2 - t^3}.$$

Seek polynomial pairs $P_j, Q_j \in \mathbb{Q}[\ell, x]$ and scalars $\sigma_j \in \mathbb{Q}[\ell]$ with the gauges (127) and

$$s(t) = 2 + \ell t^2 + \sum_{j \geq 3} \sigma_j t^j, \quad \mathfrak{F}(t, x) = \sum_{j \geq 0} t^j [P_j(x)F(x) + Q_j(x)F'(x)],$$

satisfying formally the one-step equation (128) with these weights. The orders t^0, t^1 cancel, and the t^2 equation is the Airy equation. The new profile and scalar at order t^k solve (129) with $c = 2/3$, uniquely, by the algorithm (130); on polynomials of degree at most m the operator $\mathcal{T}$ has diagonal $-2c(2j+1)$, and its inverse preserves $\mathbb{Q}[\ell, x]$. *[Write phase: source 102 prints the system, the operator and the algorithm in full; they are literally Part III's displays at $d = 2$ and are printed once, there.]* This proves existence, uniqueness, and compatibility to every formal depth.

Each profile $f_j = P_j F + Q_j F'$ vanishes exactly at $x = 0$; for $j \geq 1$ its derivative there also vanishes. For example at order three, $R_0 = 2$, $R_1 = 4x/3$, hence $\tilde{q} = -2$, $\sigma_3 = 4/3$, $Q_1 = 0$, $P_1 = -x^2/3$. The next order gives $\sigma_4 = \ell^2/12$. These are outputs of the recurrence, not additional assumptions. *[Write phase: here $R_0 = 2$ and $\sigma_3 = 4/3$, whereas Part III at $d = 2$ has $R_0 = 2c = 4/3$ and $\sigma_3 = 2\eta_f = 2/3$: the factorial gauge shifts the weights. $\sigma_4 = \ell^2/12$ in both.]*

38.2 A compatible scalar carrier and the endpoint normalization

Define h_j by the formal difference identity

$$\log \frac{s(t)}{2} = \frac{3\ell}{2t} [1 - (1 - t^3)^{1/3}] - \frac{2}{3} \log(1 - t^3) + \sum_{j \geq 1} h_j t^j [1 - (1 - t^3)^{-j/3}]. \quad (208)$$

Its coefficients at t^2, t^3 agree with $\sigma_2 = \ell$, $\sigma_3 = 4/3$. The new h_j first has coefficient $-j/3$ at order t^{j+3} , so is determined uniquely. No additive integration constant is introduced; for example $h_1 = \ell^2/4$.

For fixed $M \geq 1$, set

$$\mathcal{H}_N^{[M]} = 2^N e^{(3\ell/2)N^{1/3}} N^{2/3} \exp \left(\sum_{j=1}^M h_j N^{-j/3} \right), \quad (209)$$

$$\mathfrak{F}^{[M]}(t, x) = \sum_{j=0}^M t^j f_j(x).$$

Every carrier has leading multiplicative constant exactly one. Its exact consecutive ratio has the common expansion s through at least order t^{M+3} . Coefficients in the carrier may use profiles beyond depth M ; their existence has already been proved. This avoids inconsistent finite-product initialization constants.

At even time let $\varepsilon = n^{-1/3}$, $X = (k + 1/2)\varepsilon$, so $N = 2n$, $h = 2k$, $t = 2^{-1/3}\varepsilon$, $x = 2^{2/3}X$. Define

$$E_n^{[M]}(k) = \frac{3^{4/3}}{2^{2/3} \text{Ai}'(z)} \mathcal{H}_{2n}^{[M]} \mathfrak{F}^{[M]}(2^{-1/3}\varepsilon, 2^{2/3}X). \quad (210)$$

The leading bulk factor is

$$\frac{3^{4/3}}{\text{Ai}'(z)} 4^n e^{Bn^{1/3}} n^{2/3} \text{Ai}(z + 2 \cdot 3^{-1/3}X).$$

At $k = 0$, the Airy factor equals $\text{Ai}'(z)3^{-1/3}\varepsilon(1 + O(\varepsilon^2))$. Hence

$$E_n^{[M]}(0) = 3 \cdot 4^n e^{Bn^{1/3}} n^{1/3} (1 + O(\varepsilon)), \quad (211)$$

with leading coefficient 3 at every depth. The canonical endpoint correction series is therefore the expansion of

$$\frac{E_n^{[M]}(0)}{3 \cdot 4^n e^{Bn^{1/3}} n^{1/3}} \left(1 + \frac{1}{3n} \right)^{-1}. \quad (212)$$

The last exact factor comes from (171), contributes from order ε^3 , and must be retained. All its coefficients stabilize with increasing depth.

The coefficient field. Here is why cube roots of 2 disappear from the endpoint coefficients. Write $\mathfrak{B}_\ell(x)$ for the solution of $\mathfrak{B}'' = (cx + \ell)\mathfrak{B}$ normalized by $\mathfrak{B}_\ell(0) = 0$ and $\mathfrak{B}'_\ell(0) = 1$. If $\omega^3 = 1$, its power-series differential equation gives $\mathfrak{B}_{\omega\ell}(\omega x) = \omega \mathfrak{B}_\ell(x)$. The simultaneous substitution

$(t, x, \ell) \mapsto (\omega t, \omega x, \omega \ell)$ preserves the exact weights, shifts, and recurrence. Uniqueness of the polynomial recursion and gauges therefore gives, for each monomial $\ell^i x^s$, the congruences

$$i + s + j \equiv 0 \pmod{3} \quad \text{in } P_j, \quad i + s + j \equiv 1 \pmod{3} \quad \text{in } Q_j.$$

Likewise $i + j \equiv 0 \pmod{3}$ in σ_j and h_j . Evaluation at $x = t$, division by the leading endpoint t , and the factor $(1 + (2/3)t^3)^{-1}$ preserve invariance. Thus a term $\ell^i t^j$ in the endpoint correction has $i + j \equiv 0 \pmod{3}$. Under $t = 2^{-1/3}\varepsilon$ and $\ell = (2/3)^{2/3}z$, it becomes $2^{(2i-j)/3}3^{-2i/3}z^i\varepsilon^j$. Its exponent of 2 is integral, proving the asserted field $\mathbb{Q}[z, 3^{1/3}]$.

38.3 Uniform residuals and all order transfer

Fix M and $C_0 > 0$. In the exact one-step recurrence, the residual of $\mathcal{H}_N^{[M]} \mathfrak{F}^{[M]}(N^{-1/3}, (h+1)N^{-1/3})$, divided by $\mathcal{H}_N^{[M]}$, is bounded by

$$Ct^{M+3}(1+x)^{D'}e^{-c_0x^{3/2}}, \quad 0 \leq h \leq C_0\sqrt{N}, \quad (213)$$

for large N . Constants may depend on the fixed depth and C_0 ; $c_0 > 0$ here is unrelated to the slope $c = 2/3$. The proof is that of (141) in Part III, with these weights: $x = O(t^{-1/2})$, the rescaled denominators stay separated from zero, finite rational Taylor expansions and polynomial–Airy envelopes control the remainders, and formal cancellation through order t^{M+2} does the rest. This argument uses no convergence of an infinite expansion.

At $h = 0$, the ghost predecessor has height -1 , hence $x = 0$, where every finite profile vanishes exactly before expansion. The bottom boundary is therefore exact. Eliminating the odd step gives the first row $D_n(0) = (3n+1)/(3n)$ and $U_n(0) = (3n+1)/(3n-1)$ already used above. The even-time residual is a bounded linear combination of the one-step residuals because the weights and carrier ratios are bounded in the cutoff region.

Choose fixed smooth χ with $0 \leq \chi \leq 1$, $\chi = 1$ on $[0, 1/2]$, and $\chi = 0$ on $[1, \infty)$. Replace $E_n^{[M]}$ by $\tilde{E}_n^{[M]}(k) = \chi(k/\sqrt{n})E_n^{[M]}(k)$. This leaves the endpoint unchanged. In the transition region X is comparable to $n^{1/6}$. Every fixed profile and needed neighbor is bounded by a polynomial in n times $e^{-cn^{1/4}}$, after removing the scalar carrier. Every cutoff error, including change of cutoff from $n-1$ to n , is thus smaller than every power. Application of M_n extends the residual support at most to $k \leq \sqrt{n} + 1$.

The exact edge ratios of Σ_n give $-\log \Sigma_n(k) = O(k^2/n + k/n) = O(1)$ on this support, so Σ_n^{-1} is uniformly bounded there. Write $\Theta_n = 4^n e^{Bn^{1/3}} n^{2/3}$. The squared polynomial–Airy envelope has lattice sum $O(\varepsilon^{-1})$, whence

$$\left\| \tilde{E}_n^{[M]} - M_n \tilde{E}_{n-1}^{[M]} \right\|_2 = O(\Theta_n \varepsilon^{M+5/2}).$$

Since $\Theta_n/P_n = O(\varepsilon^{1/2})$ by (202), the normalized defect obeys

$$\left\| P_n^{-1} \Sigma_n^{-1} (\tilde{E}_n^{[M]} - M_n \tilde{E}_{n-1}^{[M]}) \right\|_2 = O(\varepsilon^{M+3}) = O(n^{-(M+3)/3}). \quad (214)$$

Increasing the depth gives an arbitrarily large forcing exponent.

Set $V_n^{[M]} = P_n^{-1} \Sigma_n^{-1} \tilde{E}_n^{[M]}$. Comparing the leading bulk factor in (210) with the sampled unit Airy profile (186) gives

$$\left\| V_n^{[M]} - \frac{\sqrt{27/2}}{C_\lambda} \psi_n \right\|_2 = O(\varepsilon). \quad (215)$$

In detail, the leading constant is $3^{4/3}/(\sqrt{c_A}C_\lambda) = \sqrt{27/2}/C_\lambda$. The finite carrier differs from its leading carrier by $1 + O(\varepsilon)$, the finitely many profile corrections have weighted ℓ^2 norm $O(\varepsilon)$, and the cutoff loses exponentially little. On its support, $\Sigma_n^{-1} - 1 = O(\varepsilon(1 + X^2))$ in this weighted norm; this follows from the logarithmic edge-ratio estimate and the polynomial–Airy decay. Equation (192) contributes $O(\varepsilon^2)$ and (202) contributes $O(\varepsilon)$. These prove (215).

By (205), $\gamma_2\sqrt{27/2}/C_\lambda = p_\infty$. Consequently $\mathbf{z}_n = u_n - \gamma_2 V_n^{[M]}$ has norm $O(n^{-1/3})$, its Perron projection tends to zero, and it obeys the exact signed recurrence

$$\mathbf{z}_n = \mathbf{T}_n \mathbf{z}_{n-1} + \varrho_n, \quad \|\varrho_n\|_2 = O(n^{-p}), \quad p = (M + 3)/3. \quad (216)$$

Neither the residual nor the approximate profiles need be positive.

Lemma 38.2 (Forced product; source 102). *For the exact operators $\mathbf{T}_n$ above, suppose (216) holds with $p > 4/3$, $\|\mathbf{z}_n\|_2 = O(n^{-1/3})$, and $\langle \psi_n, \mathbf{z}_n \rangle \rightarrow 0$. Then*

$$\|\mathbf{z}_n\|_2 = O(n^{-(p-1)}), \quad \mathbf{z}_n(0) = O(n^{-1/2-(p-1)}).$$

Proof. This is Lemma 27.1 at $d = 2$, with the operators of this Part, and its proof is literally the same: project as in (198) and (199), now with forcing; improve the perpendicular bound by the stability fact $x_n \leq (1 - cn^{-2/3})x_{n-1} + O(n^{-r}) \Rightarrow x_n = O(n^{-r+2/3})$; sum the scalar increments from $n + 1$ to infinity using the zero limit; repeat finitely many times to reach $p - 1$; and finish at the endpoint with (197) over $\Delta = \lfloor n^{2/3} \rfloor$ steps, which gives $O(n^{-p+1/2}) + O(n^{-p+1/6})$. Source 102 adds three observations: no sign assumption enters; no logarithmic boundary is encountered, since the exponent improves to $\min(q + 1/3, p - 1)$ at each round; and the factor $1 + C/n$ in (193) is already included in the boundary estimate and introduces no additional loss. $\square$

Proof of Theorem 38.1. Take $M \geq 2$ so $p > 4/3$. Lemma 38.2 applies to the normalized error identified above. Since $\Sigma_n(0) = 1$ and

$$\frac{P_n}{3 \cdot 4^n e^{Bn^{1/3}} n^{1/3}} = O(n^{1/2}),$$

its endpoint conclusion gives

$$\frac{e_n(0) - \gamma_2 \tilde{E}_n^{[M]}(0)}{3 \cdot 4^n e^{Bn^{1/3}} n^{1/3}} = O(n^{-(p-1)}) = O(\varepsilon^M).$$

The common exact factor $3^n n!/(3n + 1)$ transfers this to a_n without changing the relative error. Given J , take $M \geq \max(2, J + 1)$ and expand (212) through order ε^J . Its coefficients are independent of the deeper truncation, its tail is $O(\varepsilon^{J+1})$, and the transferred error is no larger. This proves (206); formal logarithm gives (207). $\square$

The expansion is Poincaré asymptotics in one dominant Airy sector. It does not establish convergence of the infinite series or contributions smaller than every algebraic power.

39 Exact coefficients, the total first correction and reproducible checks

The first six logarithmic coefficients from (208) and (212) are

$$\begin{aligned} L_1 &= \frac{3^{2/3}z^2}{18} = \frac{B^2}{18}, & L_2 &= 0, \\ L_3 &= -\frac{1}{9}, & L_4 &= -\frac{3^{2/3}z^2}{162} = -\frac{B^2}{162}, \\ L_5 &= \frac{3^{1/3}z(36 - z^3)}{2916} = \frac{B(108 - B^3)}{8748}, & L_6 &= \frac{2z^3}{729} - \frac{1}{162} = \frac{2B^3}{2187} - \frac{1}{162}. \end{aligned} \quad (217)$$

For example $c_1 = L_1 = z^2/(2 \cdot 3^{4/3})$. The multiplicative expansion begins

$$1 + \frac{3^{2/3}z^2}{18}n^{-1/3} + \frac{3^{1/3}z^4}{216}n^{-2/3} + \left(\frac{z^6}{3888} - \frac{1}{9}\right)n^{-1} + O(n^{-4/3}). \quad (218)$$

All remaining c_j follow by formal exponentiation of $\sum L_j \varepsilon^j$.

Remark 39.1 (The fifth and sixth coefficients; write phase). $L_1, \dots, L_4$ and c_1, c_2, c_3 are Part II's (45): $c_2 = 3^{1/3}z^4/216 = B^4/648$ and $c_3 = z^6/3888 - 1/9 = B^6/34992 - 1/9$. The B -forms in (217) are write-phase conversions ($B = 3^{1/3}z$, $z^3 = B^3/3$), checked symbolically. L_5 and L_6 are new: Part II's shipped coefficient file `data/69-binary-coef-coefficients.json` stops at L_4 . At the intake (5 October 2026) Part II's own generator `code/69-binary-coef-derive_coefficients.py`, written independently of source 102, was rerun on a copy with `-order 9` (392 seconds); its output `logforward_n` ends with $3^{1/3}z(36 - z^3)/2916$ and $2z^3/729 - 1/162$, exactly L_5 and L_6 . So two independent symbolic computations agree. Numerically $L_5 \approx -0.056413$ and $L_6 \approx -0.041240$; direct fits of the exact sequence for $n \leq 1500$ cannot resolve them, which is why the confirmation is symbolic.

39.1 The first correction for total networks

The first correction transfers to the *total* network count. Put $\gamma_T = \gamma_2 \sqrt{e}/12$. The exact maximal-count identity, Theorem 38.1, and Proposition 34.3 give

$$\text{TC}_n = \gamma_T (n!)^2 12^n e^{Bn^{1/3}} n^{-5/3} \left(1 + L_1 n^{-1/3} + O(n^{-2/3})\right). \quad (219)$$

Indeed shifting n to $n - 1$ first changes the stretched exponential at order $n^{-2/3}$; the factorial, polynomial, and L_1 shifts contribute no new term at order $n^{-1/3}$. The ratio error is also $O(n^{-2/3})$. Its bound does not identify the next total-count coefficient, even though the maximal-count coefficients are available to all orders. [5 October 2026, batch 102: Part II identifies it. Its Theorem 16.2 gives every finite order of TC_n , through the identity (T1); by (46) the logarithmic coefficient at $n^{-2/3}$ is $-23B/72$, so the multiplicative second coefficient in (219) is $B^4/648 - 23B/72$. Source 102 could not know this; see Section 41.]

As a separate shift check, the normalized successive ratio has

$$\begin{aligned} \frac{a_n}{12na_{n-1}} &= 1 + \frac{B}{3}n^{-2/3} - \frac{2}{3}n^{-1} + \frac{B^2}{27}n^{-4/3} - \frac{B}{9}n^{-5/3} \\ &\quad + \frac{2B^2}{243}n^{-7/3} - \frac{B(B^3 + 216)}{6561}n^{-8/3} + \frac{2B^3}{2187}n^{-3} + O(n^{-10/3}). \end{aligned} \quad (220)$$

The coefficients at $n^{-1/3}$ and n^{-2} vanish. To justify (220), apply the all-order theorem at a sufficiently deeper truncation before shifting and dividing. Unknown log coefficients of orders seven, eight, and nine enter the difference only at orders ten, eleven, and twelve. No arbitrary big- O remainder has been differentiated. *[Write phase: source 102 writes the coefficients in z : $3^{1/3}z/3$, $3^{2/3}z^2/27$, $-3^{1/3}z/9$, $2 \cdot 3^{2/3}z^2/243$, $-3^{1/3}z(z^3 + 72)/2187$ and $2z^3/729$; the B -forms were checked symbolically, and the intake checked that (220) follows exactly from $L_1, \dots, L_6$.]*

39.2 What is verified by finite computation

The accompanying source package separates exact identities from analytic theorems. Rational finite-index checks compare the original row recurrence, the rotated one-step recurrence, the two-step evolution, the diagonal shift, the symmetrizer, and the positive bidiagonal factorization. These detect algebra and startup errors; they do not prove the spectral asymptotics.

The exact symbolic checks reconstruct the formal shifted polynomial–Airy recurrence through t^9 , clear the rational denominators, and test both Airy coordinates. They check every profile’s boundary and derivative gauge. A separate endpoint reconstruction solves the normalized Airy differential equation as a formal power series, evaluates the profiles at $x = t$, integrates the scalar ratio by (208), includes the factor $(1 + (2/3)t^3)^{-1}$, and matches (217), the multiplicative coefficients through order six, and (220) through order nine. These are exact finite consistency certificates. The arbitrary-depth assertion rests on the polynomial inversion and analytic transfer proved above.

All verification gates in the release checker use explicit exceptions rather than Python assertions, and the full checks are executed under ordinary Python and under `python -O`. Deliberately altered input values and coefficient tables are required to fail in both modes. The package includes the exact inputs, build and replay instructions, source provenance, and the verification summary; it does not require access to the development workspace.

[Write phase: in the repository the checker, harness, reference derivation snapshots and diagnostic script are `code/102-tableaux-*`; inputs, results, replays and provenance are `data/102-tableaux-x-*`; the package documentation and review are `102-tableaux-checks-README.md`, `102-tableaux-x-checks-RESULTS.md`, `102-tableaux-diag-README.md`, `102-tableaux-SOURCES.md` and `102-tableaux-qa-REVIEW.md`. The manuscript, its PDF, the release verifier and manifest, the generic requirements file and an in-package duplicate of `endpoint_9.json` are not shipped. The intake reran everything on copies of the delivered layout on 5 October 2026: the release verifier passed; the suite harness kills each run after 300 seconds, so its first run failed by timeout on the loaded laptop (302 seconds; the delivered record took 136 seconds in total), and a rerun with the machine quieter passed, with both baselines (4,999 gates each) and 44 of 44 rejections, in 690 seconds; direct runs of the checker passed in both modes in 114 seconds each; and `reproduce_dp.py` 100 reproduced the delivered replay. Regenerated results equal the delivered ones apart from timestamps, durations and the Python version.]

39.3 Numerical diagnostics without certified amplitude digits

For context only, exact integer dynamic programming through $n = 3000$, followed by high-precision floating-point evaluation, gives the following diagnostics. The second column is the uncorrected normalized count in (206); the third removes the six displayed logarithmic corrections.

n	Uncorrected normalization	After six log corrections
100	2.3145214818	2.0226159639
500	2.1896829266	2.0226419866
1000	2.1542867259	2.0226420109
2000	2.1265206841	2.0226420146

These values suggest a limit near 2.022642, but no decimal of γ_2 is certified here. Exact integer recurrence evaluation does not turn an asymptotic remainder with unspecified constants into an interval bound. The table is a consistency diagnostic and is unnecessary for the proofs. The optional reproduction script and saved diagnostic values make its scope explicit. *[Write phase: the uncorrected column agrees with Part I's diagnostics at $n = 500, 1000, 2000$, and the corrected values are consistent with Part II's four-correction diagnostics of Section 19 (2.0226420138 at $n = 50000$). The saved values are `data/102-tableaux-diag-saved_dp_3000.json`.]*

40 Inverse asymptotics

There are two different questions: recovering the index from an exact sequence value, and finding a floor or ceiling index for an arbitrary threshold. An asymptotic approximation alone need not resolve a threshold arbitrarily close to a count. *[Write phase: this section is Part III's Section 29 at $d = 2$, with its proofs; it is printed with its binary constants because source 102 adds a first-correction total inverse. Remark 11.1 applies, and Part II's Section 17 is stronger for totals.]*

40.1 An eventual inverse without knowing the amplitude

For large positive y , define

$$x_0(y) = \frac{y}{W_0(12y/e)}, \quad L_0(y) = \log(12x_0(y)), \quad (221)$$

$$\hat{x}(y) = x_0(y) - \frac{z(3x_0(y))^{1/3} - \frac{1}{6} \log x_0(y) + \frac{1}{2} \log(2\pi)}{L_0(y)}, \quad (222)$$

where W_0 is the principal positive real branch of Lambert's W function.

Proposition 40.1 (source 102). *The known two-sided bound (43) alone implies*

$$\hat{x}(\log a_n) = n + O(1/\log n).$$

Thus nearest-integer rounding recovers n for all sufficiently large exact values a_n .

Proof. This is Proposition 29.1 with $(p, r) = (1, -2/3)$, $\Gamma = 12$: Stirling's formula yields $\log a_n = \Phi(n) + O(1)$ with $\Phi(x) = x(\log(12x) - 1) + z(3x)^{1/3} - \frac{1}{6} \log x + \frac{1}{2} \log(2\pi)$, and the Taylor and mean-value steps there apply verbatim. The error tends to zero, so rounding eventually succeeds. $\square$

There is no effective cutoff in this assertion. The omitted $\log \gamma_2$ is a bounded constant and explains why its error is only $O(1/\log n)$.

For $n \geq 2$, the original recurrence gives $a_n \geq b_{n,n} = (3n - 2)a_{n-1} > a_{n-1}$. Also differentiating (222) gives $\hat{x}'(y) \sim 1/\log x_0 > 0$. Therefore for a sufficiently large threshold Y , if

$$k_{\leq}(Y) = \max\{n : a_n \leq Y\}, \quad j_0 = \text{round}(\hat{x}(\log Y)),$$

then $j_0 \in \{k_{\leq}(Y), k_{\leq}(Y) + 1\}$, and

$$k_{\leq}(Y) = \begin{cases} j_0, & a_{j_0} \leq Y, \\ j_0 - 1, & a_{j_0} > Y. \end{cases} \quad (223)$$

Indeed $\hat{x}(\log Y)$ lies between $k_{\leq}(Y) + o(1)$ and $k_{\leq}(Y) + 1 + o(1)$. One exact or rigorously certified comparison selects the floor inverse. For the least index with $a_n \geq Y$, return j_0 if $a_{j_0} \geq Y$, and $j_0 + 1$ otherwise. Equality belongs to the first branch. Unverified floating-point comparison near equality is insufficient.

40.2 Arbitrary order inverses for the tableau and maximal counts

With the true canonically defined amplitude, put

$$\mathcal{F}_J(x) = \log \Gamma(x + 1) + x \log 12 + z(3x)^{1/3} - \frac{2}{3} \log x + \log \gamma_2 + \sum_{j=1}^J L_j x^{-j/3}. \quad (224)$$

Its derivative is eventually positive and asymptotic to $\log x$. Let $x_J(y)$ be the unique sufficiently large root of $\mathcal{F}_J(x) = y$. Taking logarithms in the all-order theorem and using the mean value theorem gives

$$x_J(\log a_n) = n + O\left(\frac{n^{-(J+1)/3}}{\log n}\right). \quad (225)$$

This is a mathematical inverse statement with the true constant. Replacing that constant by uncertified decimal digits does not give a certified inverse. A prescribed finite number of Newton steps, $m_J = \lceil \log_2(J + 4) \rceil - 1$, started at (221), suffices asymptotically, by the quadratic estimate of Part III's (158); here

$$\mathcal{F}'_J(x) = \Psi(x + 1) + \log 12 + \frac{B}{3} x^{-2/3} - \frac{2}{3x} - \sum_{j=1}^J \frac{j}{3} L_j x^{-j/3-1},$$

and the elementary Stirling replacement (159), with the same R , keeps the inverse order and Newton count.

For the maximal-reticulation sequence, the exact identity in (166) gives the model $\mathcal{M}_J(x) = \log \Gamma(x + 1) + \mathcal{F}_J(x - 1)$. Its root at $y = \log \text{TC}_{n,n-1}$ differs from n by $O(n^{-(J+1)/3}/\log n)$, since $\mathcal{M}'_J(x) \sim 2 \log x$. The appropriate initial Lambert approximation is

$$x_0^{\text{net}}(y) = \frac{y}{2W_0(\sqrt{12}y/(2e))}. \quad (226)$$

The same fixed number m_J of Newton steps has the desired order: the initial error remains $O(x_0^{1/3}/\log x_0)$ and the same quadratic estimate applies.

40.3 A first correction inverse for total networks

Let $\gamma_T = \gamma_2 \sqrt{e}/12$ and define

$$\begin{aligned} \mathcal{T}_0(x) &= 2 \log \Gamma(x + 1) + x \log 12 + z(3x)^{1/3} - \frac{5}{3} \log x + \log \gamma_T, \\ \mathcal{T}_1(x) &= \mathcal{T}_0(x) + L_1 x^{-1/3}. \end{aligned} \quad (227)$$

Equation (219) gives $\log \text{TC}_n = \mathcal{T}_1(n) + O(n^{-2/3})$. The derivative is asymptotic to $2 \log n$, so the unique sufficiently large inverse satisfies

$$\mathcal{T}_1^{-1}(\log \text{TC}_n) = n + O\left(\frac{n^{-2/3}}{\log n}\right). \quad (228)$$

With $y = \log(\text{target})$, starting at (226), two exact Newton steps for $\mathcal{T}_1(x) = y$ suffice for this order. The initial error is $O(x_0^{1/3}/\log x_0)$, while the first and second Newton root errors are respectively $O(x_0^{-1/3}/\log^3 x_0)$ and $O(x_0^{-5/3}/\log^7 x_0)$, by the same quadratic estimate as above. The second is smaller than the model error in (228).

For comparison, using only the leading model gives $\mathcal{T}_0^{-1}(\log \text{TC}_n) = n + O(n^{-1/3}/\log n)$. An explicit simplified one-step expression, with $x_0 = x_0^{\text{net}}(y)$, is

$$\hat{x}_T(y) = x_0 - \frac{z(3x_0)^{1/3} - \frac{2}{3} \log x_0 + \log(2\pi\gamma_T)}{2 \log(\sqrt{12} x_0)}.$$

Taylor's theorem gives $\hat{x}_T(\log \text{TC}_n) = n + O(n^{-1/3}/\log n)$. Its analytic root error is $O(x_0^{-1/3}/\log^2 x_0)$; omitting L_1 limits its model accuracy. It is not an implementation of the sharper two-step estimate. If $\log \gamma_T$ is omitted as well, only $O(1/\log n)$ is asserted.

The leading equivalent implies $\text{TC}_{n+1}/\text{TC}_n \sim 12n^2$, hence eventual monotonicity. The same two-candidate threshold rule applies, but still needs an exact or certified comparison and an eventual cutoff. The constants must be evaluated accurately enough for any numerical use. No all-order total-count inverse is claimed; more terms require more than the ratio bound proved here. *[Write phase: Part II's (77) and (78) already include the $n^{-2/3}$ and n^{-1} total corrections, and its transfer gives every order; (228) is the identity-free counterpart. $\hat{x}_T$ is exactly the truncation displayed in Part II's (78): there $x = x_0^{\text{net}}(y)$, $s = 2 \log(2\sqrt{3}x)$ and $C_T = \log(2\pi\gamma_T)$.]*

41 Further questions and research (Part IV)

Source 102's closing section, "Further questions and boundaries of the result", is printed first, with dated notes; the write-phase items follow. No claim of source 102 was found to be wrong.

Certified amplitude computation. The product and scalar series in (203) and (204) give a route to numerical enclosure. Effective constants in the spectral, drift, projection, and smoothing estimates would be needed to bound the tails explicitly. Interval computations of finite matrices could then certify digits of γ_2 and effective inverse cutoffs. The present proof supplies existence and qualitative rates, not such numerical enclosures. *[Write phase: still open; Part I's question 2 (Section 12).]*

Corrections for total networks. The quantitative ratio and total-variation bound transfer the first correction. Identifying the total-count coefficient at order $n^{-2/3}$, and then an all-order total-count expansion, requires sharper information on the free-root-child counts $F_{n,r}$ or their aggregate weights. A big- O bound for their contribution cannot determine those coefficients. *[5 October 2026, batch 102: answered in the repository. Part II's Theorem 16.2 gives the all-order total-count expansion, and (46) gives the coefficient asked for: $-23B/72$ in logarithmic form, $B^4/648 - 23B/72$ in the multiplicative form of (219). It uses the Pons–Batle identity (T1), proved in [4], which source*

102 cites but does not use, rather than information on $F_{n,r}$. Whether the free-root-child route of Proposition 34.3 can itself be pushed to all orders remains open; see item 1 below.]

Beyond all algebraic orders. The Airy spectrum suggests additional decaying components, but a transseries would require their separate amplitudes, interactions, and uniform remainders. Neither the all-depth polynomial algorithm nor a finite numerical residual proves an exponentially small sector.

Other network regimes. Fixed, sparse, and maximal reticulation counts occupy different asymptotic regimes. The exact all-order statement here is for the maximal diagonal through its word/tableau normalization. A uniform bridge to sparse-reticulation results would need a separate two-parameter analysis.

Scope of the proof and source package. The leading proof includes its own form compactness, localization, dimension-change treatment, boundary kernel, positivity argument, and amplitude product. The all-order proof includes its own forced-product lemma and does not require a companion relaxed-tree article. Exact finite checks are reproducibility safeguards, not a machine verification of the analytic proof. The finite literature inspection supports precise attribution of the prior scale and ratio theorem, but is not a claim of exhaustive novelty. *[Write phase: the “companion relaxed-tree article” is presumably the relaxed-tree manuscript of the same delivery series, bundle Report 100, placed as Part V of the sibling report ../a082161-airy-amplitudes by 6ab1f1979; the two share the lemma family of Parts I–IV.]*

Write-phase items.

1. *An identity-free all-order transfer.* Proposition 34.3 obtains $\mathcal{R}_{n-1} = \sqrt{e}(1 + O(n^{-2/3}))$ from Lemmas 3.17–3.20 of [3] alone. Whether expansions of the free-root-child ratios f_k could give the all-order transfer of Theorem 16.2 without the identity (T1) is open; such a route might extend to $d \geq 3$, where no analogue of (T1) is known (Part III, Section 32, item 2). (Write phase.)
2. *Certified digits.* The diagnostics 2.0226420146 (here) and 2.02264201 (Part II) are uncertified; see the first paragraph. (Sources 102 and 69.)
3. *The next logarithmic coefficients.* L_7, L_8, L_9 enter (220) only beyond n^{-3} ; they are computable by the same algorithm (Part II’s generator at higher order) but were not computed. (Write phase.)

Bibliography note for Parts III–IV. *[Write phase: source 104’s entries cfllyw, fyz, walls26, oeis213863 and source 102’s fyz, bw, cfllyw, walls26, oeis are the existing entries [3], [1], [4], [6] and [2]; their remaining entries are added at the end of the bibliography. Source 104’s entry for the AofA paper and source 102’s agree; source 102 also gives its arXiv number.]*

References

- [1] M. Fuchs, G.-R. Yu and L. Zhang, *On the Asymptotic Growth of the Number of Tree-Child Networks*, European Journal of Combinatorics **93** (2021), 103278. Author PDF dated 16 September 2020: <https://web.math.nccu.edu.tw/mfuchs/Number-of-TCNs-final.pdf>.
- [2] C. Banderier and M. Wallner, *Young tableaux with periodic walls: counting with the density method*, Séminaire Lotharingien de Combinatoire **85B** (2021), Article 47, 12 pp.; Theorem 4.1. <https://www.mat.univie.ac.at/~slc/wpapers/FPSAC2021/47.html>.
- [3] Y.-S. Chang, M. Fuchs, H. Liu, M. Wallner and G.-R. Yu, *Enumerative and Distributional Results for d -combining Tree-Child Networks*, Advances in Applied Mathematics **157** (2024), 102704; author version (primary manuscript) dated 25 March 2024. <https://web.math.nccu.edu.tw/mfuchs/d-comb-journal-rev.pdf>.
- [4] Z. Lin, F. Liu, J. Liu, J. Liu and G. Xin, *Proof of a Conjecture on Young Tableaux with Walls*, arXiv:2601.09551v3, 6 September 2026; Theorem 1.1. <https://arxiv.org/html/2601.09551v3>.
- [5] Y.-S. Chang, *An Enumerative and Probabilistic Study of Various Phylogenetic Network Classes*, dissertation catalogued (listed by the author) in 2026; title page and inspected PDF compile date 31 December 2025. Printed pp. 100 and 137 (Remark 35). <https://web.math.nccu.edu.tw/mfuchs/Yu-Sheng-The-sis.pdf>.
- [6] OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, entry A213863. <https://oeis.org/A213863>.
- [7] *Transseries and inversion*, a volume of the ProveIt repository: `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` (theorems `p0:thm:lambert-core`, `p0:thm:perturbed-inversion`, `p0:thm:staircase`). Added in writing (batch 77).
- [8] H. Liu, M. Wallner and G.-R. Yu, *A combinatorial framework for the Pons–Batle identity: Young tableaux, lattice paths, and limit laws*, LIPIcs **381**, AofA 2026, Article 13, pp. 13:1–13:20, 13 July 2026. <https://doi.org/10.4230/LIPIcs.AofA.2026.13>; arXiv:2605.07587v1. Added in writing (batch 102), from sources 104 and 102.
- [9] H. Yu and L. Zhang, *Asymptotic counting of binary phylogenetic networks*, arXiv:2605.23126v2, 13 June 2026. <https://arxiv.org/abs/2605.23126>. Added in writing (batch 102).
- [10] M. Fuchs and H. Yu, *A cube-root phase transition in tree-child networks and the enumeration threshold for galled networks*, arXiv:2608.20860v1, 21 August 2026. <https://arxiv.org/abs/2608.20860>. Added in writing (batch 102).
- [11] H. Yu and L. Zhang, *A short combinatorial proof of the Pons–Batle identity for counting tree-child networks*, arXiv:2609.04979v1, 4 September 2026. <https://arxiv.org/abs/2609.04979>. Added in writing (batch 102).
- [12] H. Liu, *Trident tableaux for tree-child networks with one reticulation node: A bijection with two-wall tableaux*, arXiv:2609.28892v1, 24 September 2026. <https://arxiv.org/abs/2609.28892>. Added in writing (batch 102), from source 104.
- [13] A. Elvey Price, W. Fang, B. Louf and M. Wallner, *Bivariate asymptotics via random walks: application to large genus maps*, arXiv:2506.06924v2, April 2026, Section 10.1. <https://arxiv.org/abs/2506.06924>. Added in writing (batch 102), from source 102.
- [14] OEIS Foundation, sequence A213864. <https://oeis.org/A213864>. Accessed by source 104 on 2 October 2026; added in writing (batch 102).
- [15] OEIS Foundation, sequence A213275. <https://oeis.org/A213275>. Accessed by source 104 on 2 October 2026; added in writing (batch 102).

- [16] OEIS Foundation, sequence A167484. <https://oeis.org/A167484>. Accessed by source 104 on 2 October 2026; added in writing (batch 102).